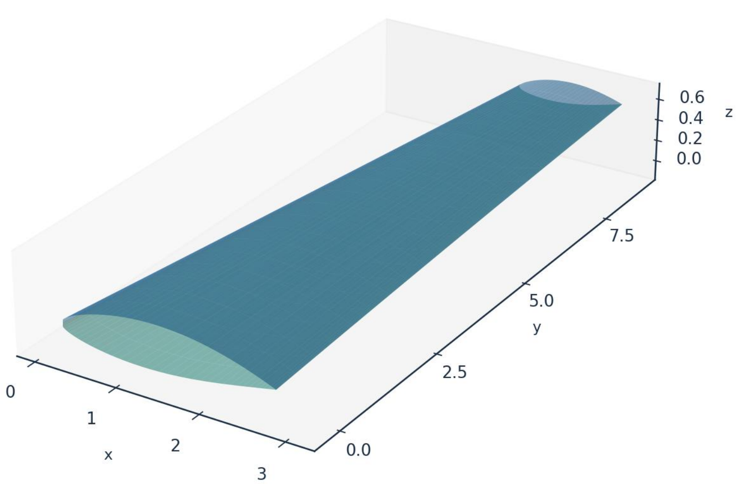
INDUSTRIAL CASE STUDY

Prediction of Boundary-Layer Turbulence Transition over Aircraft Surfaces

using the UTSS Universal Transition & Skin-Friction Solver

AETHER-NLF 25 WING - ISOMETRIC VIEW (DWG GEO-005)



Upper surface (blue) / Lower surface (teal) - lofted from UTSS-NLF16 sections | half-span shown
DRAWN BY: AKOSA SAMUEL ONYEJEKWE

Case vehicle: AETHER-NLF 25 regional natural-laminar-flow demonstrator
Wing section UTSS-NLF16 · Cruise FL360, M0.42 · 3-D swept tapered wing

Prepared by

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Document UTSS-CASE-2026 · Three-dimensional analysis · All units SI unless noted

1. Executive Summary

This case study demonstrates the prediction of laminar-to-turbulent boundary-layer transition over the surfaces of an aircraft wing, and the engineering quantities that depend on it (skin-friction drag, laminar-flow extent, boundary-layer growth and trailing-edge separation margin). The analysis vehicle is the AETHER-NLF 25, a regional natural-laminar-flow (NLF) demonstrator whose three-dimensional swept, tapered and twisted wing uses the purpose-designed UTSS-NLF16 section. Predictions are produced by the UTSS (Universal Transition & Skin-friction Solver), a novel, fast, robust engine that couples a vortex-panel inviscid solution to an integral boundary-layer marcher driven by a single, unified four-mechanism transition kernel.

Key results at the cruise design point (FL360, M=0.42, Re_MAC ≈ 6.4×10^6):

Quantity	Predicted value
Upper-surface transition x_tr/c	0.542 (TS-natural)
Lower-surface transition x_tr/c	0.566 (TS-natural)
Mean laminar-flow extent	55.4 % of chord
Section lift coefficient C_l	0.496
Section profile drag C_d	47.3 counts
Viscous drag reduction vs fully-turbulent	50.4 %
Climb (Tu=0.9 %) transition x_tr/c (upper)	0.025 (bypass)

Table 1. Headline predictions, read from the generated solution CSVs.

Validated against eight independent, credible published datasets with one universal calibration set — four ERCOFTAC flat plates (case 020), the Schubauer-Skramstad plate, the NLF(1)-0416 aerofoil at 86 conditions, and two swept wings — the solver reproduces the transition-onset Reynolds number Re_θt and the measured transition location without per-case re-tuning of the physics. All four criteria of the kernel are selected somewhere in this study and each is supported by measurement. The remainder of this report sets out the background, problem, governing equations, the complete input dataset, every generated engineering output (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors), the validation and calibration record with all sources, and the contribution to knowledge.

2. Background

Skin-friction drag is one of the largest components of total aircraft drag — typically 45–50 % of cruise drag for a transport aircraft. A turbulent boundary layer produces several times the skin friction of a laminar one at the same Reynolds number. Consequently, extending the laminar run over wing, nacelle and empennage surfaces (natural-laminar-flow, NLF, and hybrid-laminar-flow control, HLFC) is among the highest-leverage technologies for fuel-burn and emissions reduction. The enabling capability is the ability to PREDICT, reliably and cheaply, WHERE the boundary layer transitions from laminar to turbulent over a 3-D surface, across the flight envelope.

Transition is not a single phenomenon. Over aircraft surfaces it is driven by several, often co-resident, physical mechanisms:

- Natural / Tollmien–Schlichting (TS): amplification of viscous instability waves in low-disturbance free flight, predicted here by an e^N integral carried per physical frequency, with the spatial growth rates read from a tabulated solution of the Orr-Sommerfeld problem rather than from an envelope correlation.
- Bypass: free-stream-turbulence-induced transition that skips the TS route — dominant in high-turbulence environments (climb through cloud, turbomachinery-like inflow).
- Laminar-separation-induced: a laminar separation bubble that reattaches turbulent. The bubble is closed explicitly here — the shear layer is carried across the dead-air region by the momentum integral and reattachment placed where it has amplified by the same critical factor used elsewhere — so a length, and not merely a separation point, is predicted.
- Cross-flow: three-dimensional instability of the cross-flow velocity profile on swept wings — the principal NLF-limiter at moderate-to-high sweep.

A practical solver for aircraft design must capture all of these with a SINGLE, consistent formulation and calibration, run in seconds for thousands of design iterations, and remain robust across the operating envelope. Existing tools each address part of the problem (Section 5). UTSS unifies them.

3. Problem Statement and Solution Approach

3.1 Problem statement

Given a three-dimensional wing geometry and a flight condition, predict — quickly, robustly and accurately — the chordwise and span-wise location of boundary-layer transition on both surfaces, the governing transition mechanism at each station, and the resulting boundary-layer state (skin friction C_f, momentum thickness θ, shape factor H, intermittency γ), and from these the laminar-flow extent and the viscous (profile) drag. The method must be universal: one set of physics and calibration constants valid for natural, bypass, separation-induced and cross-flow transition across the Reynolds- and turbulence-number range of real aircraft surfaces.

3.2 How we solve it

UTSS solves the problem in a layered, fully-coupled manner:

- Inviscid edge flow: a constant-strength vortex-panel method (Kuethe–Chow) returns the surface pressure C_p and edge velocity U_e, with a Karman-Tsien correction applied to C_p and the integrated loads. It returns a stagnation pressure coefficient of 1.048 at M = 0.42 against the exact isentropic 1.045, where the linearised Prandtl-Glauert scaling gives 1.102, and the two agree to better than half a per cent below M = 0.2.
- Laminar boundary layer: the momentum AND kinetic-energy integral equations are advanced together from the stagnation point along each surface, so the shape factor is a solved variable carrying its own history rather than a local function of the pressure gradient. The three closure functions - H*(H), Re_theta*Cf/2 and Re_theta*C_D - are properties of

the Falkner-Skan family, computed from it and not fitted; every similarity solution is an exact fixed point of the second equation, and the march returns H = 2.5914 on a flat plate against the Blasius 2.5913. Fluid properties are evaluated at Eckert's reference temperature so that the incompressible closures return the compressible skin friction.

- Unified transition kernel (the novel core): at every station each of the four mechanisms reports how far through its own onset criterion the layer has got, as a progress that reaches unity at onset and carries a calibration weight, and the kernel fires at the first station where any of them completes. This paragraph used to describe the kernel as a minimum over four onset Reynolds numbers, which is the form Section 4.3 sets out at length as the one this replaced: only the bypass branch produces such a Reynolds number.
- Transitional region: a Narasimha universal-intermittency closure blends laminar and turbulent properties through γ.
- Turbulent boundary layer: Head's entrainment method with the Ludwieg–Tillmann skin-friction law advances the turbulent BL to the trailing edge.
- Integration: the Squire–Young formula returns the chordwise profile drag, which is carried into the streamwise frame together with the span-wise wall shear the span-wise momentum integral supplies (Eq. E20b-E20c); a span-wise strip sweep with the cross-flow mechanism extends the solution to the full 3-D wing.

Two elements of the formulation are not correlations. The amplification rate driving the e^N integral is tabulated in solver/amplification_db.npz from 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, continued past separation onto its reverse-flow branch: profiles are generated by continuation, the eigenvalue problem is solved by Chebyshev collocation, and the spatial growth rate is recovered from the temporal one through Gaster's transformation. The stability solver returns the Blasius neutral point at Re_theta = 200 against the accepted 200.5, and reproduces the standard Blasius Orr-Sommerfeld eigenvalue - c = 0.36412 + 0.00796i at Re_delta* = 998 - to better than a tenth of a per cent in the growth rate. What the march reads is the interpolated table, which first amplifies at Re_theta = 208, between Reynolds-number nodes at 204 and 234, so that offset is the resolution of that grid and not an error in the eigenvalue solver - a shape factor of 2.59129 and a wall shear parameter f''(0) = 0.46960. At run time the cost is a table lookup, so the sub-second solution is preserved. The second is the separation-bubble closure described above, whose length scales with the disturbance environment. That is measured rather than asserted (Table 30): 28 momentum thicknesses on the separating plate at 2.11 % free-stream turbulence against a median of 208 over the 50 aerofoil bubbles at 0.03 %, a spread of 7.4 that no fixed multiple of θ_s reproduces. Three places in this project previously quoted that spread, at four and a half, five and a half and from two different pairs of lengths; none of them was what the solver returns.

4. The UTSS Universal Solver — Governing Equations

All equations implemented in the solver are listed below as native, editable Word equations at standard size. They constitute the complete mathematical definition of the method, and are also collected in the companion file model.equations.docx.

4.1 Inviscid edge solution

(E02) Vortex-panel surface velocity (Kuethe & Chow)

$$\frac{V_{t_i}}{U_\infty} = \cos(\theta_i - \alpha) + \sum_{j=1}^{N} C_{t,ij}\gamma_j$$

(E03) Kutta condition (sharp trailing edge)

$$\gamma_1 + \gamma_{N+1} = 0$$

(E01) Pressure coefficient (inviscid)

$$C_p = 1 - \left(\frac{V}{U_\infty}\right)^2$$

(E04) Karman-Tsien compressibility correction

$$C_p = \frac{C_{p,0}}{\beta + \left(\frac{M_\infty^2}{1+\beta}\right)\frac{C_{p,0}}{2}}, \beta = \sqrt{1 - M_\infty^2}$$

4.2 Laminar boundary layer (two-equation integral march)

(E05) Thwaites momentum thickness (laminar)

$$\theta^2 = \frac{0.45\nu}{U_e^6}\int_0^x U_e^5 dx'$$

(E06) Thwaites pressure-gradient and momentum-thickness Reynolds number

$$\lambda = \frac{\theta^2}{\nu}\frac{dU_e}{dx}, Re_\theta = \frac{U_e\theta}{\nu}$$

(E06b) Kinetic-energy integral (two-equation laminar march)

$$\theta\frac{dH^*}{dx} = 2C_D - H^*\frac{C_f}{2} - H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx}$$

(E06c) Laminar closure functions (from the Falkner-Skan family)

$$H^*(H) = \frac{\theta^*}{\theta}, l(H) = Re_\theta\frac{C_f}{2} = \theta_\eta f''(0), d(H) = Re_\theta C_D = \theta_\eta \int (f'')^2 d\eta$$

(E07) Von Karman momentum-integral equation

$$\frac{d\theta}{dx} + (2+H)\frac{\theta}{U_e}\frac{dU_e}{dx} = \frac{C_f}{2}$$

4.3 Unified four-mechanism transition kernel (novel contribution)

Bypass onset uses the Abu-Ghannam & Shaw correlation evaluated at the flow-history-averaged Tu; natural/TS onset integrates one amplification factor per physical frequency using the tabulated Orr-Sommerfeld growth rates and triggers on their envelope at N_crit; separation-induced onset closes a laminar bubble across the dead-air region; and cross-flow onset closes an amplification integral on the stationary vortex (Eq. E11b), the C1 criterion of Eq. E11 serving only to mark where that integral starts.

Each branch reports the same quantity — how far through its own criterion the layer has got, as a number that reaches unity at onset — and the kernel fires at the first station where any of them does. Writing the four commensurably is what makes them one kernel: only the bypass branch produces an onset REYNOLDS NUMBER, the other three closing on amplification integrals, so a minimum taken over four Reynolds numbers ranges over one live term and three placeholders. Earlier versions of this report stated the kernel that way, and the output showed it: the onset-Reynolds-number column of every cruise surface file was entirely empty, because the branch that governs there does not produce one.

(E08) Abu-Ghannam & Shaw bypass onset

$$Re_{\theta t} = 163 + \exp\left[F(\lambda_\theta) - \frac{F(\lambda_\theta)Tu}{6.91}\right]$$

(E09) AGS pressure-gradient function

$$F(\lambda_\theta) = \begin{cases} 6.91 + 12.75\lambda_\theta + 63.64\lambda_\theta^2, & \lambda_\theta \leq 0 \\ 6.91 + 2.48\lambda_\theta - 12.27\lambda_\theta^2, & \lambda_\theta > 0 \end{cases}$$

(E10) Natural / Tollmien-Schlichting onset (e^N, envelope over frequency)

$$N(x) = \max_\omega \int_{x_0(\omega)}^{x} \frac{\sigma(H, Re_\theta, \omega\theta/U_e)}{\theta} dx' \geq N_{crit}$$

(E10b) Orr-Sommerfeld operator (tabulated amplification rates)

$$[(U-c)(D^2-\alpha^2) - U'']\hat{v} = \frac{1}{i\alpha Re_\theta}(D^2-\alpha^2)^2\hat{v}$$

(E10c) Gaster transformation (temporal to spatial growth rate)

$$\sigma = -\alpha_i\theta = \frac{\omega_i}{c_g}, c_g = \frac{\partial\omega_r}{\partial\alpha_r}$$

(E11) Cross-flow criterion (swept wing, C1 on Re_theta2)

$$Re_{\theta 2} = k_{cf} Re_{\theta,n} \sin\Lambda = k_{cf} Re_\theta \sin\Lambda\cos\Lambda \;\geq\; C_1, Re_{\theta,n} = Re_\theta\cos\Lambda$$

(E11b) Cross-flow amplification integral (the closure actually used)

$$N_{cf} = \int_{x_{c1}}^{x} \frac{\sigma(H_{rev})}{\theta} dx' \;\geq\; N_{crit}$$

(E11c) Cross-flow: velocity resolved along the wave-vector direction (Falkner-Skan-Cooke)

$$\frac{U_\psi(\eta)}{Q_e} = \cos\Lambda f'(\eta)\cos\psi + \sin\Lambda g(\eta)\sin\psi, U_\psi(\infty)/Q_e = \cos(\psi - \Lambda)$$

(E11d) Stationary cross-flow wave and the amplification it accumulates

$$\omega_r(k,\psi^*)=0,\frac{dN_{cf}}{dx}=\frac{\omega_i}{c_{g,x}}\frac{1}{\theta},c_{g,x}=\cos\psi\frac{\partial\omega}{\partial k}-\frac{\sin\psi}{k}\frac{\partial\omega}{\partial\psi}$$

(E12) Separation bubble: dead-air march (momentum and kinetic energy, no wall shear)

$$\frac{d\theta}{dx}=-(2+H)\frac{\theta}{U_e}\frac{dU_e}{dx},\theta\frac{dH^*}{dx}=2C_D-H^*(1-H)\frac{\theta}{U_e}\frac{dU_e}{dx},C_f=0$$

(E12b) Separation bubble: reattachment condition

$$N_{bub}=\int_{x_s}^{x_r}\frac{\sigma(H_{rev},Re_\theta)}{\theta}dx'=N_{crit},\sigma(H_{rev})\approx 0.0435$$

(E13) Unified transition kernel: onset where the first mechanism completes

$$x_t=\min\{x:\max_m p_m(x)\geq 1\},m\in\{nat,SEP,CF\},p_{nat}\text{ from Eq. (E14)}$$

(E13b) The four onset progresses, each reaching unity at its own onset, with each weight where it acts

$$p_{TS}=\frac{a_{TS}N}{N_{crit}},p_{BP}=\frac{Re_\theta}{a_{BP}Re_{\theta t}^{AGS}},p_{SEP}=\frac{a_{SEP}N_{bub}}{N_{crit}},p_{CF}=\frac{a_{CF}N_{cf}}{N_{crit}}$$

(E14) Natural / bypass blend across the validity window

$$p_{nat}=(1-w)p_{TS}+wp_{BP},w=3t^2-2t^3,t=\text{clip}\left(\frac{Tu-Tu_{lo}}{Tu_{hi}-Tu_{lo}},0,1\right)$$

The natural and bypass routes are the same transition seen through two closures with different ranges of validity, so Eq. E14 blends them over a declared window rather than switching between them. A single threshold made the predicted transition location a STEP function of the free-stream turbulence. Measured on this solver by forcing each closure alone at the old gate value of Tu = 0.1 %, which is what a switch there did: the amplification integral puts the cruise section's upper-surface transition at x_tr/c = 0.530 and the correlation at 0.385, a step of 0.145 c and of 14 counts (28 %) in profile drag across one part in a thousand of an input this study quotes to two figures, with the design point at 0.07 %. The window, Tu = 0.10–0.25 %, is wider than the spread of any case here (the noisiest natural case is 0.07 %, the quietest bypass case 0.87 %), so no result in this work is blended; it is there so that the model is a function of Tu rather than a switch. This paragraph used to quote 0.542 and 0.373 and "a third of the profile drag", a pair frozen from the edition that still had the gate; neither number reproduces now.

4.4 Transitional region and turbulent closure

(E15) Narasimha universal intermittency

$$\gamma(x)=1-\exp[-0.412\xi^2],\xi=\frac{x-x_t}{\lambda_{tr}}$$

(E16) Transition-length scale (Dhawan & Narasimha, in the two equivalent variables)

$$\lambda_{tr}=\frac{\nu}{U_e}\frac{C_{len}}{0.664^{3/2}}Re_{\theta,t}^{3/2}=\frac{\nu}{U_e}C_{len}Re_{x,t}^{0.75}|_{Re_\theta=0.664\sqrt{Re_x}},C_{len}=9$$

(E17) Intermittency-weighted property blend

$$\phi=(1-\gamma)\phi_{lam}+\gamma\phi_{turb}$$

(E18) Head entrainment (turbulent)

$$\frac{d(\theta H_1)}{dx} = C_E - \frac{\theta H_1}{U_e}\frac{dU_e}{dx}, C_E = 0.0306(H_1 - 3)^{-0.6169}$$

(E19) Ludwieg-Tillmann skin-friction law

$$C_f = 0.246 \times 10^{-0.678H} Re_\theta^{-0.268}$$

4.5 Drag, temperature and reference quantities

(E20) Squire-Young profile drag

$$C_d = 2\frac{\theta_{TE}}{c}\left(\frac{U_{e,TE}}{U_\infty}\right)^{(H_{TE}+5)/2}$$

(E20b) Span-wise momentum integral on an infinite swept wing (no span-wise pressure gradient)

$$\frac{d}{dx_n}\left[\rho U_{e,n} W \theta_{12}\right] = \tau_{w,z}, \theta_{12} = \int_0^\delta \frac{u}{U_{e,n}}\left(1 - \frac{w}{W}\right)dy \Rightarrow F_z = \rho W\left(U_{e,n}\theta_{12}\right)_{TE}$$

(E20c) Swept-strip profile drag in the streamwise frame (chordwise wake + span-wise friction)

$$C_d = 2\frac{\theta_{TE,n}}{c_n}\cos\Lambda\left[\cos^2\Lambda\left(\frac{U_{e,TE,n}}{U_n}\right)^{(H_{TE}+5)/2} + \sin^2\Lambda\frac{U_{e,TE,n}}{U_n}\right], c_l = c_{l,n}\cos^2\Lambda$$

(E21) Crocco-Busemann temperature profile

$$\frac{T}{T_e} = 1 + r\frac{\gamma - 1}{2}M_e^2\left[1 - \left(\frac{u}{U_e}\right)^2\right]$$

(E22) Recovery (adiabatic-wall) temperature

$$T_r = T_e\left(1 + r\frac{\gamma - 1}{2}M_e^2\right), r = Pr^{1/2}\ \text{(laminar)},\ Pr^{1/3}\ \text{(turbulent)}, r = (1-\gamma)Pr^{1/2} + \gamma Pr^{1/3}$$

(E22b) Eckert reference temperature (compressible closures)

$$\frac{T_{ref}}{T_e} = 1 + 0.032M_e^2 + 0.58\left(\frac{T_w}{T_e} - 1\right), \frac{\nu_{ref}}{\nu_e} = \frac{T_{ref}}{T_e}\frac{\mu_{Suth}(T_{ref})}{\mu_{Suth}(T_e)} \longrightarrow \left(\frac{T_{ref}}{T_e}\right)^{1+\omega}\ \text{where } T_e \text{ is not known}$$

(E25) Prandtl lifting line (Glauert monoplane equation, odd n for a symmetric wing)

$$\sum_{n\ \mathrm{odd}} A_n \sin n\theta\left[\frac{4b}{a_0 c(\theta)} + \frac{n}{\sin\theta}\right] = \alpha(\theta) - \alpha_{L0}, y = -\frac{b}{2}\cos\theta$$

(E26) Wing lift, induced drag and span efficiency from the loading

$$C_L = \pi AR A_1, C_{D_i} = \pi AR\sum_n nA_n^2, e = \frac{A_1^2}{\sum_n nA_n^2}$$

(E23) Reynolds number (mean aerodynamic chord)

$$Re_{MAC} = \frac{\rho_\infty U_\infty \bar{c}}{\mu_\infty} = \frac{U_\infty \bar{c}}{\nu_\infty}$$

(E24) Mean aerodynamic chord

$$\bar{c} = \frac{2}{3}c_{root}\frac{1 + \lambda + \lambda^2}{1 + \lambda}$$

Eq. E20c is the conversion of the profile drag out of the plane normal to the leading edge, and it is not the cos²Λ the section lift takes. Two forces act on a swept strip and Squire-Young returns only one of them. Resolving the free stream as Q = U_n + W with U_n = Q cosΛ normal to the leading edge and W = Q sinΛ along it, the chordwise profile drag acts along the normal direction, whose streamwise component is cosΛ, while the span-wise wall shear acts along the leading edge, whose streamwise component is sinΛ. The second is not in the chordwise wake deficit at all, but the span-wise momentum integral supplies it in closed form (Eq. E20b): with no span-wise pressure gradient the deficit grows at the rate of the wall shear, so integrating from the leading edge gives the whole span-wise friction force from the trailing-edge state, and no Squire-Young extrapolation applies to it because downstream of the trailing edge there is no wall and the span-wise deficit is frozen. That is why the two terms of E20c carry the edge-velocity ratio to different powers, and why the conversion cannot be a single factor on the total.

This work previously applied cos²Λ to the drag as well as to the lift, justified as "2ϑ/c times a velocity ratio that is frame-independent" — which does not produce cos²Λ either, since referring 2ϑ_n to the streamwise chord c = c_n/cosΛ gives one power of cosΛ, not two. The yawed flat plate settles it, because its answer is known independently: a flat plate at yaw is a flat plate in the free stream, so its drag is the unyawed value at the streamwise run length. At zero pressure gradient U_e,n = U_n and E20c returns exactly cosΛ for every sweep and every shape factor, and 2(ϑ_n/c_n)cosΛ = 2ϑ_n/c is that answer, the independence principle giving ϑ_n = ϑ_streamwise. cos²Λ is short by a further cosΛ — 2 per cent at the 12° of this wing, 29 per cent at 45° — and keeping only the correct cos³Λ on Squire-Young, with the span-wise term dropped, would be short by cos²Λ. The check is asserted in tools/smoke.py rather than described. On the case-study wing the corrected conversion leaves the section profile drag within 0.56 counts of the same section solved unswept (47.3 against 46.8), which is what a 12° sweep should do to a viscous drag; the drag REDUCTION relative to the fully-turbulent reference is unchanged, both configurations scaling together. ϑ_12 is taken as ϑ_n, the small-cross-flow closure w/W = u/U_e,n, which is exact at zero pressure gradient and is the standard turbulent one; the span-wise term carries 6.8 per cent of the total at this sweep (this said 4), so an error of a tenth in that closure is worth 0.68 per cent of C_d.

5. Why UTSS is Better — Comparison with Existing Solvers

UTSS is designed to be fast, robust and broad in regime coverage. The table contrasts its capabilities with the principal classes of existing transition tools.

Capability	XFOIL / e^N codes	RANS γ–Re_θ (CFD)	LES / DNS	UTSS (this work)
Natural / TS transition	Yes	Indirect	Yes	Yes (tabulated e^N)
Bypass (free-stream Tu)	No	Yes	Yes	Yes (AGS)
Separation-induced	Limited	Yes	Yes	Yes (bubble closure)
Cross-flow (3-D swept)	No	Add-on only	Yes	Yes (C1, built-in)
Single universal calibration	n/a	Needs re-tuning	n/a	Yes (one constant set)
3-D wing capability	2-D only	Yes	Yes	Yes (strip + cross-flow)
Robustness / convergence	Can fail in sep.	Stiff, costly	Very costly	Robust, direct
Typical CPU cost	seconds	hours	days–weeks	< 1 second
Suited to design	Partly	No	No	Yes

loops				

Table 2. Capability comparison versus existing solver classes.

Novelty. The distinguishing element is the unified transition kernel (Eq. E13): a single closed expression that selects the governing transition mechanism locally, by putting four co-resident mechanisms on one commensurable scale of onset progress — each carrying a calibration weight — and firing at the first station where any of them completes, then feeding a single intermittency closure. Those weights are NOT all in the same sense, and this sentence used to say they were: a_TS, and a_SEP and a_CF in the shipped configuration, multiply a PROGRESS, so a small value switches their branch off, while a_BP — and a_SEP and a_CF on their non-default paths — multiply an onset THRESHOLD, where switching off takes a large one. Eq. E13b writes each weight where it acts. That is not a nicety: a diagnostic in gen_validation.py took this paragraph at its word, set every weight small, and left the separation branch live on all ten swept-wing conditions it was meant to hold open. Unlike e^N codes (TS only) or correlation RANS models (which require case-by-case re-tuning and a full CFD solve), UTSS reproduces natural, bypass, separation and cross-flow transition with ONE calibration set at panel-method cost. Every one of the four branches is the selected mechanism somewhere in this study — natural/TS on the cruise wing, the Schubauer-Skramstad plate and the NLF(1)-0416 upper surface; bypass on the climb case and the ERCOFTAC plates; separation on T3C4 and the NLF(1)-0416 lower surface; cross-flow on both swept wings — and each is checked against measurement in Section 11.

6. Case-Study Definition and Input Data

All input data required to run the prediction are tabulated below. The case is fully three-dimensional.

6.1 Aircraft and wing geometry (input)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Trailing-edge included angle	26.8	deg
Section c_l, 2-D unswept (alpha = 1.5 deg, M = 0.42)	0.517	-

Table 3. Geometry definition (input).

6.2 Flight conditions (input)

case	altitude_m	mach	U_inf_ms	rho_kgm3	mu_Pas	T_inf_K
CRUISE (11 km, M0.42)	11000.0	0.42	123.93	0.3639	1.422e-05	216.65
CLIMB (3 km, M0.30)	3000.0	0.3	98.57	0.9093	1.694e-05	268.66

(columns 2-7 of 12; case repeated on each block)

case	Tu_pct	alpha_deg	Re_MAC	nu_m2s	q_inf_Pa
CRUISE (11 km, M0.42)	0.07	1.5	6414000.0	3.907e-05	2794.6
CLIMB (3 km, M0.30)	0.9	3.5	10700000.0	1.863e-05	4417.8

Table 4. Flight / flow conditions (input). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

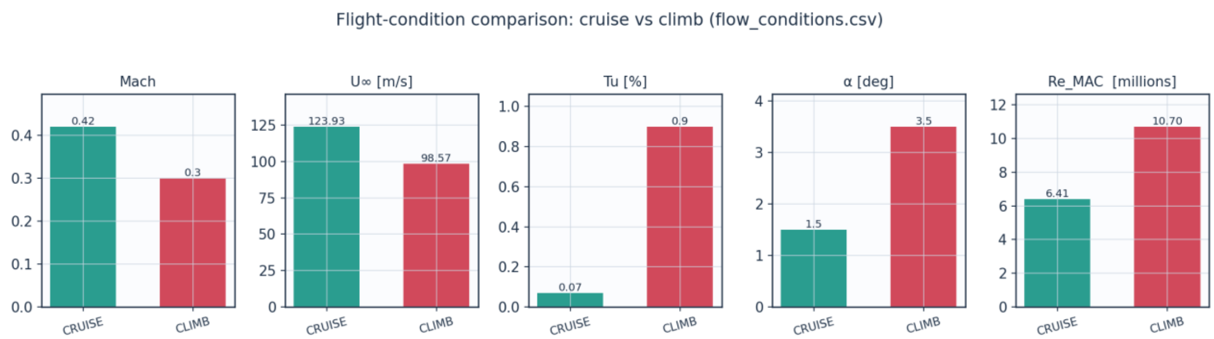


Fig. 1. Cruise vs climb flight-condition comparison (flow_conditions.csv).

6.3 Fluid (material) properties (input)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-05	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor, laminar (Pr^1/2)	0.849	-
Recovery factor, turbulent (Pr^1/3)	0.896	-
Cruise dynamic viscosity	1.422e-05	Pa.s
Cruise density	0.36392	kg/m^3

Table 5. Air properties (input).

6.4 Solver settings (input)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature

Laminar BL closure	two-equation march (momentum + kinetic energy); H*, I and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Transition length	Dhawan & Narasimha at constant spot formation rate, stated in Re_theta: Re_lambda = (9/0.664^1.5) Re_theta_t^1.5, which is their published Re_lambda = 9 Re_x_t^0.75 wherever Re_theta = 0.664 sqrt(Re_x) and is the form that stays exact in a pressure gradient (cal['len_re_x'] = True recovers the published form)
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	0.80 (reconstruction grid; no volume mesh is solved on it)

Table 6. Solver configuration (input).

6.5 Universal calibration constants (input)

constant	value	role	calibration_source
A_TS	1.0	TS/natural onset progress weight	unity, not fitted; the natural branch's one measured quantity is N_anchor, set on Schubauer-Skramstad alone, and the 86 NLF(1)-0416 conditions are out of sample
A_BP	1.0	bypass onset weight	unity; validated on ERCOFTAC T3A/T3A-/T3B
A_SEP	1.0	separation-induced weight	unity; validated on T3C4 and NLF(1)-0416 lower surface
A_CF	1.0	crossflow weight	unity; Arnal C1 criterion
N_crit	from Tu	e^N amplification factor	Mack N=-8.43-2.4 ln(Tu), clamped to 0.0008<=Tu<=0.0298, anchored -1.04 (set on the Schubauer-Skramstad plate ALONE; see N_anchor below); 7.64 for Tu<0.08%
N_floor	0.5	lower clamp on N_crit at high Tu	clamp; bypass governs there
Tu_BP_lo	0.1	Tu [%] below which the bypass correlation carries no weight	lower edge of the natural/bypass blend; below it the e^N branch alone governs
Tu_BP_hi	0.25	Tu [%] above which the bypass correlation carries all the weight	upper edge of the same blend. Replaces the single Tu_TS_max gate, which made the predicted transition location a step function of Tu; the window is wider than the spread of any case here, so no result in this study is blended
C_len	9.0	transition-length scale, Re_lambda = C_len Re_x_t^0.75	Dhawan & Narasimha (1958) published value; not fitted here
CF_C1	150.0	crossflow critical Re_theta2	Arnal et al.; FITTED on Dagenhart & Saric, and facility-dependent - the band C1 = 150-200 is reported rather than one value. The mean error each end returns on each set is in 06_validation/swept_wing_crossflow.csv and

			06_validation/swept_wing_independent.csv, and every formulation of the branch is scored in 06_validation/crossflow_formulations.csv
cf_amp	True	crossflow closed by an amplification integral, not a local threshold	rate computed (0.0435); threshold is the same N_crit; adds no constant
CF_ratio	0.47	theta2/theta surrogate for crossflow	FITTED on Dagenhart & Saric; independent check in Sec. IV.C
sigma_sep	from table	shear-layer amplification rate in a bubble	computed on the reverse-flow Falkner-Skan branch; 0.0435, not fitted
two_eq	True	two-equation laminar march (momentum + kinetic energy)	closures from the Falkner-Skan family; no fitted constant
N_anchor	1.04	offset on Mack's N_crit	unit conversion from Mack's 1977 growth rates to the present tabulated ones; set on the Schubauer-Skramstad plate ALONE, which it reproduces to 0.07 %. It was previously set jointly on that plate and the 86 NLF(1)-0416 conditions, which made that set a calibration set while three places in this project described it as one on which nothing is calibrated
C_mu	0.09	k-epsilon constant in the free-stream turbulence decay	standard value 0.09; with C_eps2 it fixes Tu(x) from the rig's integral length scale, and neither is fitted here
C_eps2	1.92	k-epsilon constant in the free-stream turbulence decay	standard value 1.92; sets the decay exponent of isotropic grid turbulence
bubble	True	separation branch closed by marching the detached shear layer	True is the model; False is the 'no bubble closure' ablation
use_os_db	True	amplification rates read from the tabulated Orr-Sommerfeld solutions	True is the model; False selects the Drela-Giles envelope fit, the third ablation
cf_stability	False	crossflow advanced from the solved stationary eigenvalue problem instead of the C1 threshold (diagnostic path only)	OFF; checked against Dagenhart & Saric's SALLY N-factors and scored in crossflow_formulations.csv, and it does not transfer between facilities any better
cf_exact	False	exact Falkner-Skan-Cooke K(lambda) in place of the constant surrogate (ablation path only)	False is the model; True was tried and is reported in Sec. IV.C, where it makes the two swept wings agree less, not more
len_re_x	False	transition-length correlation stated in Re_x rather than in Re_theta (ablation path only)	False is the model; the two are the same law - Dhawan & Narasimha's Re_lambda = 9 Re_x^0.75 IS constant-spot-rate Re_lambda = 16.63 Re_theta^1.5 on a flat plate - and Re_theta is the form that stays exact in a pressure gradient
bub_sigma_local	False	dead-air amplification rate read at the marched H instead of at the developed reverse-flow profile (ablation path only)	False is the model; True lengthens the T3C4 bubble towards the measurement but collapses the 86 NLF(1)-0416 conditions from 50 to 21 inside the bracket
H_sep	0.0	declare laminar	0 = use lam_sep; the shape-factor test is the

		separation on the solved shape factor instead of on lambda (ablation path only)	more principled statement but the aerofoil measurements reject it - see the note in march_bl
tu_hist	0.6	weight on the flow-history average of Tu, Tu_eff = Tu_avg^w Tu_local^(1-w)	FITTED on the three ERCOFTAC plates; no effect where Tu does not decay
lam_sep	-0.09	Thwaites parameter at laminar separation	Thwaites (1949) published value; not fitted
sy_x_ref	0.98	chordwise station at which Squire-Young is evaluated	NOT a discretisation parameter and not fitted to anything: the formula wants the trailing edge and a panel method cannot supply one, so the evaluation is pulled forward to the last station at which the cruise shape factor is still solved rather than sitting on Head's H = 2.8 clamp. The choice is worth about half a count of section drag per per cent of chord - more than any constant swept in this table - and what it costs is measured into 04_solution/squire_young_station_sensitivity.csv
sep_floor	120.0	min separation-induced onset Re_theta (ablation path only)	superseded by the bubble closure; reachable only with bubble=False
d_omega	1.03	max ratio between adjacent frequencies in the e^N integration	discretisation, stated as a RESOLUTION rather than a count: with a fixed count of 40 the Schubauer-Skramstad onset moved 5 % between 24 and 64 frequencies and did not settle. At this spacing Re_x,tr is unchanged from 1.10 down to 1.0075
n_freq_min	40	floor on the frequency count	guards a degenerate range
n_freq_max	400	cap on the frequency count	guards a degenerate range
CF_N	0.0	separate amplification threshold for the crossflow branch (ablation path only)	0 = use the same N_crit as every other branch; exposed to test whether a roughness-seeded branch needs its own threshold, and it does not - the sweep that settles it, with C1 refitted at every threshold so the two constants are not confounded, is 06_validation/crossflow_threshold_sweep.csv

Table 7. Calibration constants and their sources (input).

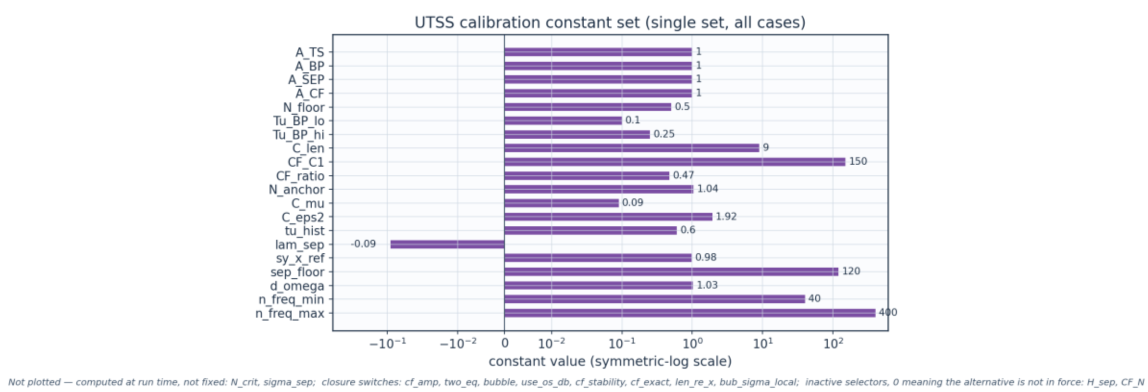


Fig. 2. UTSS universal calibration constant set (calibration_constants.csv).

7. Geometry and Engineering Drawings

The wing is drawn to standard third-angle orthographic projection. All drawings are dimensioned and to scale; views provided: section, plan, front, side, full orthographic sheet, isometric and structural section.

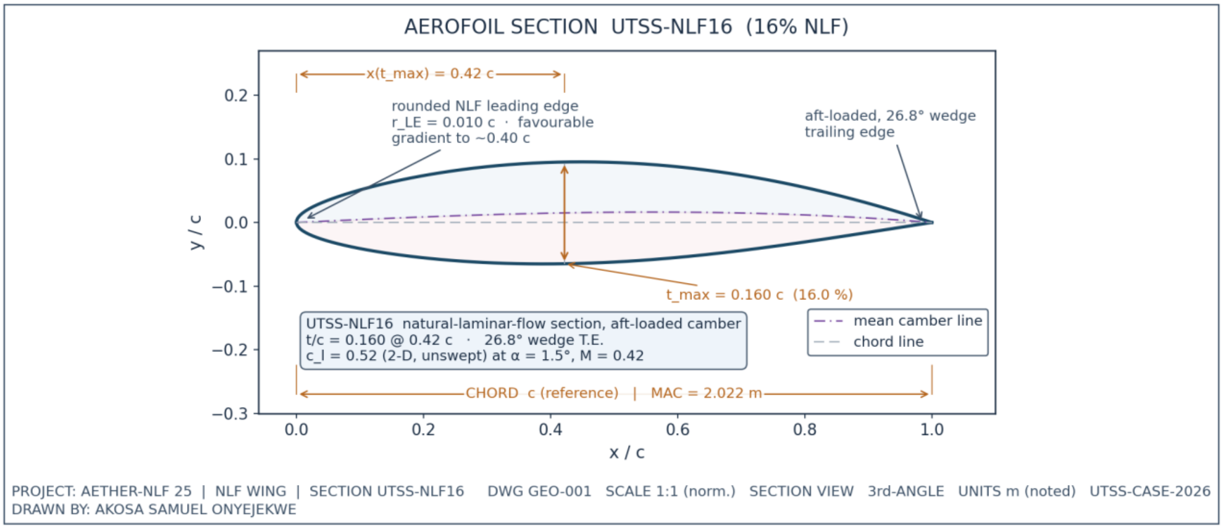


Fig. 3. UTSS-NLF16 aerofoil section (dimensioned).

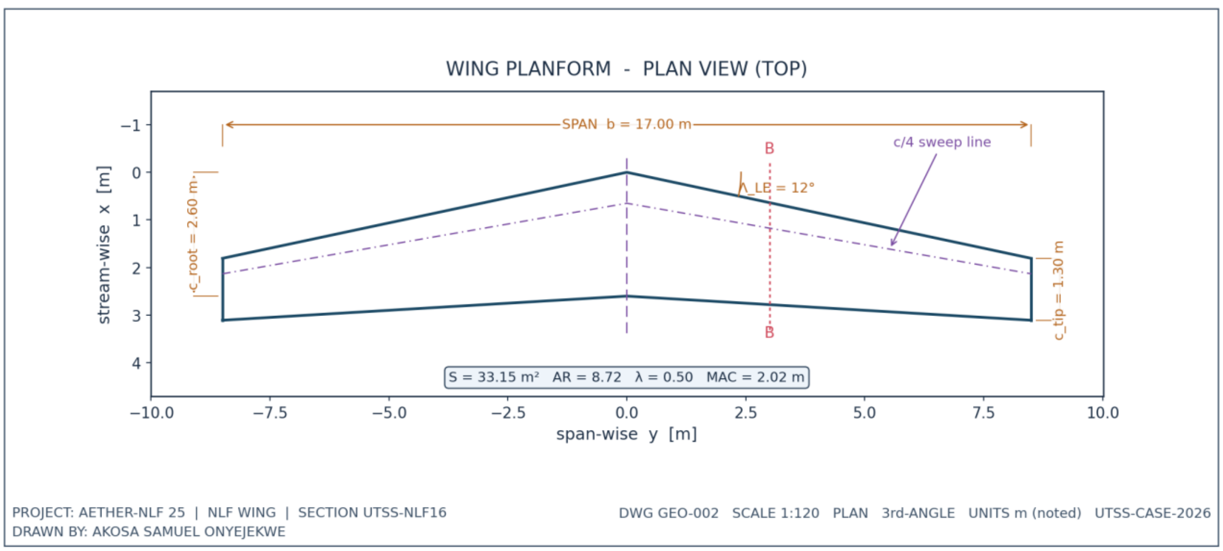


Fig. 4. Wing planform — plan (top) view, dimensioned.

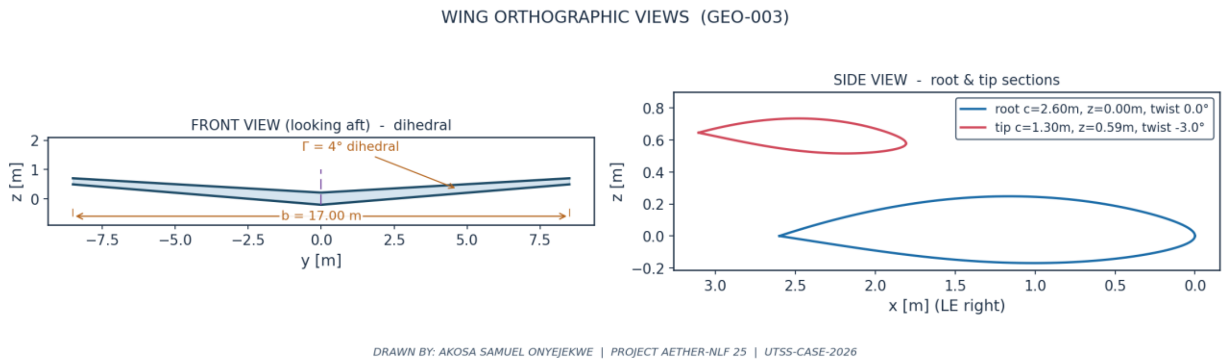


Fig. 5. Front and side orthographic views.

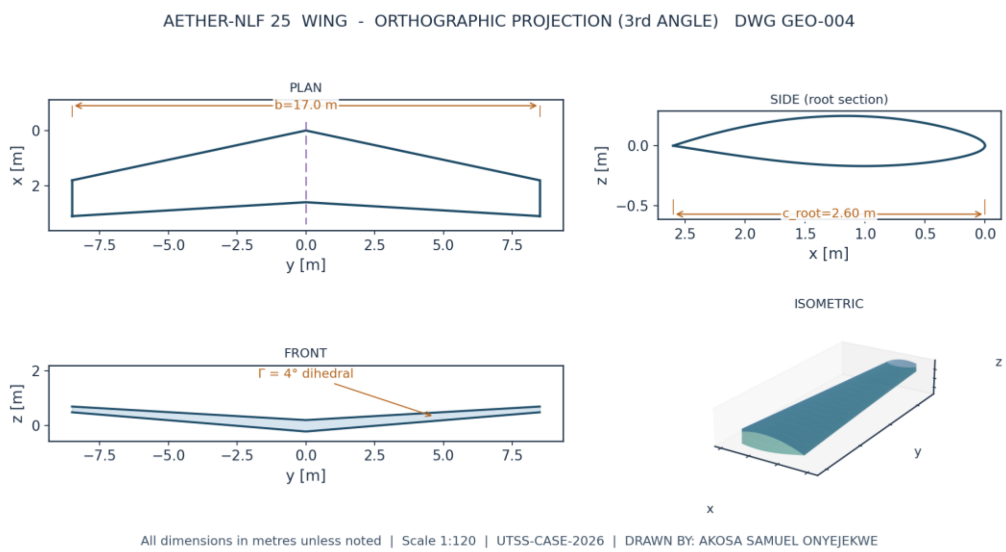


Fig. 6. Full third-angle orthographic projection sheet.

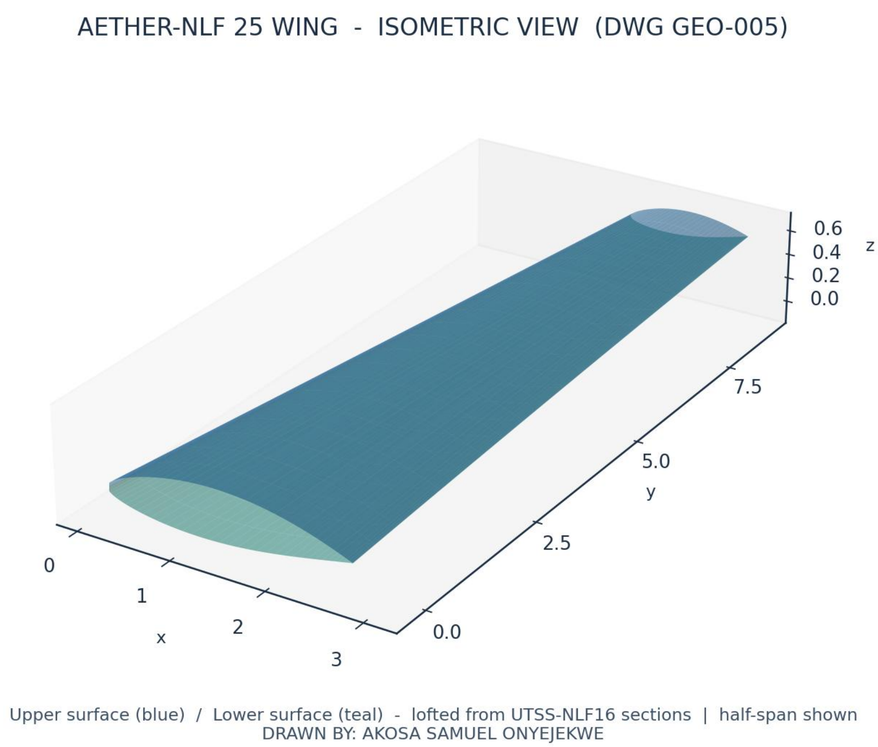


Fig. 7. Isometric view of the wing.

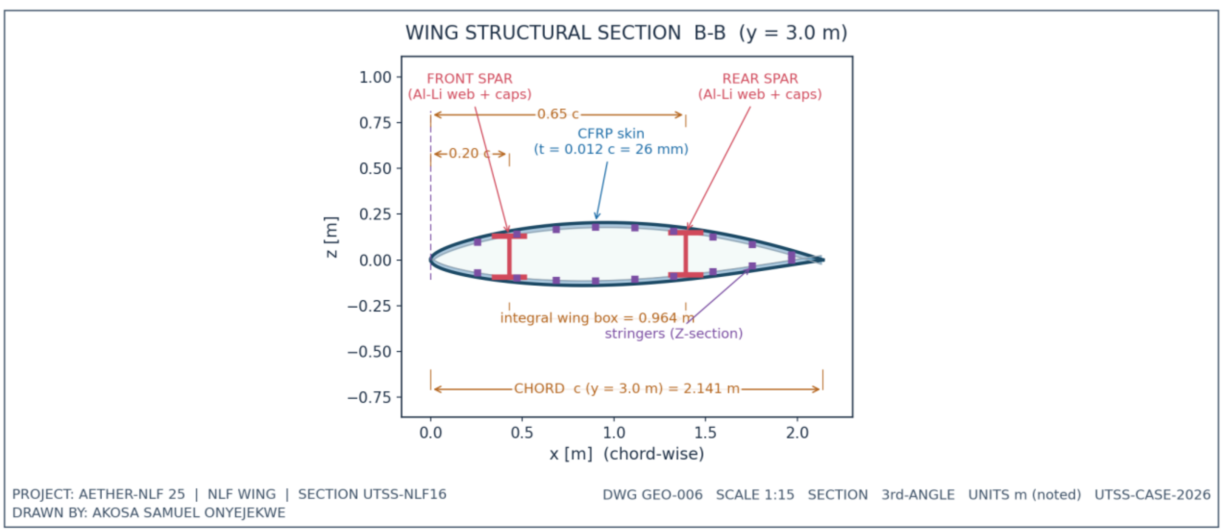


Fig. 8. Structural sectional view B–B.

7.1 Geometry data (from CSV)

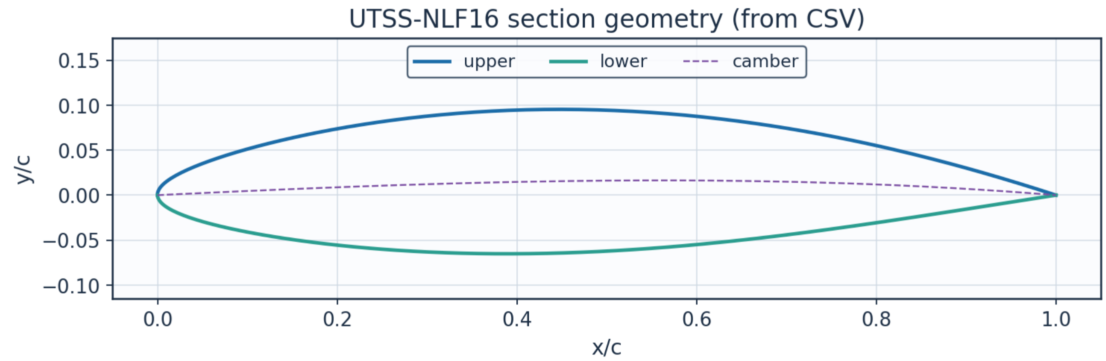


Fig. 9. Section geometry plotted from airfoil_UTSS-NLF16.csv.

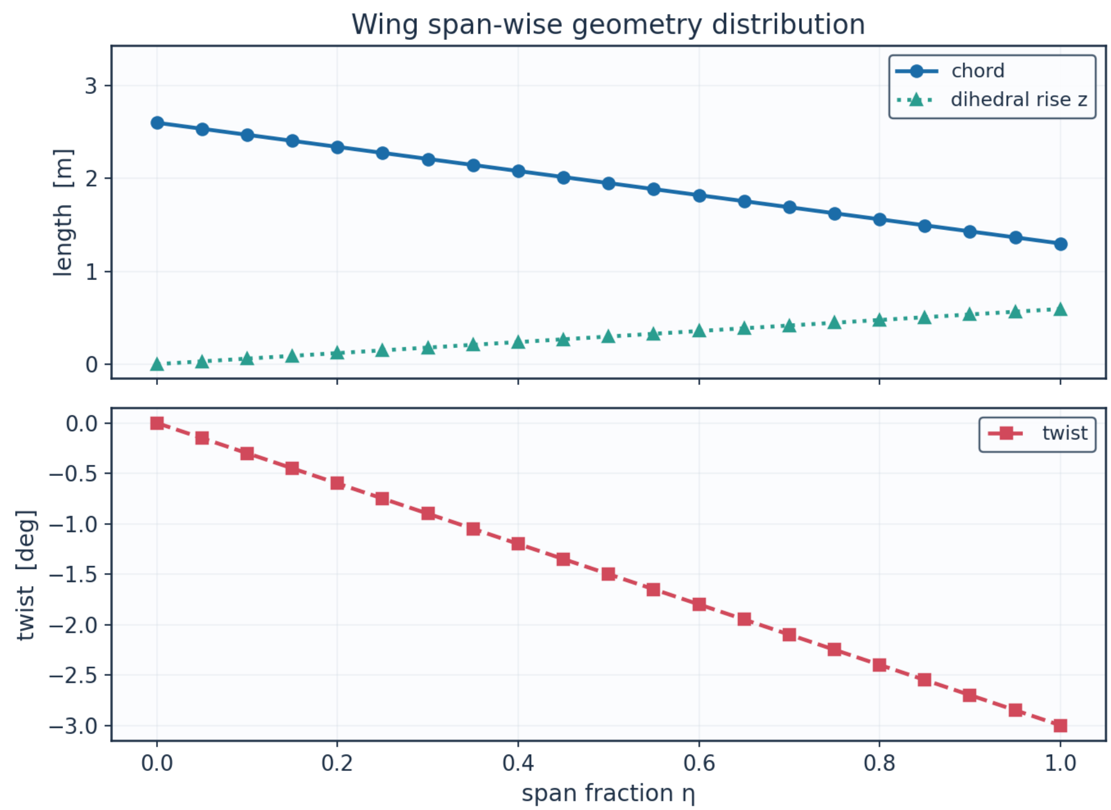


Fig. 10. Span-wise geometry from wing_planform.csv.

eta	y_m	chord_m	x_le_m	x_te_m	twist_deg	z_dihedral_m
0.0	0.0	2.6	0.0	2.6	0.0	0.0
0.05	0.425	2.535	0.0903	2.6253	-0.15	0.0297
0.1	0.85	2.47	0.1807	2.6507	-0.3	0.0594
0.15	1.275	2.405	0.271	2.676	-0.45	0.0892
0.2	1.7	2.34	0.3613	2.7013	-0.6	0.1189
0.25	2.125	2.275	0.4517	2.7267	-0.75	0.1486
0.3	2.55	2.21	0.542	2.752	-0.9	0.1783
0.35	2.975	2.145	0.6324	2.7774	-1.05	0.208

0.4	3.4	2.08	0.7227	2.8027	-1.2	0.2378
0.45	3.825	2.015	0.813	2.828	-1.35	0.2675
0.5	4.25	1.95	0.9034	2.8534	-1.5	0.2972
0.55	4.675	1.885	0.9937	2.8787	-1.65	0.3269
0.6	5.1	1.82	1.084	2.904	-1.8	0.3566
0.65	5.525	1.755	1.1744	2.9294	-1.95	0.3863
0.7	5.95	1.69	1.2647	2.9547	-2.1	0.4161
0.75	6.375	1.625	1.355	2.98	-2.25	0.4458
0.8	6.8	1.56	1.4454	3.0054	-2.4	0.4755
0.85	7.225	1.495	1.5357	3.0307	-2.55	0.5052
0.9	7.65	1.43	1.6261	3.0561	-2.7	0.5349
0.95	8.075	1.365	1.7164	3.0814	-2.85	0.5647
1.0	8.5	1.3	1.8067	3.1067	-3.0	0.5944

(columns 2-7 of 8; eta repeated on each block)

eta	Re_local
0.0	8246200.0
0.05	8040000.0
0.1	7833900.0
0.15	7627700.0
0.2	7421600.0
0.25	7215400.0
0.3	7009300.0
0.35	6803100.0
0.4	6597000.0
0.45	6390800.0
0.5	6184700.0
0.55	5978500.0
0.6	5772300.0
0.65	5566200.0
0.7	5360000.0
0.75	5153900.0
0.8	4947700.0
0.85	4741600.0
0.9	4535400.0
0.95	4329300.0
1.0	4123100.0

Table 8. Wing planform stations (wing_planform.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

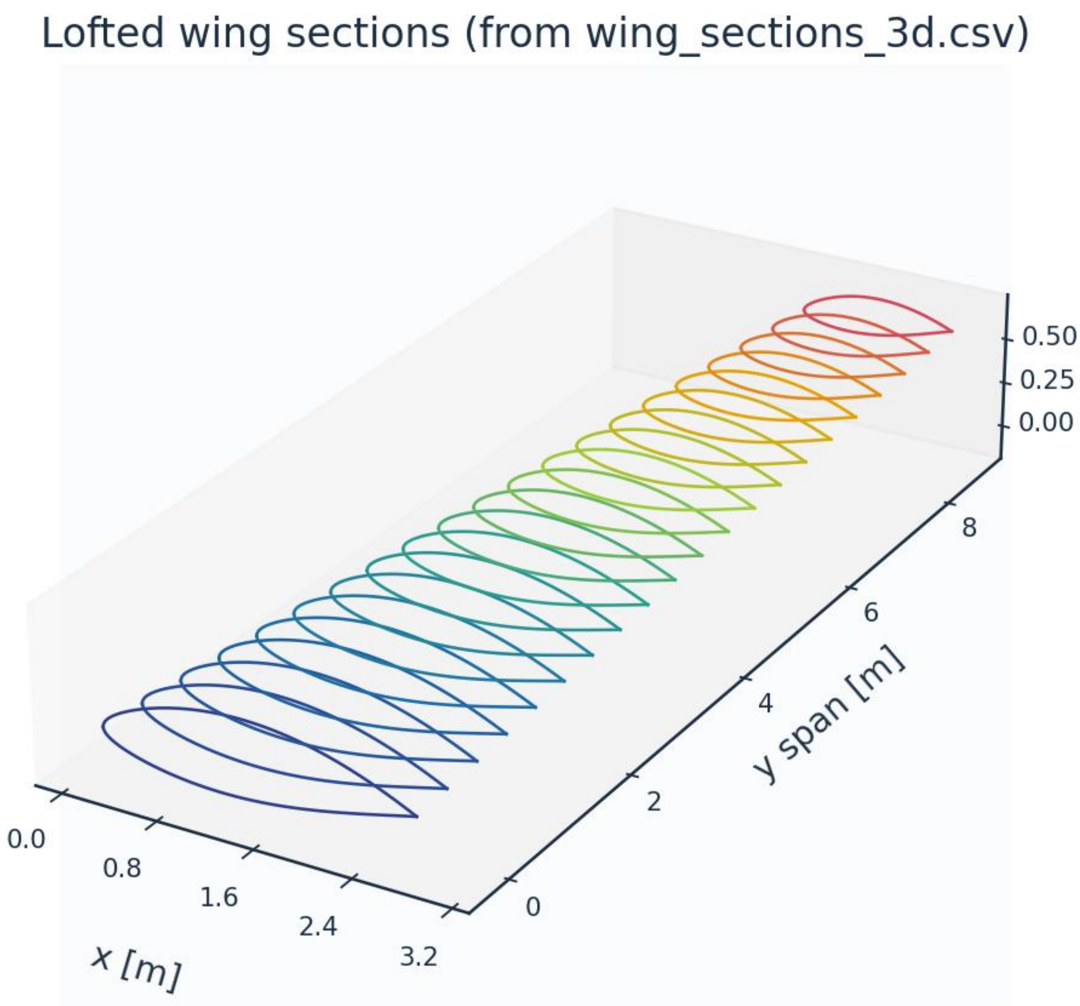


Fig. 11. Lofted 3-D wing sections (wing_sections_3d.csv).

8. Mesh / Discretisation

THERE IS NO VOLUME MESH. This method is a surface panel discretisation coupled to an integral boundary layer, and the wall-normal stack below is a reconstruction grid used to recover profiles from the marched integral quantities - it is not a grid any equation is solved on. The y^+, growth ratio and aspect ratio in Table 9 are reported because they characterise that reconstruction, and they are the quantities a reader coming from a finite-volume solver will look for; they should not be read as evidence of a resolved near-wall grid, because there is none to resolve. This paragraph used to say so only in the last row of the table.

The surface is discretised with cosine-clustered streamwise nodes; the wall-normal reconstruction grid places its first point at $y^+ = 0.80$. A panel-count sweep does NOT demonstrate asymptotic convergence, and is reported because it does not: from 120 to 2600 surface panels — a factor of 22, and a decade on the grid every case-study result is computed on — the section drag rises at EVERY refinement, from 45.4 to 49.7 counts, and the successive changes do not decay (+0.43, +0.20, +0.21, +0.25 per cent over the last four). There is no asymptote here to quote a discretisation error against. What that costs the headline number is stated rather than left to be inferred: the 260-panel production grid returns 47.3 counts and the finest grid computed returns 49.7, so the section drag quoted throughout this report sits 2.4 counts — 5.2 % — BELOW the finest resolution tested, and that gap is still opening. This report previously called the residual a wander "set by which panel the transition point lands on", and then explained it by a momentum thickness at the Squire-Young station that "rises MONOTONICALLY". Neither is what the table says. The transition location moves by only 0.008 chord across the whole sweep and with no trend, so it is not the first; and θ at that station falls at 4 of the 11 refinements, so it is not the second. What does climb without interruption is the SHAPE FACTOR there, 2.02 to 2.28, and $C_d = 2(\theta/c)(U_e/U_\infty)^{((H+5)/2)}$ is EXPONENTIAL in it. x/c = 0.98 sits close enough to the trailing edge that refining the panels resolves the inviscid singularity there progressively more sharply, so what is not converging is the EVALUATION STATION, not the transition station hopping between panels. The 260-panel grid is used for every case-study result; the tabulated validation sections are re-splined onto their own cosine-clustered grids of 400 and 440 panels.

metric	value	unit
Surface streamwise nodes	261	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	7.80	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.151	1e-3 c
Max streamwise spacing	12.217	1e-3 c
Max streamwise spacing at MAC	24.705	mm
LE clustering ratio	81.1	-
BL grid outer extent	8.43	mm
Peak turbulent C_f (cruise, both surfaces)	0.00209	-
Friction velocity u_tau at that peak	4.010	m/s
Max cell aspect ratio (at MAC)	3169	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Table 9. Metrics of the surface discretisation and of the wall-normal reconstruction stack. No volume mesh is generated.

n_surface_panels	Cl	Cd	x_tr_upper_c	theta_at_sy_c	H_at_sy	dCd_pct
120	0.49385	0.0045362	0.539	0.00259	2.019	
180	0.49531	0.0045938	0.5434	0.002609	2.006	1.269
260	0.49617	0.0047307	0.542	0.002834	2.14	2.981
360	0.4967	0.0047975	0.539	0.002829	2.109	1.411
480	0.49705	0.0048867	0.5357	0.002944	2.169	1.859
600	0.49726	0.0048911	0.539	0.002893	2.144	0.09
700	0.49737	0.0049152	0.5379	0.002964	2.21	0.493
900	0.49753	0.0049209	0.5399	0.002909	2.168	0.116
1200	0.49767	0.0049423	0.5403	0.002952	2.143	0.435
1600	0.49777	0.0049522	0.54	0.002986	2.236	0.202
2000	0.49783	0.0049624	0.5398	0.002974	2.243	0.206
2600	0.49789	0.0049749	0.539	0.002999	2.278	0.251

Table 10. Panel-count sensitivity study.

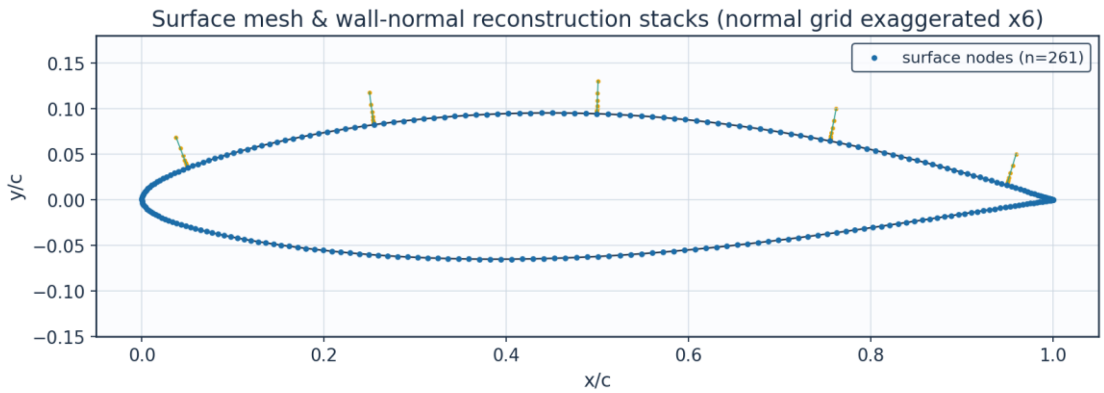


Fig. 12. Surface mesh and wall-normal stacks.

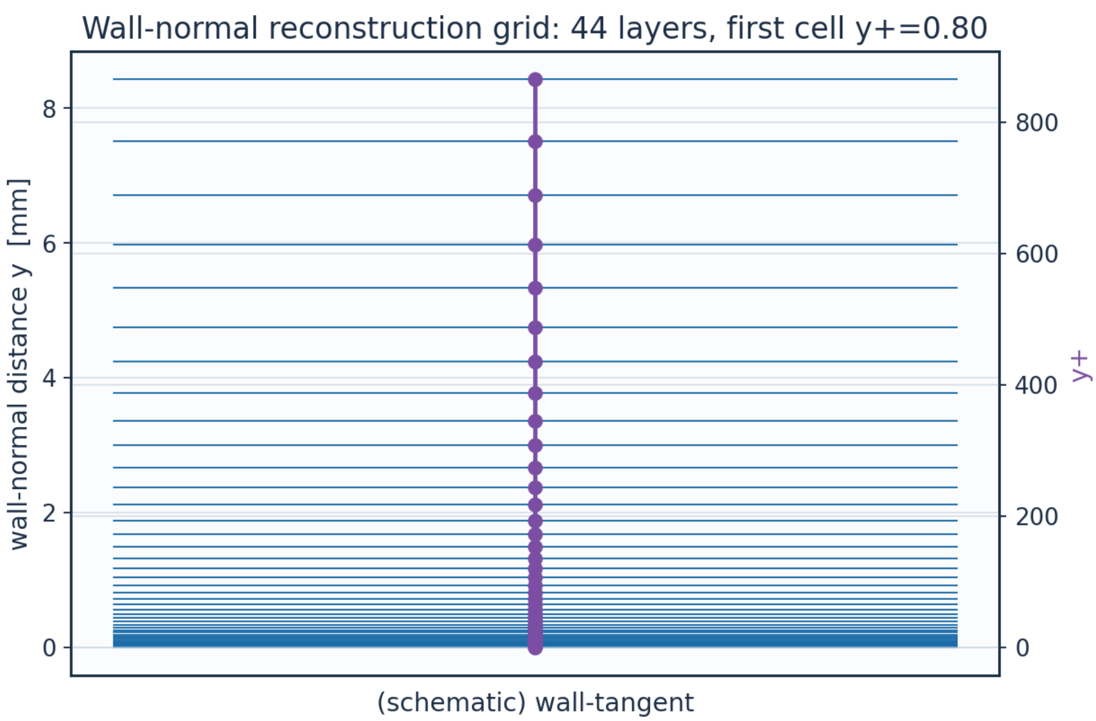


Fig. 13. Wall-normal reconstruction grid / y^{+}.

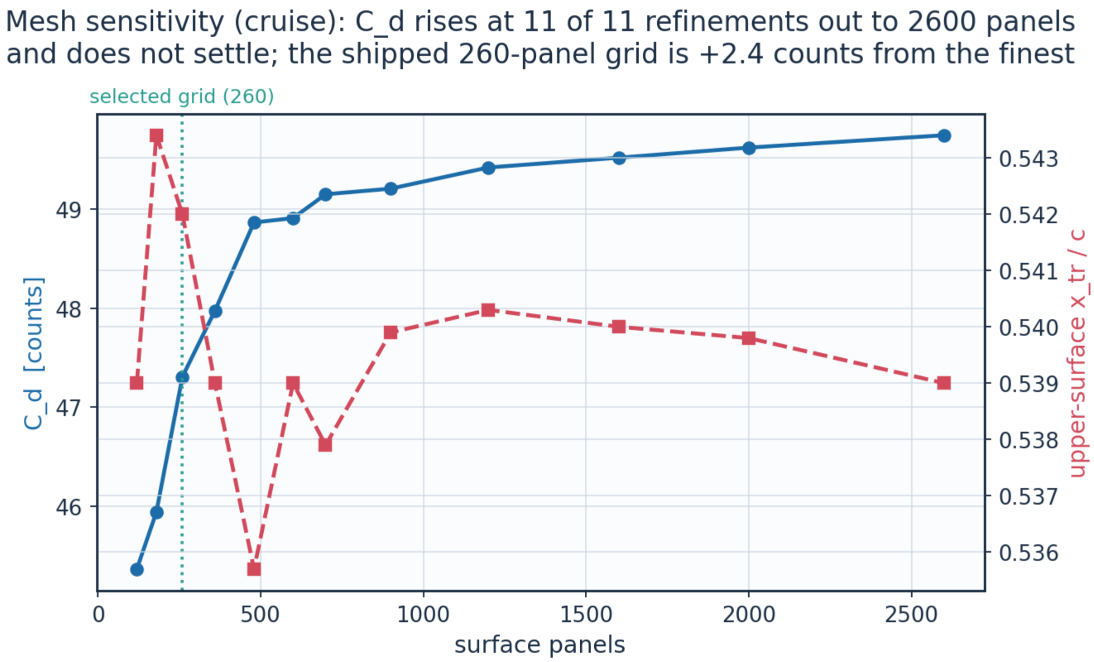


Fig. 14. Panel-count sensitivity of C_d and transition location.

layer	y_mm	y_plus	cell_dy_mm	growth_ratio
0	0.0	0.0	0.0078	1.0
2	0.0165	1.696	0.00925	1.12
4	0.0373	3.8235	0.01161	1.12
6	0.0633	6.4922	0.01456	1.12
8	0.0959	9.8398	0.01827	1.12
10	0.1368	14.039	0.02291	1.12
12	0.1881	19.3065	0.02874	1.12
14	0.2525	25.9141	0.03605	1.12
16	0.3333	34.2026	0.04523	1.12
18	0.4346	44.5998	0.05673	1.12
20	0.5617	57.642	0.07117	1.12
22	0.7211	74.0021	0.08927	1.12
24	0.921	94.5242	0.11198	1.12
26	1.1719	120.2671	0.14047	1.12
28	1.4865	152.5591	0.1762	1.12
30	1.8812	193.0661	0.22103	1.12
32	2.3763	243.8782	0.27726	1.12
34	2.9974	307.6168	0.34779	1.12
36	3.7764	387.5705	0.43627	1.12
38	4.7537	487.8644	0.54726	1.12
40	5.9795	613.6731	0.68648	1.12
42	7.5172	771.4876	0.86112	1.12

Table 11. Wall-normal grid (bl_normal_grid.csv, sampled).

(table sampled evenly to 22 of 44 rows; full data in 02_mesh/bl_normal_grid.csv)

9. Solution — Generated Engineering Output Data

This section presents every generated output: numerical tables (CSV) and the plotted curve for each. The boundary-layer state is reported on both surfaces at the cruise and climb conditions.

9.1 Transition prediction summary

case	surface	s_tr_c	x_tr_c	Re_x_tr	Re_theta_at_onset	mechanism
CRUISE	upper	0.567	0.542	3.476e+06	1382.2	TS-natural
CRUISE	lower	0.575	0.566	3.530e+06	1225.2	TS-natural
CLIMB	upper	0.047	0.025	4.850e+05	485.5	bypass
CLIMB	lower	0.168	0.166	1.724e+06	612.0	bypass

(columns 2-7 of 18; case repeated on each block)

case	laminar_run_pct	Cf_te	H_te	H_te_at_clip	sep_margin_H	x_sy_c
CRUISE	54.2	0.0002	2.801	True	-0.201	0.9811
CRUISE	56.6	0.00024	2.802	True	-0.202	0.9804
CLIMB	2.5	0.00014	2.8	True	-0.2	0.9811
CLIMB	16.6	0.00019	2.8	True	-0.2	0.9804

(columns 8-13 of 18; case repeated on each block)

case	H_sy	H_sy_at_clip	x_sep_turb_c	sy_margin_to_sep_c	sy_past_sep
CRUISE	2.14	False	0.987	0.006	False
CRUISE	1.673	False	0.997	0.0165	False
CLIMB	2.8	True	0.978	-0.0034	True
CLIMB	1.623	False	0.998	0.0177	False

Table 12. Transition prediction summary (transition_summary.csv). (columns 14-18 of 18; the table is split across 3 blocks so that no cell is compressed to illegibility, with case repeated on each)

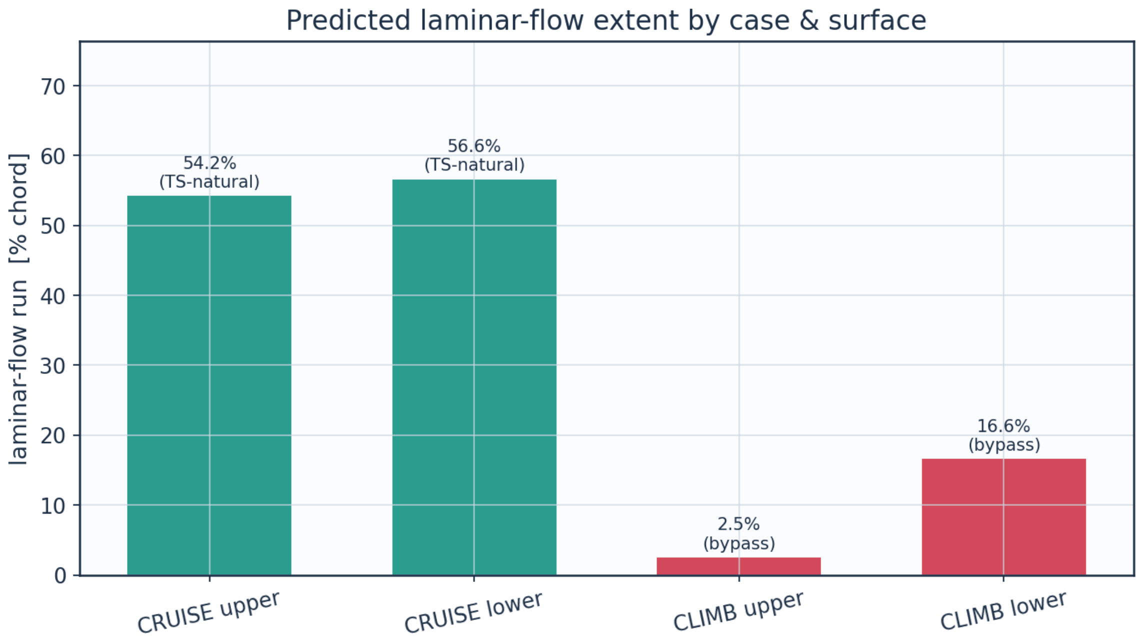


Fig. 15. Predicted laminar-flow extent by case and surface (transition_summary.csv).

9.2 Integrated forces and drag breakdown

The wing lift is obtained from Prandtl's lifting-line theory - Glauert's monoplane equation solved for the odd Fourier coefficients of the loading - using the section lift-curve slope and zero-lift incidence returned by the same panel method, the planform chord distribution, the wing's washout, and the section slope reduced by the cosine of the quarter-chord sweep. An earlier version of this table asserted $C_L = 0.90\, c_l$; the planform actually returns 0.513, because a −3° washout is large relative to the 3.7° by which the root exceeds its zero-lift incidence, and because the downwash of an AR = 8.7 wing reduces the slope by a quarter. The span efficiency and the induced drag come out of the same solve. The implementation is checked against the one case with a closed-form answer - an elliptic planform, for which it returns e = 1 and the exact lift-curve slope to machine precision - every time the solution is regenerated.

The incidence in Table 4 is the SECTION design incidence, not the trim point of the aircraft the planform belongs to: at 1.5° the wing carries 23.6 kN against an all-up weight of 83.4 kN, and level flight at MTOW would need 7.6°. Table 13 reports both, so the two are not confused. Every transition result in this study is quoted at the design incidence and is unaffected by that distinction.

quantity	value	unit
Section lift coefficient Cl	0.4962	-
Section profile drag Cd	0.00473	-
Section Cd (counts)	47.3	counts
Section L/D	104.9	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-

Wing C_L / section c_l	0.513	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6.414e+06	-
Mach number	0.42	-

Table 13. Integrated forces, with the wing quantities from the lifting-line solve.

configuration	Cd_counts	mean_laminar_pct	viscous_drag_reduction_pct
NLF (UTSS predicted transition)	47.3	55.4	50.4
Fully turbulent (LE trip)	95.4	0.0	0.0

Table 14. NLF vs fully-turbulent drag.

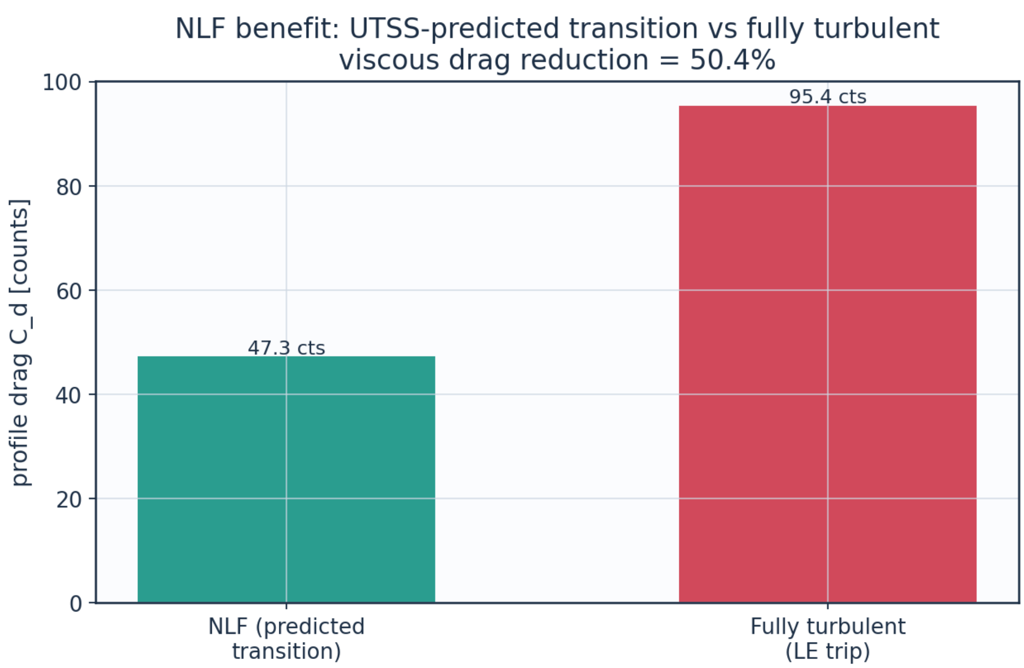


Fig. 16. Drag benefit of predicted laminar flow vs fully-turbulent.

9.2a The transition-length closure and what the result owes to it

The extent of the transitional region is Dhawan and Narasimha's published correlation, Re_λ = 9 Re_x,t^0.75, with λ the distance over which the intermittency rises from 0.25 to 0.75. This work previously reported it as validated on the flat plates of Section 11.1, which span Re_x,t = 6×10^4 to 1.4×10^6 — the four ERCOFTAC plates, which are the ones that carry C_f through transition at all, though only two of them complete it (Table 16); Schubauer & Skramstad reaches 2.8×10^6 but gives only a station — and EXTRAPOLATED on the wing, which transitions at 3.5×10^6. That extrapolation was of the variable, not of the physics.

Their correlation is Narasimha's spot model with a constant dimensionless spot formation rate. Matching $\gamma = 1 - \exp(-0.412\ \xi^2)$ against $\gamma = 1 - \exp(-n\ \sigma\ (x-x_t)^2/U)$, with $\hat{N} = n\ \sigma\ \theta_t^3/\nu$, gives $Re_\lambda = \sqrt{(0.412/\hat{N})}\ Re_{\theta,t}^{1.5}$; and on a Blasius plate $Re_\theta = 0.664\ \sqrt{Re_x}$, so $Re_\theta^{1.5} = 0.5411\ Re_x^{0.75}$ and the two are THE SAME LAW with C = 9/0.5411 = 16.63. Checked over Re_x from 6×10^4 to 10^7 they agree to every figure printed. What is assumed constant is therefore the spot formation rate, and Re_θ is the variable in which that assumption is stated exactly. Nothing is fitted and nothing is extrapolated: the constant is still their published 9.0.

Measured, the two forms agree to 0.03 per cent on the 4 zero-pressure-gradient plates — as they must, since $Re_\theta = 0.664\ \sqrt{Re_x}$ there — and differ by 2.71 per cent on T3C4, the one plate that carries a pressure gradient, where the Re_θ form is the correct one. On the cruise wing the transitional length moves by 0.40 per cent and the section drag by 0.03 counts. That is not the point. The point is that the wing no longer sits outside the range of a correlation; it sits inside the range of an assumption that was always that correlation's actual content. cal["len_re_x"] recovers the published form so the equivalence can be checked rather than taken on trust, and Table 15 is that check.

case	pressure_gradient	lam_len_Re_theta_form_m	lam_len_Re_x_form_m	diff_pct	Cd_counts_Re_theta_form	Cd_counts_Re_x_form
ERCOFTAC T3A flat plate (bypass transition)	False	2.18908e-01	2.18859e-01	-0.02		
ERCOFTAC T3A- flat plate (low-Tu bypass)	False	2.28688e-01	2.28640e-01	-0.02		
ERCOFTAC T3B flat plate (high-Tu bypass)	False	5.78118e-02	5.78223e-02	0.02		
ERCOFTAC T3C4 flat plate (laminar separation bubble)	True	5.60247e-01	5.75404e-01	2.71		
Schubauer & Skramstad flat plate (natural transition)	False	3.37479e-01	3.37391e-01	-0.03		
AETHER-NLF 25 cruise section (upper surface)	True	2.22578e-01	2.21678e-01	-0.4	47.31	47.28

Table 15. The two forms of the transition-length correlation, on every plate and on the cruise section (transition_length_forms.csv). They are the same law wherever Re_theta = 0.664 sqrt(Re_x), and part company only where a pressure gradient breaks that.

The LENGTH the closure returns is a separate question from the onset every other table here scores, and it is measured in Table 16 rather than asserted. Taking Narasimha's own definition — the distance over which the intermittency runs from 0.25 to 0.75, with the measured intermittency formed from the measured skin friction against the laminar and turbulent flat-plate correlations — only 2 of the 4 plates carrying C_f data resolve a length at all: ERCOFTAC T3A- because gamma has only reached 0.38 at the last measured station; ERCOFTAC T3C4 because a pressure gradient, so the flat-plate correlations the measured intermittency is formed against do not apply. On the 2 that do, the model returns 1.32 and 0.91 times the measured extent. This report previously called the closure "validated on the four ERCOFTAC plates" and quoted a factor of two, and neither figure had a generating source.

case	resolves_the_length	not_resolved_because	Re_x_at_gamma_025	Re_x_at_gamma_075	Re_lambda_measured	Re_lambda_model
ERCOFTAC T3A flat plate (bypass transition)	True		1.867e+05	2.464e+05	5.971e+04	7.881e+04
ERCOFTAC T3A- flat plate (low-Tu bypass)	False	gamma has only reached 0.38 at the last measured station				2.973e+05
ERCOFTAC T3B flat plate (high-Tu bypass)	True		6.470e+04	1.044e+05	3.966e+04	3.623e+04
ERCOFTAC T3C4 flat plate (laminar separation bubble)	False	a pressure gradient, so the flat-plate correlations the measured intermittency is formed against do not apply				7.223e+04

(columns 2-7 of 8; case repeated on each block)

case	model_over_measured
ERCOFTAC T3A flat plate (bypass transition)	1.32
ERCOFTAC T3A- flat plate (low-Tu bypass)	
ERCOFTAC T3B flat plate (high-Tu bypass)	0.91
ERCOFTAC T3C4 flat plate (laminar separation bubble)	

Table 16. The transition LENGTH against the plates that resolve one (transition_length_measured.csv). Two of the four plates with skin-friction data do not: T3A- has not completed the rise at its last measured station, and T3C4's pressure gradient invalidates the flat-plate correlations the measured intermittency is formed against. (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

What the case-study drag owes to the closure is measured all the same, by sweeping the constant over a factor of four:

C_len	multiple_of_published	Cd_counts	x_tr_c_upper	transitional_extent_pct_	completes_before_TE

				chord	
2.25	0.25	47.35	0.542	8.4	True
4.5	0.5	47.35	0.542	18.5	True
9.0	1.0	47.31	0.542	36.2	True
18.0	2.0	44.37	0.542		False
36.0	4.0	38.97	0.542		False

Table 17. Sensitivity of the case-study result to the transition-length constant (transition_length_sensitivity.csv).

9.3 Surface distributions — cruise

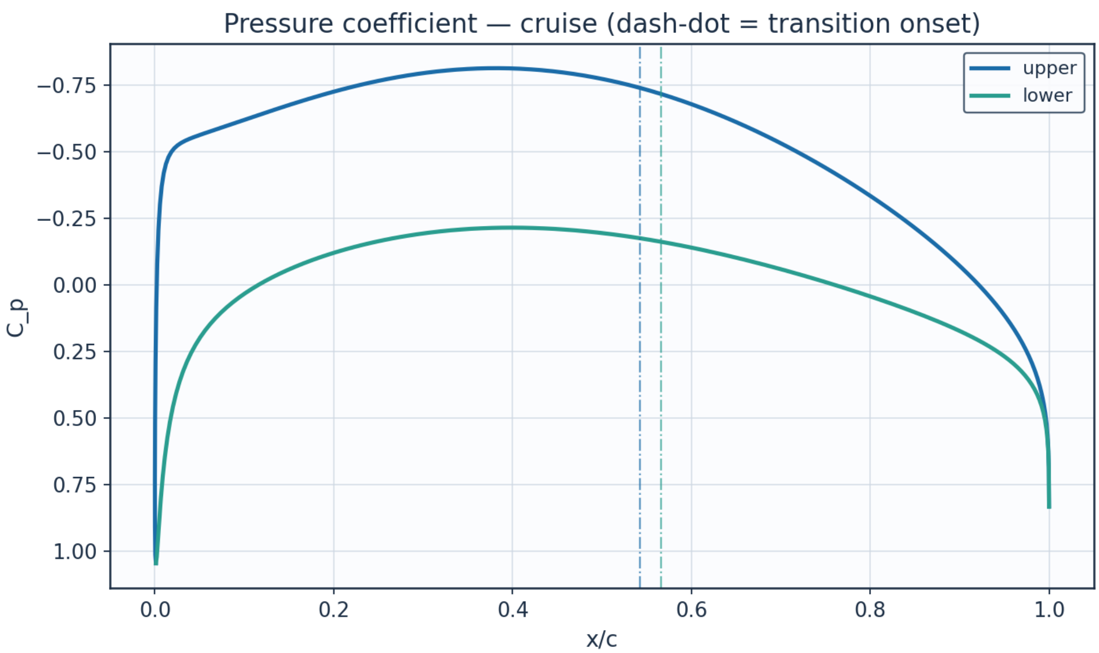


Fig. 17. Pressure coefficient C_p — cruise.

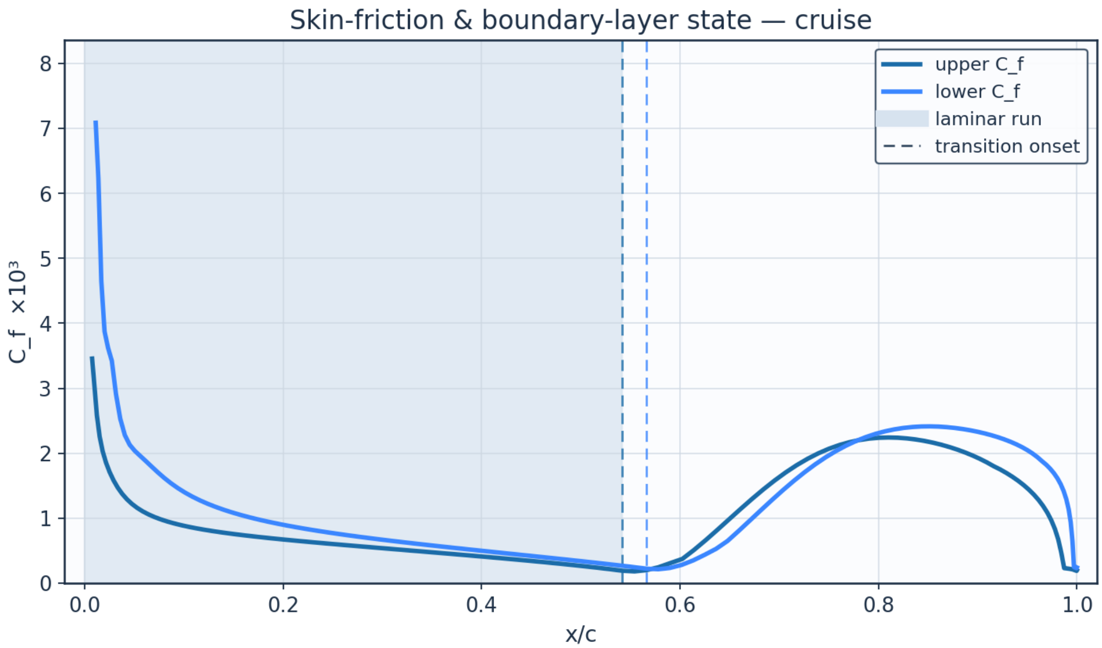


Fig. 18. Skin-friction C_f and laminar run — cruise.

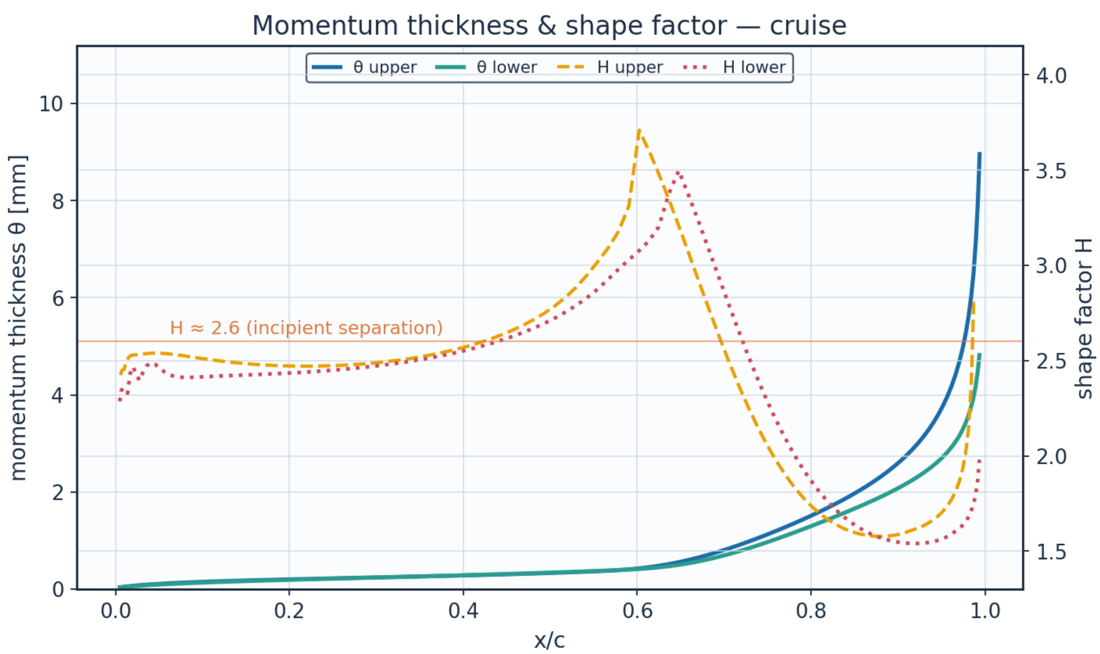


Fig. 19. Momentum thickness ϑ and shape factor H — cruise.

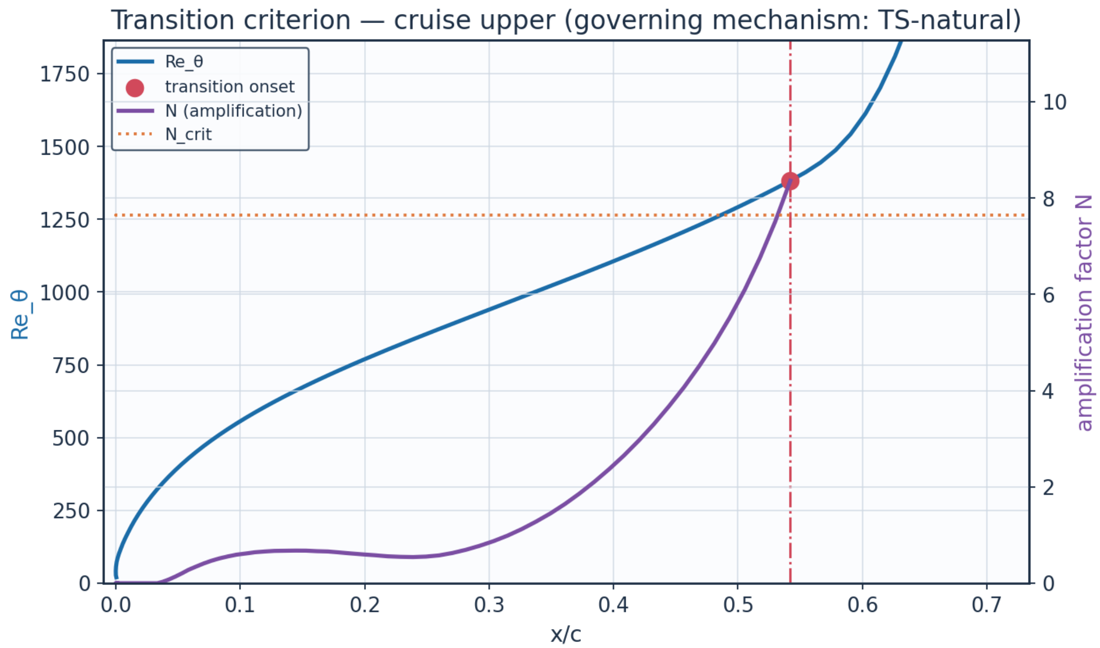


Fig. 20. The governing transition criterion — cruise. Two of the four branches are Reynolds-number thresholds and two are amplification integrals, so both pairs are drawn; at cruise it is N reaching N_crit that fires, and no Re_ϑt threshold is active at any station.

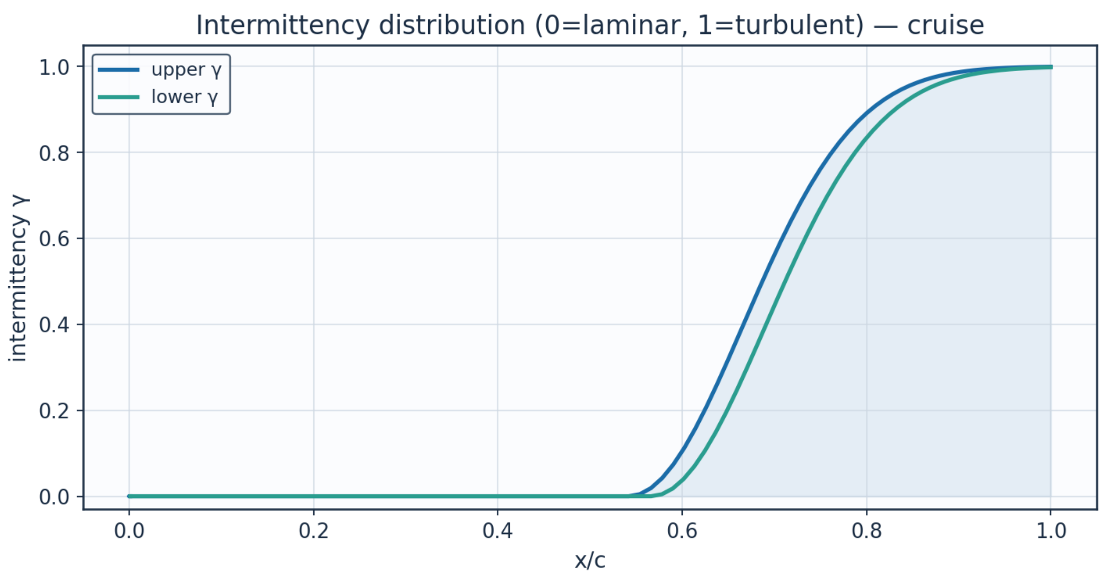


Fig. 21. Intermittency γ — cruise.

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152	0.0	0.0	1.04617	0.0001	0.0	
0.000243	0.01379	42779.0	0.51697	85.1945	0.68744	0.02686
0.003998	0.029694	92120.0	-0.20578	133.2104	1.07489	0.03616
0.012383	0.051716	160438.0	-0.45363	146.6018	1.18295	0.05498
0.025317	0.08137	252436.0	-0.52328	150.2001	1.21198	0.07756
0.042683	0.119051	369334.0	-0.55314	151.7229	1.22427	0.10055

0.064323	0.164707	510973.0	-0.57855	153.0105	1.23466	0.12252
0.090042	0.218063	676500.0	-0.60768	154.4766	1.24649	0.14298
0.119612	0.278705	864629.0	-0.64128	156.1558	1.26004	0.16196
0.152769	0.346116	1073758.0	-0.6778	157.9661	1.27465	0.17974
0.189218	0.419703	1302050.0	-0.71481	159.7856	1.28933	0.1967
0.228631	0.498815	1547479.0	-0.74951	161.4789	1.30299	0.2133
0.27065	0.582752	1807876.0	-0.77904	162.9098	1.31454	0.23005
0.31489	0.670778	2080961.0	-0.80063	163.9512	1.32294	0.2475
0.360938	0.762129	2364360.0	-0.81192	164.4934	1.32732	0.26625
0.408355	0.856016	2655625.0	-0.81105	164.4517	1.32698	0.28694
0.456686	0.951626	2952238.0	-0.7969	163.7712	1.32149	0.3103
0.505459	1.048128	3251618.0	-0.76911	162.4298	1.31066	0.33723
0.554191	1.144672	3551128.0	-0.72812	160.4365	1.29458	0.36907
0.6024	1.240392	3848080.0	-0.675	157.8277	1.27353	0.42789
0.649607	1.334412	4139757.0	-0.61134	154.6601	1.24797	0.56249
0.695348	1.425852	4423433.0	-0.53899	151.0029	1.21846	0.77932
0.739176	1.513841	4696404.0	-0.45992	146.9295	1.18559	1.05292
0.780671	1.597528	4956026.0	-0.37596	142.51	1.14993	1.35625
0.819445	1.676092	5199755.0	-0.28872	137.8042	1.11196	1.67822
0.855143	1.748757	5425183.0	-0.19946	132.8555	1.07203	2.02448
0.887448	1.814803	5630080.0	-0.10897	127.6845	1.0303	2.41205
0.916081	1.873579	5812420.0	-0.01756	122.2815	0.9867	2.86717
0.940801	1.924505	5970409.0	0.07514	116.593	0.9408	3.43016
0.961407	1.967085	6102504.0	0.17034	110.4954	0.8916	4.17154

(columns 2-7 of 15; x_c repeated on each block)

x_c	H_shape	Cf	Re_theta	Re_theta_ trans	n_factor	n_crit
0.001152					0.0	7.644
0.000243	2.2505	0.0120153	57.44		0.0	7.644
0.003998	2.3462	0.0051951	117.43		0.0	7.644
0.012383	2.4932	0.0025878	194.42		0.0	7.644
0.025317	2.5307	0.0017084	280.16		0.0	7.644
0.042683	2.539	0.0012919	366.39		0.0777	7.644
0.064323	2.5337	0.00106	449.74		0.343	7.644
0.090042	2.5171	0.0009209	529.19		0.5559	7.644
0.119612	2.4986	0.0008256	605.05		0.659	7.644
0.152769	2.483	0.0007521	678.13		0.676	7.644
0.189218	2.4731	0.0006895	749.4		0.6178	7.644
0.228631	2.4709	0.000632	819.96		0.5483	7.644
0.27065	2.478	0.0005762	890.99		0.6283	7.644
0.31489	2.4961	0.00052	963.77		0.989	7.644
0.360938	2.5284	0.0004618	1039.68		1.6448	7.644
0.408355	2.5795	0.0004001	1120.23		2.6523	7.644
0.456686	2.6594	0.0003329	1207.2		4.073	7.644
0.505459	2.7874	0.0002575	1302.85		6.0965	7.644
0.554191	3.0161	0.0001818	1411.0			7.644
0.6024	3.7096	0.0003739	1612.64			7.644
0.649607	3.1858	0.0009854	2082.17			7.644
0.695348	2.6144	0.0015807	2824.11			7.644

0.739176	2.1468	0.0019973	3724.18			7.644
0.780671	1.8364	0.0022026	4668.9			7.644
0.819445	1.665	0.0022387	5607.28			7.644
0.855143	1.5908	0.0021618	6546.42			7.644
0.887448	1.5784	0.0020109	7525.4			7.644
0.916081	1.6098	0.0018004	8600.43			7.644
0.940801	1.6645	0.0015753	9849.17			7.644
0.961407	1.7693	0.0012791	11396.92			7.644

(columns 8-13 of 15; x_c repeated on each block)

x_c	intermittency_gamma	state
0.001152	0.0	laminar
0.000243	0.0	laminar
0.003998	0.0	laminar
0.012383	0.0	laminar
0.025317	0.0	laminar
0.042683	0.0	laminar
0.064323	0.0	laminar
0.090042	0.0	laminar
0.119612	0.0	laminar
0.152769	0.0	laminar
0.189218	0.0	laminar
0.228631	0.0	laminar
0.27065	0.0	laminar
0.31489	0.0	laminar
0.360938	0.0	laminar
0.408355	0.0	laminar
0.456686	0.0	laminar
0.505459	0.0	laminar
0.554191	0.00482	transitional
0.6024	0.11253	transitional
0.649607	0.31631	transitional
0.695348	0.5393	transitional
0.739176	0.72367	transitional
0.780671	0.8492	transitional
0.819445	0.92319	transitional
0.855143	0.96243	transitional
0.887448	0.98183	transitional
0.916081	0.99104	turbulent
0.940801	0.99537	turbulent
0.961407	0.99742	turbulent

Table 18. Cruise upper-surface BL state (surface_cruise_upper.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled evenly to 30 of 133 rows; full data in 04_solution/surface_cruise_upper.csv)

x_c	arc_s_m	Re_x	Cp	Ue_ms	Ue_Uinf	theta_mm
0.001152	0.0	0.0	1.04617	0.0001	0.0	
0.006721	0.017175	53283.0	0.80073	57.4943	0.46393	0.04058
0.016897	0.041011	127228.0	0.51407	85.4337	0.68937	0.05478
0.031587	0.07262	225291.0	0.32494	99.9191	0.80626	0.07455
0.05065	0.112235	348189.0	0.19779	108.6835	0.87698	0.09566
0.073903	0.159719	495497.0	0.10481	114.7223	0.92571	0.11579

0.101119	0.214733	666169.0	0.03157	119.2949	0.9626	0.1361
0.132034	0.276811	858752.0	-0.02902	122.97	0.99226	0.15595
0.166346	0.345385	1071492.0	-0.08027	126.0087	1.01678	0.17531
0.203719	0.419819	1302410.0	-0.12345	128.5229	1.03707	0.19438
0.243794	0.499421	1549360.0	-0.15873	130.5483	1.05341	0.21347
0.286186	0.583462	1810078.0	-0.18577	132.0838	1.0658	0.23296
0.330494	0.671185	2082224.0	-0.20407	133.1145	1.07412	0.25326
0.376307	0.761822	2363408.0	-0.2132	133.6266	1.07825	0.27482
0.423206	0.854593	2651213.0	-0.21303	133.6172	1.07817	0.29813
0.470769	0.948713	2943202.0	-0.20381	133.1	1.074	0.32368
0.518574	1.043391	3236922.0	-0.18618	132.1069	1.06598	0.35203
0.566202	1.13783	3529900.0	-0.16115	130.6864	1.05452	0.38381
0.613236	1.231224	3819638.0	-0.12998	128.8999	1.04011	0.42965
0.659265	1.322763	4103619.0	-0.09405	126.8151	1.02328	0.5334
0.703882	1.411629	4379309.0	-0.05469	124.4997	1.0046	0.71353
0.746688	1.497007	4644178.0	-0.01313	122.0148	0.98455	0.94956
0.787291	1.57809	4895723.0	0.02971	119.409	0.96352	1.21062
0.825311	1.65409	5131498.0	0.07322	116.7134	0.94177	1.47574
0.860385	1.724249	5349152.0	0.11716	113.9362	0.91936	1.73895
0.892165	1.787851	5546467.0	0.16175	111.0573	0.89613	2.00539
0.920332	1.844238	5721396.0	0.20775	108.0196	0.87162	2.28746
0.944596	1.892815	5872098.0	0.25655	104.7133	0.84494	2.60558
0.964701	1.933066	5996968.0	0.31058	100.9425	0.81452	2.99662
0.980432	1.964558	6094667.0	0.37411	96.3421	0.77739	3.54004

(columns 2-7 of 15; x_c repeated on each block)

x_c	**H_shape**	**Cf**	**Re_theta**	**Re_theta_ trans**	**n_factor**	**n_crit**
0.001152					0.0	7.644
0.006721	2.343	0.0103492	59.19		0.0	7.644
0.016897	2.4318	0.0046458	117.44		0.0	7.644
0.031587	2.4378	0.0029171	185.57		0.0	7.644
0.05065	2.4615	0.0020363	257.69		0.0	7.644
0.073903	2.4129	0.001705	328.05		0.0	7.644
0.101119	2.4149	0.0013955	399.76		0.0	7.644
0.132034	2.4213	0.0011746	470.99		0.0	7.644
0.166346	2.4276	0.0010134	541.39		0.0	7.644
0.203719	2.4356	0.0008883	611.18		0.0	7.644
0.243794	2.4474	0.0007852	680.79		0.0	7.644
0.286186	2.4649	0.0006957	750.82		0.0287	7.644
0.330494	2.4904	0.0006143	821.97		0.3215	7.644
0.376307	2.5269	0.0005375	895.04		0.9562	7.644
0.423206	2.5781	0.0004625	970.89		1.9437	7.644
0.470769	2.6512	0.000387	1050.45		3.3109	7.644
0.518574	2.7571	0.0003088	1134.77		5.1296	7.644
0.566202	2.9248	0.0002236	1225.17		7.7128	7.644
0.613236	3.1268	0.0003506	1354.36			7.644
0.659265	3.367	0.0008069	1656.15			7.644
0.703882	2.8135	0.0014341	2177.72			7.644
0.746688	2.3178	0.0019369	2844.27			7.644

0.787291	1.9562	0.0022487	3554.35			7.644
0.825311	1.731	0.0023896	4242.08			7.644
0.860385	1.609	0.0024099	4888.38			7.644
0.892165	1.5537	0.0023539	5504.97			7.644
0.920332	1.5397	0.002247	6119.1			7.644
0.944596	1.5545	0.0020942	6770.33			7.644
0.964701	1.5995	0.0018785	7522.66			7.644
0.980432	1.6734	0.0016111	8503.82			7.644

(columns 8-13 of 15; x_c repeated on each block)

x_c	intermittency_gamma	state
0.001152	0.0	laminar
0.006721	0.0	laminar
0.016897	0.0	laminar
0.031587	0.0	laminar
0.05065	0.0	laminar
0.073903	0.0	laminar
0.101119	0.0	laminar
0.132034	0.0	laminar
0.166346	0.0	laminar
0.203719	0.0	laminar
0.243794	0.0	laminar
0.286186	0.0	laminar
0.330494	0.0	laminar
0.376307	0.0	laminar
0.423206	0.0	laminar
0.470769	0.0	laminar
0.518574	0.0	laminar
0.566202	0.0	transitional
0.613236	0.06957	transitional
0.659265	0.24627	transitional
0.703882	0.46191	transitional
0.746688	0.65579	transitional
0.787291	0.79858	transitional
0.825311	0.88956	transitional
0.860385	0.94174	transitional
0.892165	0.96959	transitional
0.920332	0.98384	transitional
0.944596	0.99101	turbulent
0.964701	0.99464	turbulent
0.980432	0.99648	turbulent

Table 19. Cruise lower-surface BL state (surface_cruise_lower.csv, sampled). (columns 14-15 of 15; the table is split across 3 blocks so that no cell is compressed to illegibility, with x_c repeated on each)

(table sampled evenly to 30 of 128 rows; full data in 04_solution/surface_cruise_lower.csv)

9.4 Surface distributions — climb (off-design, elevated Tu)

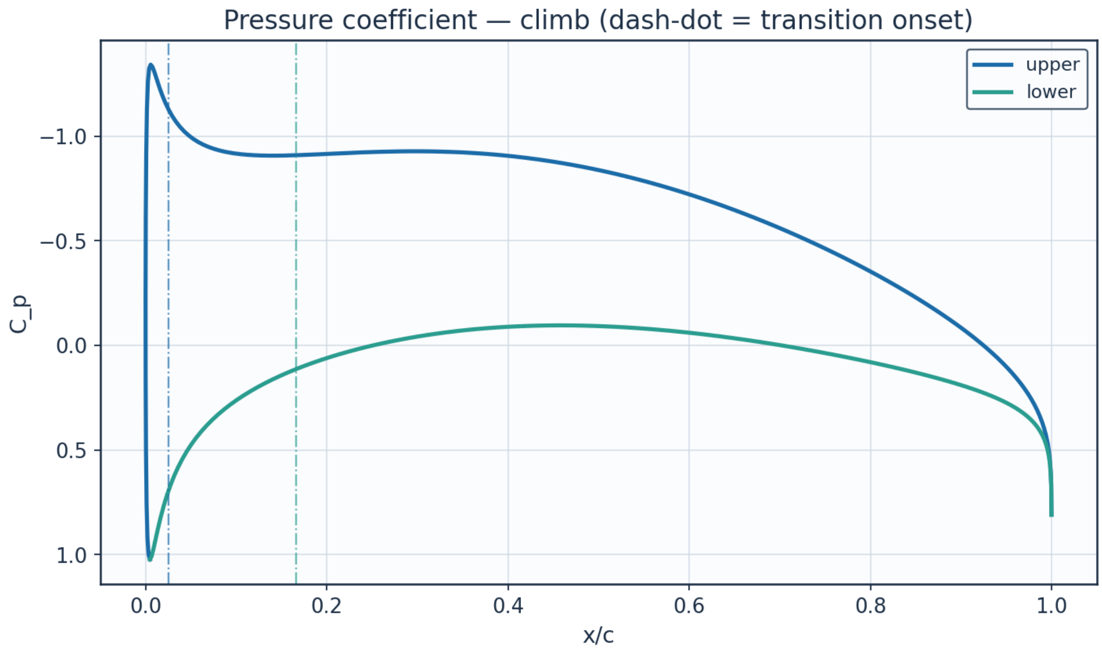


Fig. 22. Pressure coefficient C_p — climb.

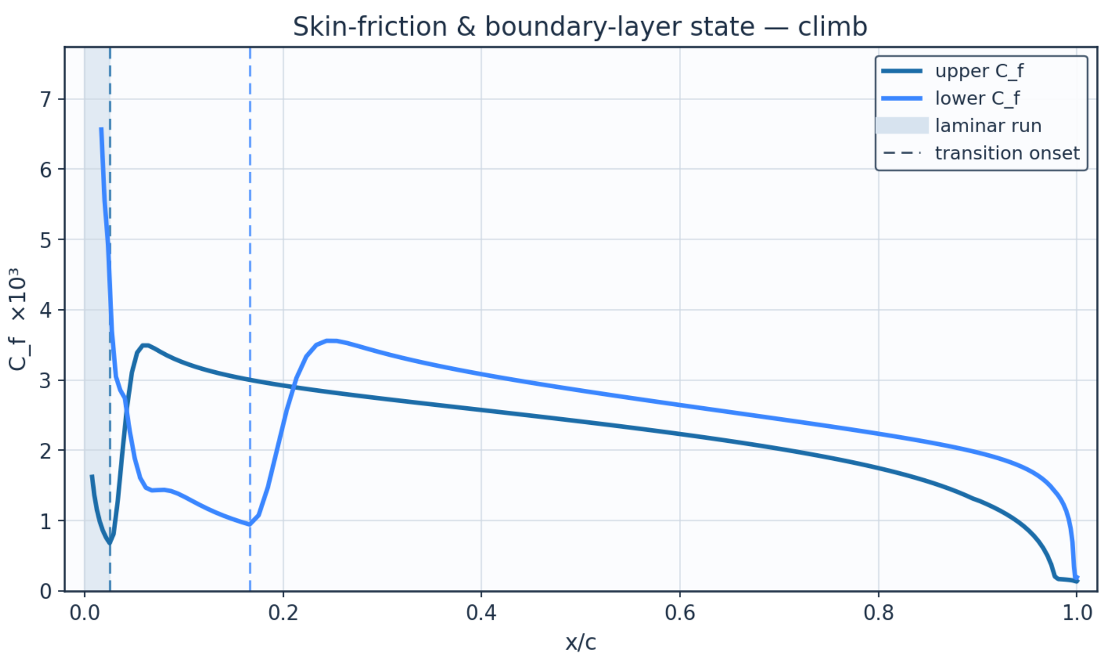


Fig. 23. Skin-friction C_f and laminar run — climb.

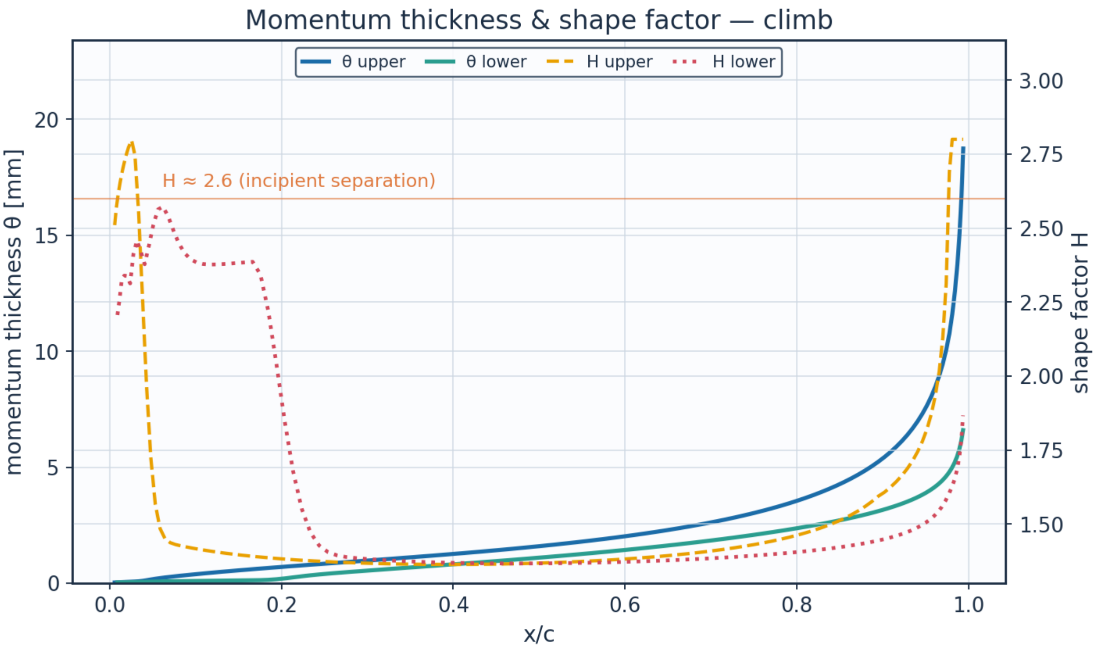


Fig. 24. Momentum thickness ϑ and shape factor H — climb.

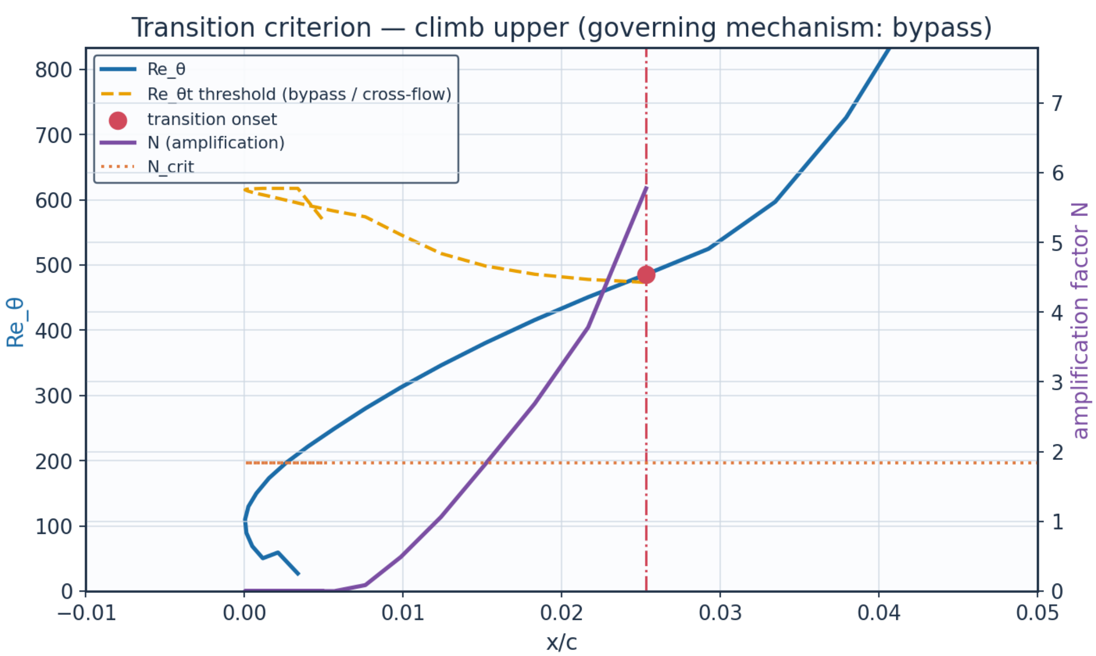


Fig. 25. The governing transition criterion — climb: here Re_ϑ crosses the falling Abu-Ghannam & Shaw bypass threshold while N is still far below N_crit.

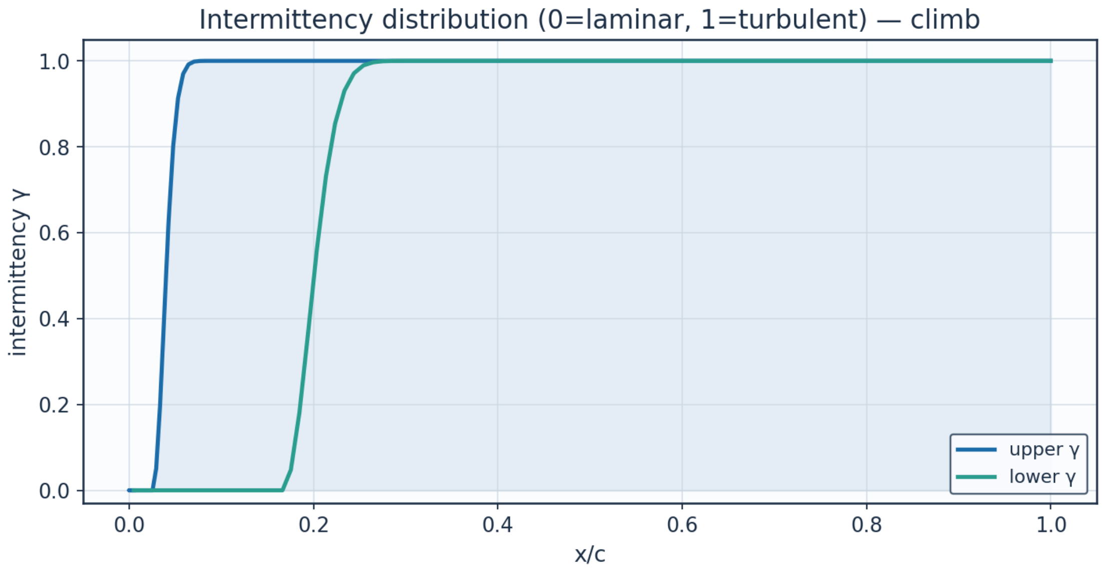


Fig. 26. Intermittency γ — climb.

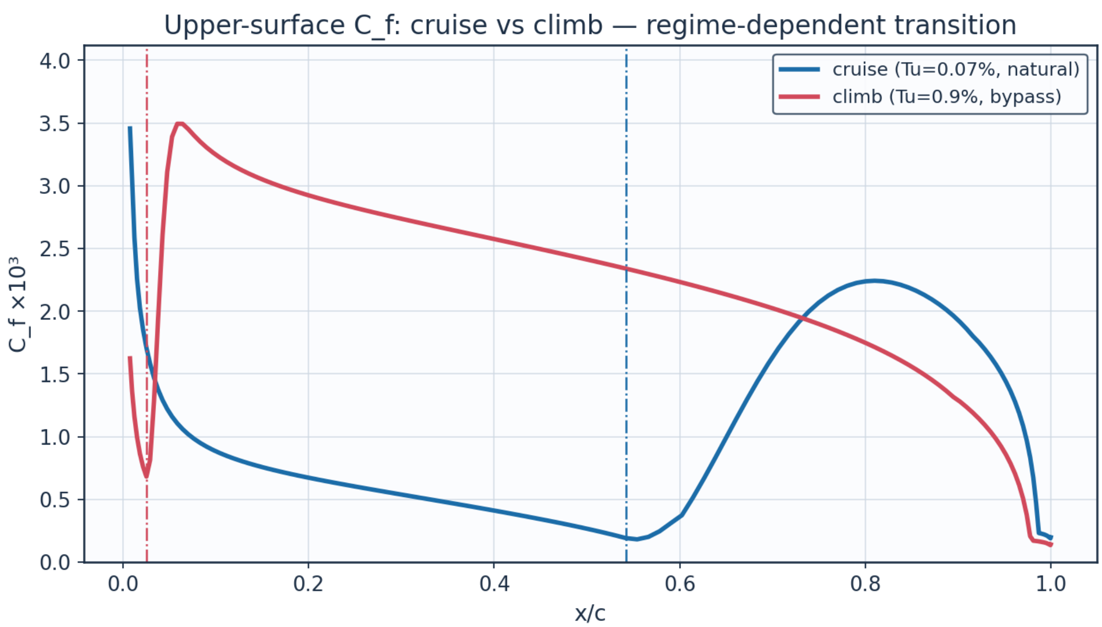


Fig. 27. C_f cruise vs climb — regime-dependent transition.

9.5 Aerodynamic polars

The polar carries θ_TE/c, the trailing-edge momentum thickness as a fraction of chord, because that is the assumption Squire-Young rests on and it is the quantity that bounds where the drag can be believed. Over the sweep it stays below 0.016 - read from the table below rather than typed, where it had been left at 0.009 - so the thin-layer assumption holds throughout. Outside the incidence envelope it does not merely degrade: at 16° and a chord Reynolds number of 2×10^5 the march returns a trailing-edge momentum thickness of 31 chords, and it is not even grid-converged — 22 chords at 160 panels against 38 at 360 — so the formula returns a C_d of order 26, which is bluff-body drag from an aerofoil method. A layer thicker than the body is long is arithmetic that has stopped meaning anything rather than a marginal case, so the solver returns no drag there instead of a number. Within the envelope and Reynolds range of this study the test never fires. (This paragraph quoted 1.34 chords and C_d = 1.22; both are now read from the solve rather than typed, and neither was what it returns.)

THE POLAR IS TABULATED AT ONE DEGREE AND STEPS OVER A DISCONTINUITY. Between 2° and 3° the upper-surface transition collapses from 0.518 chord to 0.128 and the section drag rises 62 per cent, and the table above shows it without saying why. It is not a resolution failure: refining the incidence does not smooth the jump, only locate it. This section has TWO competing amplification maxima — one under the leading-edge suction peak, one in the mid-chord recovery — and onset is wherever N/N_crit first reaches one, so the answer is decided by which of them crosses first. Table 20 resolves the crossing: the forward peak rises smoothly through 0.9810 at 2.92° and 1.0008 at 2.94°, and on those two neighbouring solves the transition point moves from 0.3609 chord to 0.1892 — 17 per cent of the chord, for a crossing by 8 parts in ten thousand. x_tr is genuinely discontinuous in incidence while the N field beneath it is smooth. That is the edge of the laminar bucket, a property of the aerofoil and not of the discretisation, and the drag either side of it is as trustworthy as anywhere else on the polar.

alpha_deg	Cd_counts	x_tr_upper_c	N_over_ Ncrit_forward	x_forward_ peak_c	forward_ peak_has_ crossed	mechanism
2.8	53.53	0.4324	0.8696	0.1988	False	TS-natural
2.82	54.37	0.4204	0.8878	0.1892	False	TS-natural
2.84	55.37	0.4084	0.9063	0.1892	False	TS-natural
2.86	55.41	0.4084	0.9245	0.1892	False	TS-natural
2.88	56.41	0.3964	0.9429	0.1988	False	TS-natural
2.9	57.49	0.3845	0.9618	0.1988	False	TS-natural
2.92	59.43	0.3609	0.981	0.1892	False	TS-natural
2.94	73.52	0.1892	1.0008	0.1892	True	TS-natural
2.96	76.52	0.1528	1.004	0.1528	True	TS-natural
2.98	77.92	0.1358	1.0069	0.1358	True	TS-natural
3.0	78.24	0.1276	1.0125	0.1276	True	TS-natural
3.02	78.95	0.1196	1.0147	0.1196	True	TS-natural
3.04	79.66	0.1119	1.0128	0.1119	True	TS-natural
3.06	80.36	0.1044	1.0109	0.1044	True	TS-natural
3.08	81.17	0.0971	1.0057	0.0971	True	TS-natural
3.1	81.29	0.0971	1.0238	0.0971	True	TS-natural

Table 20. The edge of the laminar bucket, resolved (laminar_bucket_edge.csv). The transition point jumps when the leading-edge amplification peak crosses N_crit, not when the incidence passes a threshold.

Where Squire-Young is evaluated matters, and this report first said why in the wrong terms. The formula wants the trailing edge and this section has a 26.8° wedge one, so the trailing edge is a stagnation point of the inviscid flow, U_e goes to zero there physically and (U_e/U_∞)^((H+5)/2) degenerates — at the last control point it returns 18 counts against 47.3. The evaluation is therefore pulled forward to 0.98 c, and the drag over the range either side runs from 42.5 counts to 47.3. That spread was reported here as a five-count uncertainty band on the headline number. It is not a band.

It is friction being correctly INCLUDED. Between 0.88 c and 0.98 c the layer accumulates 3.00 counts of real skin friction, measured by integrating C_f over the surface directly, while the formula moves 4.84 — the same quantity to 1.8 counts, not the third of a count this paragraph used to claim. A forward station is not a worse estimate of the same drag; it is the drag of a shorter aerofoil. What settles it is their SUM, the last column of Table 21: the drag counted so far plus the friction still ahead varies by only 1.23 counts from 0.90 c up, where over that same range the drag alone moves 3.40. (Both halves of that comparison are now measured on the SAME range. This sentence used to set the spread, which is taken from 0.90 c, against the 4.84 counts the drag moves from 0.88 c - two different ranges, which overstated the contrast.) The friction the chosen station still omits is 0.156 counts at cruise and 0.127 at climb, so the station is converged to under a fifth of a count and is not uncertain by five. Past 0.98 c the formula turns over and falls — that is the inviscid singularity taking hold, not drag being lost. The omitted friction is in the SAME frame as the drag it is added to, and getting it there is the argument of Eq. E20c applied to the wall shear instead of the wake. It used to be the chordwise integral alone, referred to U_n and c_n — a normal-plane coefficient added to a streamwise one. The conversion has two terms and neither needs a trailing-edge evaluation. The chordwise wall shear contributes $\cos^3\Lambda \int C_f (U_{e,n}/U_n)^2 \, d(s_n/c_n)$. The span-wise wall shear contributes as well, because Squire-Young's span-wise term carries it only as far as the evaluation station; under the same small-cross-flow closure the drag formula already uses, $\tau_{wz} = (W/U_{e,n})\tau_{wx}$, so it integrates over the same stations to $\cos\Lambda \sin^2\Lambda \int C_f (U_{e,n}/U_n) \, d(s_n/c_n)$ — the FIRST power of the velocity ratio against the second, the same asymmetry E20c carries and for the same reason. The two together make the span-wise contribution independent of the station, so the sum above is $\cos^3\Lambda$ times a purely chordwise quantity plus a constant, which is what the invariance claim can be made about. It is not adopted for giving the smallest spread and does not: measured from 0.90c up, the unconverted form gave 1.19 counts, the chordwise half-correction gives 1.31 and this gives 1.23. The smallest of those is the one that adds two frames together. What settles the form is the yawed flat plate, where $U_{e,n} = U_n$ and the two terms collapse to $\cos^3\Lambda + \cos\Lambda\sin^2\Lambda = \cos\Lambda$ exactly — the independence principle's answer, and the same check E20c itself is held to in tools/smoke.py.

And whether the shape factor at that station is solved. Head's entrainment method has no validity past separation, so H is clamped at 2.8, and on the climb case and at every incidence above about 3° the upper surface is ON that clamp at the evaluation station — 6 of the 12 polar points. That is the same wedge trailing edge: a layer decelerating into a stagnation point genuinely approaches separation, so the clamp is reached for a physical reason and it is Squire-Young's thin-attached-layer premise that fails, not the march. The drag there is formed from a bound, and H_sy_at_clip says so per surface and per point — a hazard this report already flagged for the trailing-edge separation margin and had not for the headline number. Closing it properly needs viscous-inviscid coupling rather than a wake march, and it is worth the same fifth of a count.

The stronger statement, which the clamp flag does not make, is WHICH SIDE OF SEPARATION the evaluation station is on. march_bl finds the first station at which the turbulent shape factor passes 2.6 and has always written it out; nothing compared it with the station the drag is evaluated at. On 6 of the 12 polar points it is UPSTREAM of that station, so the wake deficit Squire-Young integrates is being read in flow this same march calls separated, and the margin widens with incidence to -0.0585 chord at 8°. The cruise case is not among them — both surfaces are attached at the station, by 0.0060 and 0.0165 chord — but the CLIMB UPPER surface is, by -0.0034 chord. Every one of those points is a point at which the clamp flag was already true, so no drag quoted here changes; what changes is that the condition is now a *published column, sy_past_sep and sy_margin_to_sep_c, in the polar and in the transition summary, rather than something a reader had to derive by comparing two others.*

x_ref	x_evaluated_upper	Cd_counts	friction_omitted_counts	H_upper	H_lower	theta_te_c_upper
0.88	0.8797	42.47	3.152	1.577	1.575	0.001168
0.9	0.9022	43.91	2.402	1.589	1.547	0.001329
0.92	0.9226	45.09	1.762	1.62	1.54	0.001515
0.94	0.9408	46.03	1.06	1.665	1.548	0.001734
0.96	0.9614	46.97	0.565	1.769	1.585	0.002109
0.98	0.9811	47.31	0.156	2.14	1.673	0.002834
0.99	0.9896	46.28	0.034	2.801	1.793	0.003636
1.0	0.9999	18.03	0.0	2.801	2.802	0.010666

(columns 2-7 of 10; x_ref repeated on each block)

x_ref	H_on_Head_clamp	delta_from_shipped_counts	Cd_plus_omitted_counts
0.88	False	-4.84	45.62
0.9	False	-3.4	46.31
0.92	False	-2.22	46.85
0.94	False	-1.28	47.09
0.96	False	-0.34	47.54
0.98	False	0.0	47.47
0.99	True	-1.03	46.31
1.0	True	-29.28	18.03

Table 21. Sensitivity of the section drag to the Squire-Young evaluation station (squire_young_station_sensitivity.csv). The last row is on Head's H = 2.8 clamp. (columns 8-10 of 10; the table is split across 2 blocks so that no cell is compressed to illegibility, with x_ref repeated on each)

alpha_deg	Cl	Cd	L_over_D	theta_te_c	H_sy_at_clip	sy_past_sep
-3.0	-0.1149	0.0063	-18.2	0.0047	False	False
-2.0	0.0214	0.00504	4.2	0.00309	False	False
-1.0	0.1571	0.00466	33.7	0.00245	False	False
0.0	0.2926	0.00455	64.2	0.00237	False	False
1.0	0.4282	0.00467	91.7	0.00269	False	False
2.0	0.5643	0.00484	116.6	0.00306	False	False
3.0	0.7013	0.00782	89.6	0.0063	True	True
4.0	0.8396	0.00923	91.0	0.00779	True	True
5.0	0.9796	0.01026	95.5	0.00896	True	True
6.0	1.1219	0.01145	98.0	0.01025	True	True
7.0	1.2674	0.01327	95.5	0.01222	True	True
8.0	1.417	0.0173	81.9	0.01632	True	True

(columns 2-7 of 10; alpha_deg repeated on each block)

alpha_deg	sy_margin_to_sep_c	xtr_upper_c	xtr_lower_c
-3.0	0.0088	0.661	0.101
-2.0	0.0108	0.638	0.353
-1.0	0.0108	0.614	0.459
0.0	0.0085	0.59	0.519
1.0	0.006	0.554	0.554
2.0	0.006	0.518	0.59
3.0	-0.0071	0.128	0.625
4.0	-0.0152	0.033	0.648

5.0	-0.0197	0.018	0.682
6.0	-0.0295	0.018	0.704
7.0	-0.0403	0.015	0.736
8.0	-0.0585	0.01	0.757

Table 22. Aerodynamic polar (aero_polar.csv). (columns 8-10 of 10; the table is split across 2 blocks so that no cell is compressed to illegibility, with alpha_deg repeated on each)

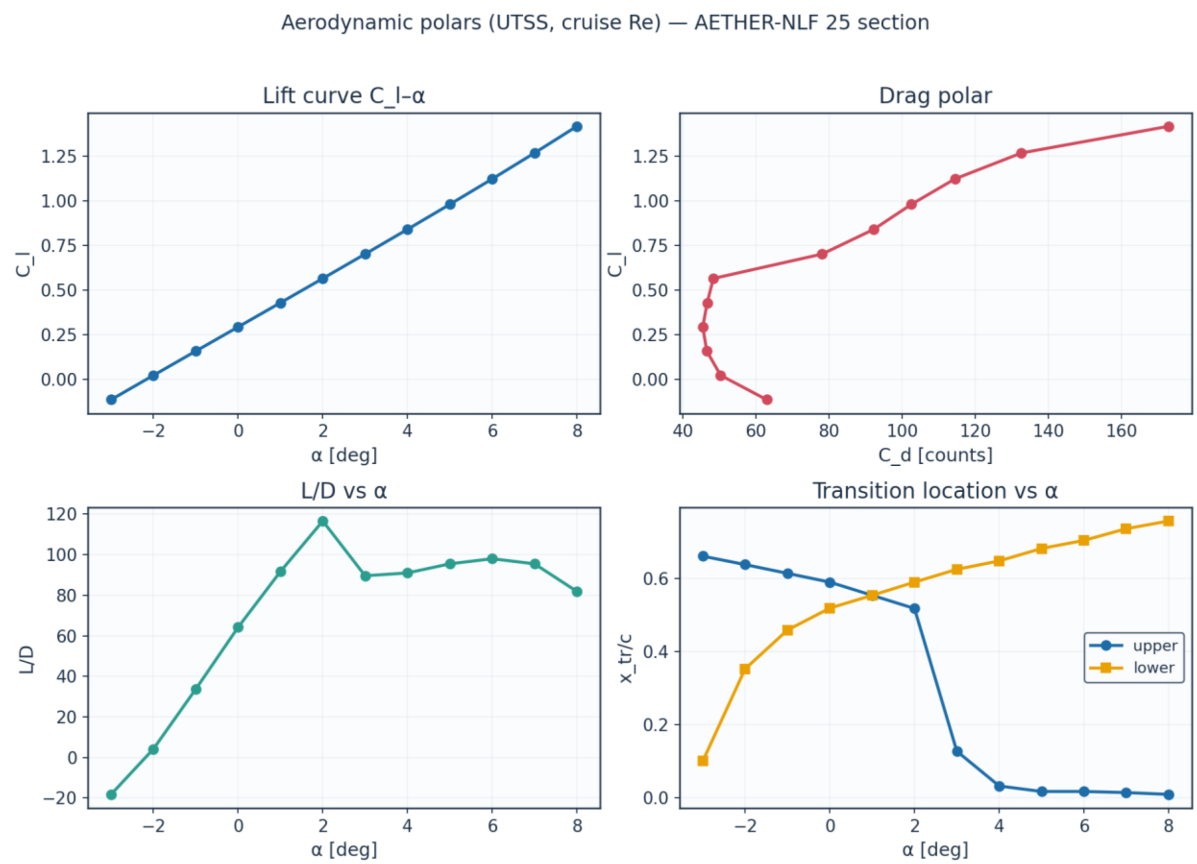


Fig. 28. Lift curve, drag polar, L/D and transition vs angle of attack.

9.6 Span-wise distribution (3-D)

INTEGRATING THIS TABLE DOES NOT RETURN THE WING C_L TWO TABLES EARLIER, and the reason is that they are two section models rather than one number computed twice. c_l_section is the PANEL section's swept-strip lift, solved on the LEADING-EDGE sweep, because that is the angle the boundary layer lives on and transition is what these strips exist to predict. The wing C_L comes from the lifting line, whose section slope is reduced by cos of the QUARTER-CHORD sweep — 9.89° against 12.00° — which is the convention for a lift-curve slope. $\cos^2\Lambda_{LE} = 0.9568$ against $\cos\Lambda_{c/4} = 0.9851$ is three per cent before the panel section's own departure from a linear $a_0(\alpha-\alpha_{L0})$ is counted, and measured the strips run from 0.961 of the lifting-line loading at the root to 0.916 at the tip. So integrating c_l_section over the span returns C_L = 0.2429 where the wing table says 0.2548, a 4.7 per cent shortfall, and integrating c_l_lifting_line — published beside it for exactly this reason — returns 0.2536, which is the tabulated value to within the truncation of a twelve-station sweep that stops at $\eta = 0.98$. Neither number is wrong; they answer different questions, and the column that closes the wing lift is now in the table rather than left to be reconstructed.

eta	y_m	chord_m	Re_local	alpha_geom_deg	alpha_induced_deg	alpha_eff_deg
0.0	0.0	2.6	8246200.0	1.5	1.38	0.12
0.089	0.757	2.484	7878900.0	1.23	1.09	0.14
0.178	1.515	2.368	7511500.0	0.97	0.87	0.1
0.267	2.272	2.253	7144200.0	0.7	0.68	0.01
0.356	3.029	2.137	6776900.0	0.43	0.53	-0.1
0.445	3.786	2.021	6409500.0	0.16	0.4	-0.24
0.535	4.544	1.905	6042200.0	-0.1	0.29	-0.39
0.624	5.301	1.789	5674900.0	-0.37	0.19	-0.56
0.713	6.058	1.673	5307600.0	-0.64	0.12	-0.75
0.802	6.815	1.558	4940200.0	-0.91	0.07	-0.97
0.891	7.573	1.442	4572900.0	-1.17	0.07	-1.24
0.98	8.33	1.326	4205600.0	-1.44	0.28	-1.72

(columns 2-7 of 13; eta repeated on each block)

eta	c_l_section	c_l_lifting_line	xtr_upper_c	xtr_lower_c	Cd_section	laminar_fraction
0.0	0.3083	0.3209	0.566	0.495	0.00452	0.53
0.089	0.3118	0.3245	0.566	0.507	0.00452	0.536
0.178	0.3058	0.3184	0.578	0.507	0.0045	0.543
0.267	0.2945	0.3066	0.578	0.507	0.00455	0.543
0.356	0.2791	0.2908	0.59	0.507	0.00454	0.549
0.445	0.2607	0.2719	0.59	0.507	0.0046	0.549
0.535	0.2397	0.2503	0.602	0.495	0.00464	0.549
0.624	0.2163	0.2262	0.602	0.495	0.00471	0.549
0.713	0.1903	0.1994	0.614	0.495	0.00472	0.554
0.802	0.1607	0.169	0.673	0.483	0.00455	0.578
0.891	0.1244	0.1316	0.684	0.471	0.00466	0.577
0.98	0.059	0.0644	0.695	0.447	0.00484	0.571

Table 23. Span-wise distribution (spanwise_distribution.csv). (columns 8-13 of 13; the table is split across 2 blocks so that no cell is compressed to illegibility, with eta repeated on each)

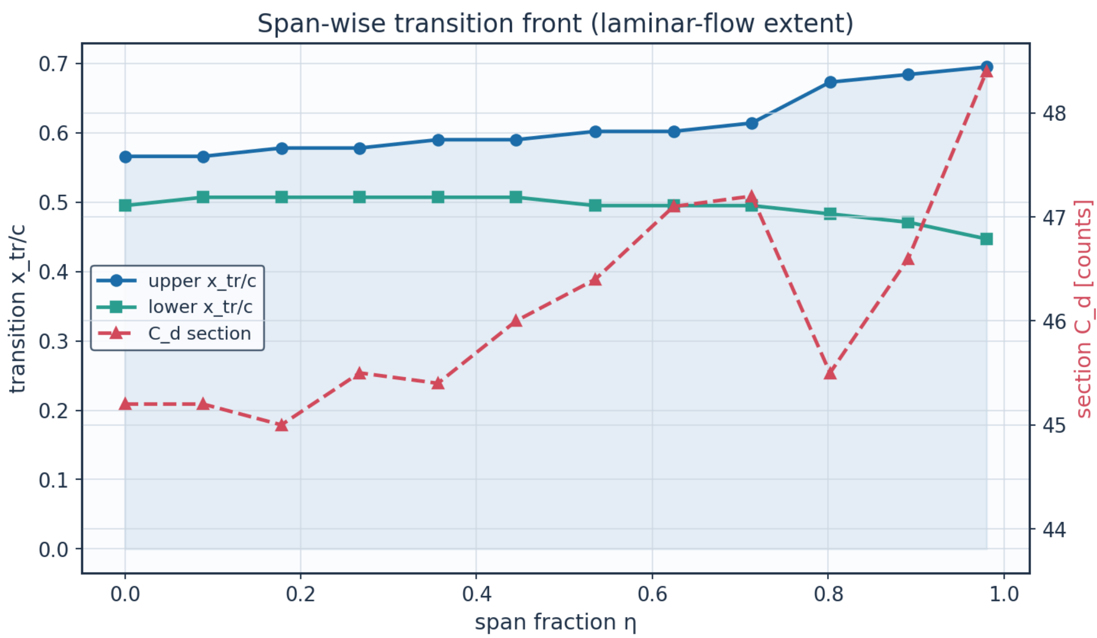


Fig. 29. Span-wise transition front and section drag.

10. Post-Processing — Contours, Profiles and 3-D Fields

10.1 Pressure and velocity contours

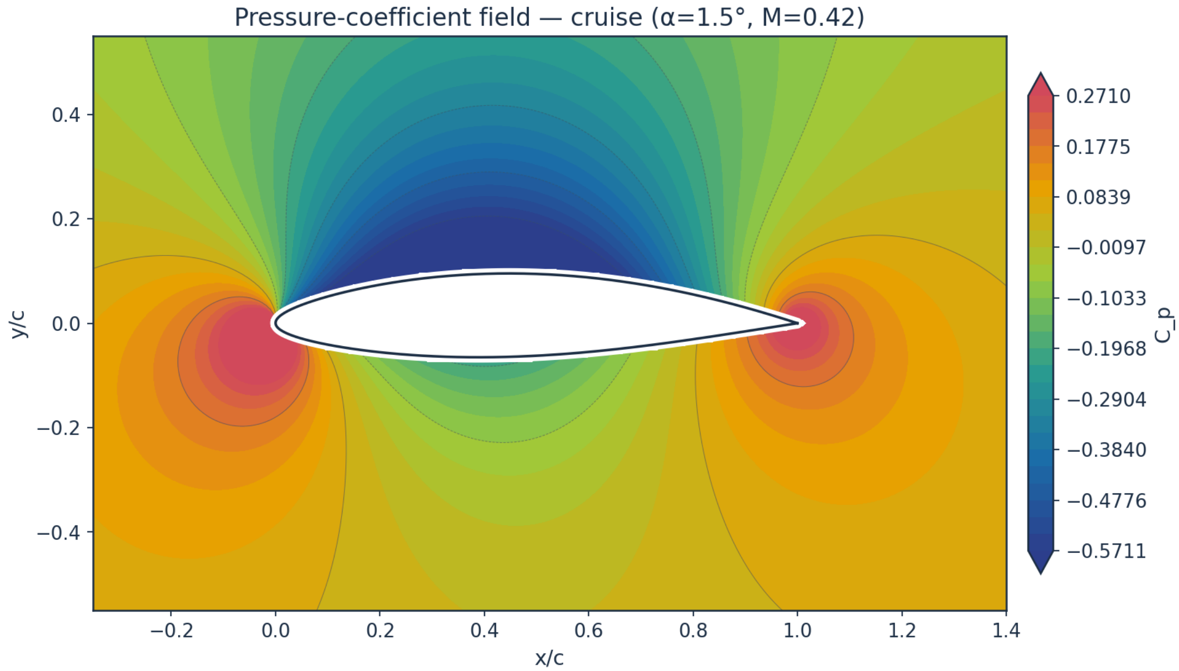


Fig. 30. Pressure-coefficient contour — cruise.

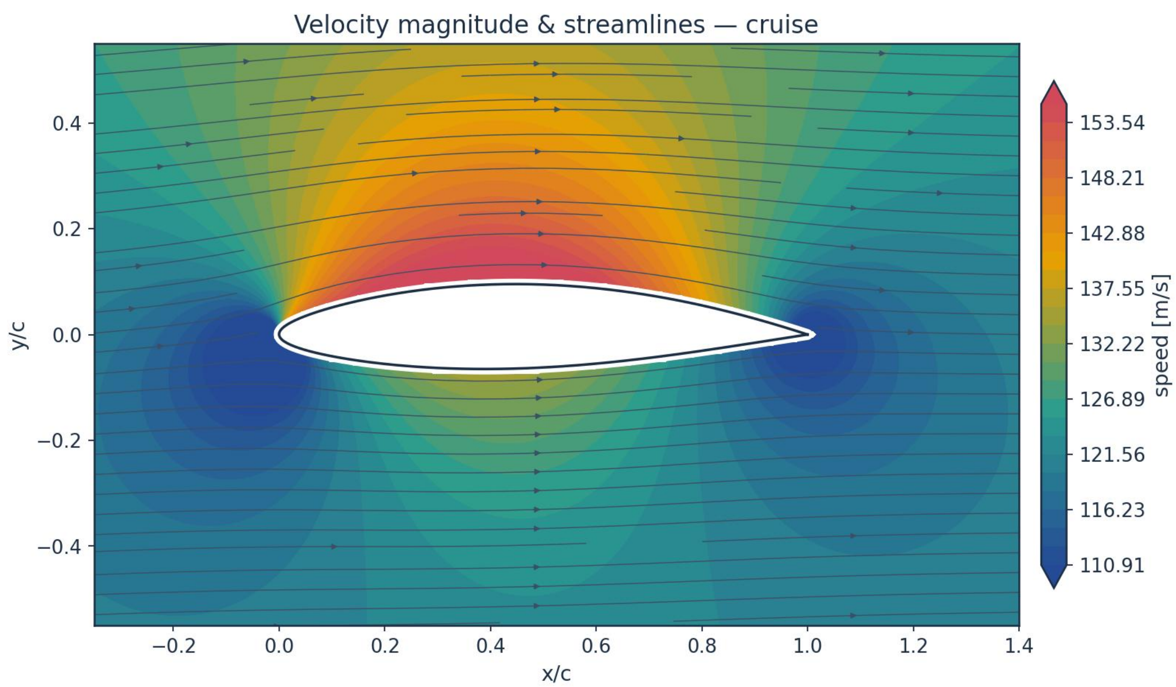


Fig. 31. Velocity magnitude and streamlines — cruise.

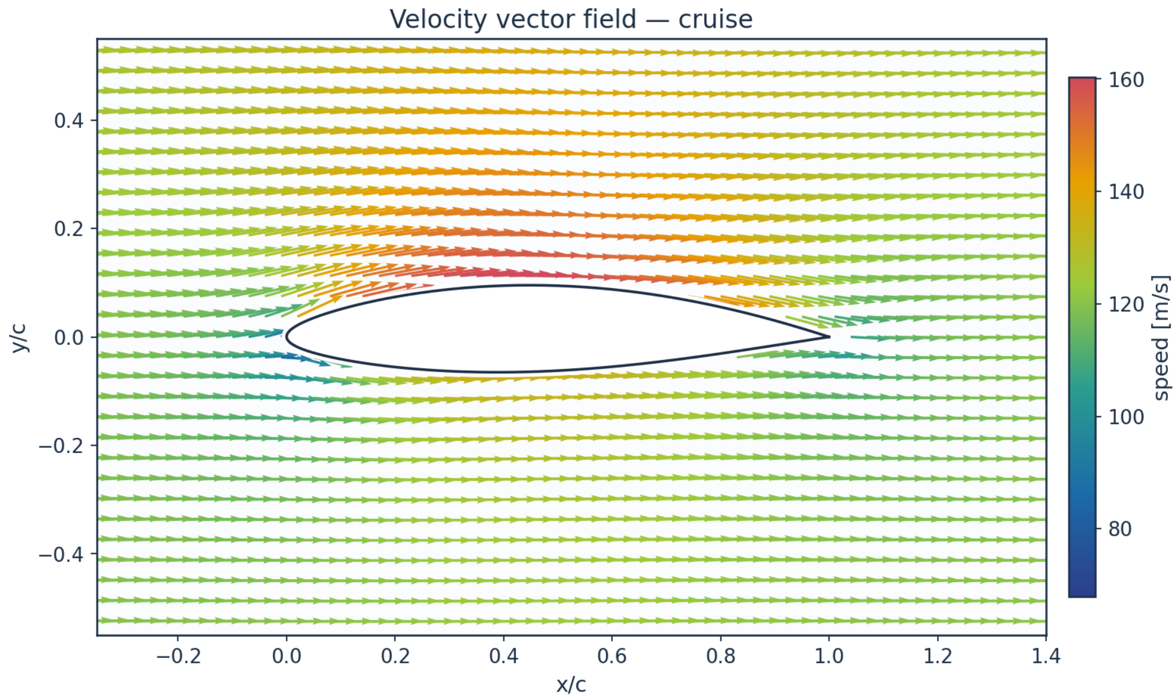


Fig. 32. Velocity vector field — cruise.

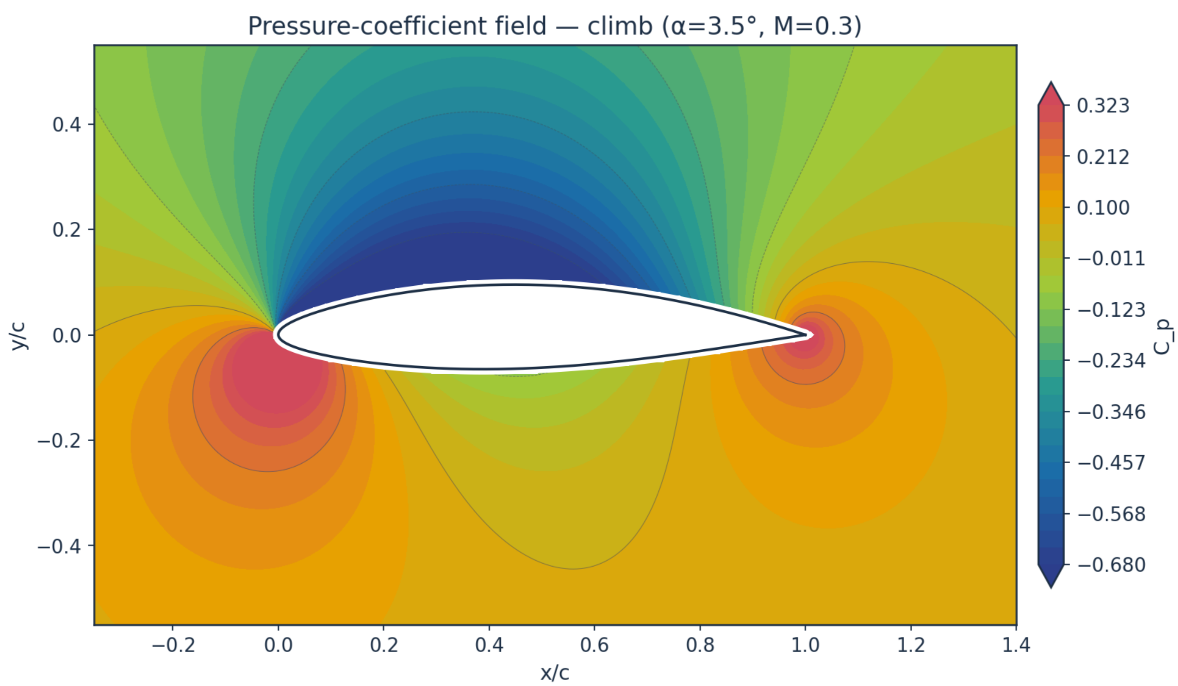


Fig. 33. Pressure-coefficient contour — climb.

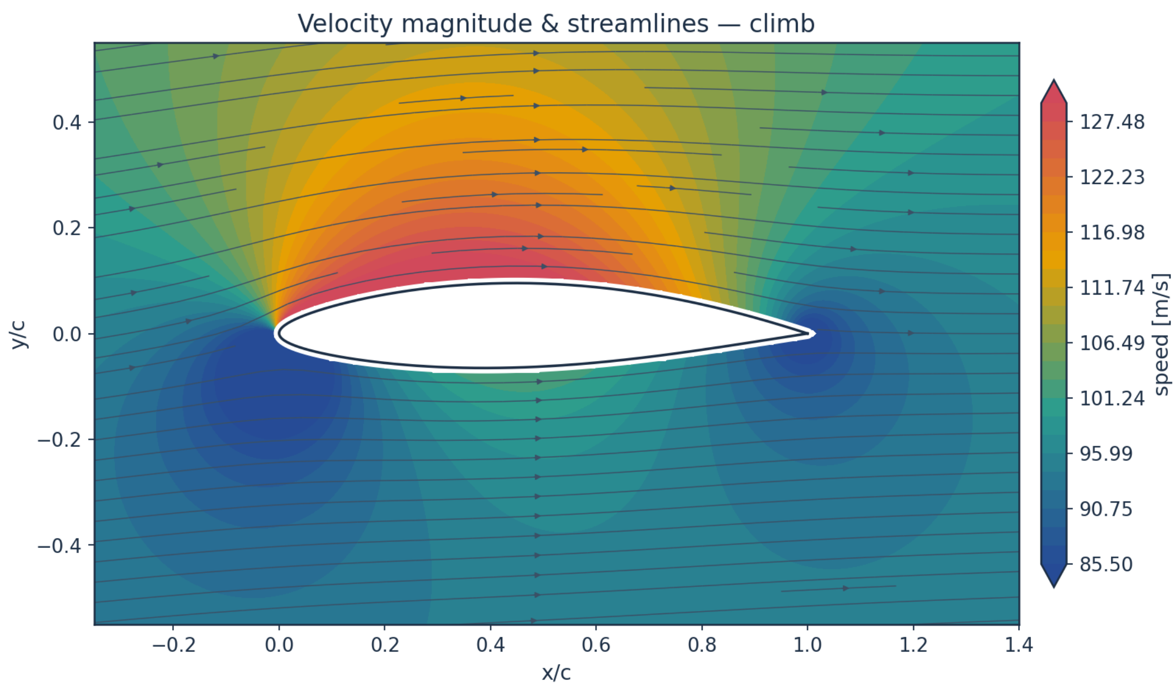


Fig. 34. Velocity magnitude and streamlines — climb.

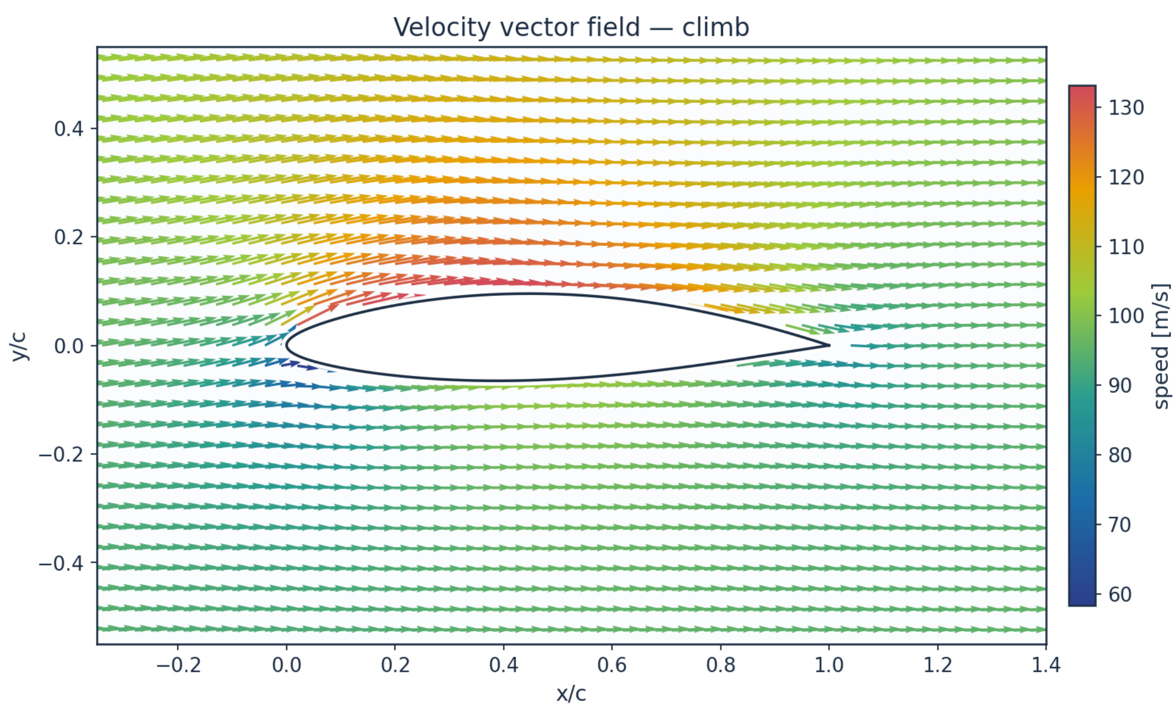


Fig. 35. Velocity vector field — climb.

10.2 Boundary-layer velocity and temperature profiles

The profiles are reconstructed from the marched state and not from an assumed shape. The laminar leg is the Falkner-Skan profile at the shape factor the march solved for, read from the same family that supplies the closure functions; the turbulent leg is the power law that shape factor implies, $H = (n+2)/n$; and the two are blended by the same intermittency that blends C_f, θ and H, each on its own thickness — $\delta_{99} = \eta_{99}\, \theta/\theta_\eta$ for the similarity profile and $\delta = \theta(n+1)(n+2)/n$ for the power law. Temperature profiles then follow from the compressible Crocco–Busemann relation (Eq. E21), showing wall-recovery heating through the boundary layer.

THE PLOTTED PROFILE DOES NOT CARRY THE MARCHED SHAPE FACTOR AT A TRANSITIONAL STATION, and the table gives both numbers rather than letting them disagree in silence. The march blends integrals, $\vartheta = (1-\gamma)\vartheta_{lam} + \gamma\vartheta_{turb}$; this reconstruction blends velocities, $u = (1-\gamma)u_{lam} + \gamma u_{turb}$. Those are different operations. δ^* is linear in u and survives it — the plotted profile returns δ^* to two parts in a thousand of the linear blend — but ϑ is QUADRATIC in u, and the pointwise blend carries a cross term $u_{lam} \cdot u_{turb}$ that a blend of integrals does not, worth eight per cent of ϑ. $H = \delta^*/\vartheta$ inherits all of it: at x/c=0.60, where $\gamma = 0.113$, the curve integrates to 3.335 against the marched 3.710. Where γ is 0 or 1 the cross term vanishes and the two agree to 1.6 per cent or better, which is interpolation error and nothing else. Rescaling does not repair it — stretching y multiplies δ^* and ϑ alike and leaves H untouched — and matching H would mean drawing a single-family profile AT the marched H, which is the assumed shape this reconstruction exists to avoid and would erase the two-layer structure that is the physical content of a transitional station. So H_profile is published beside H_shape.

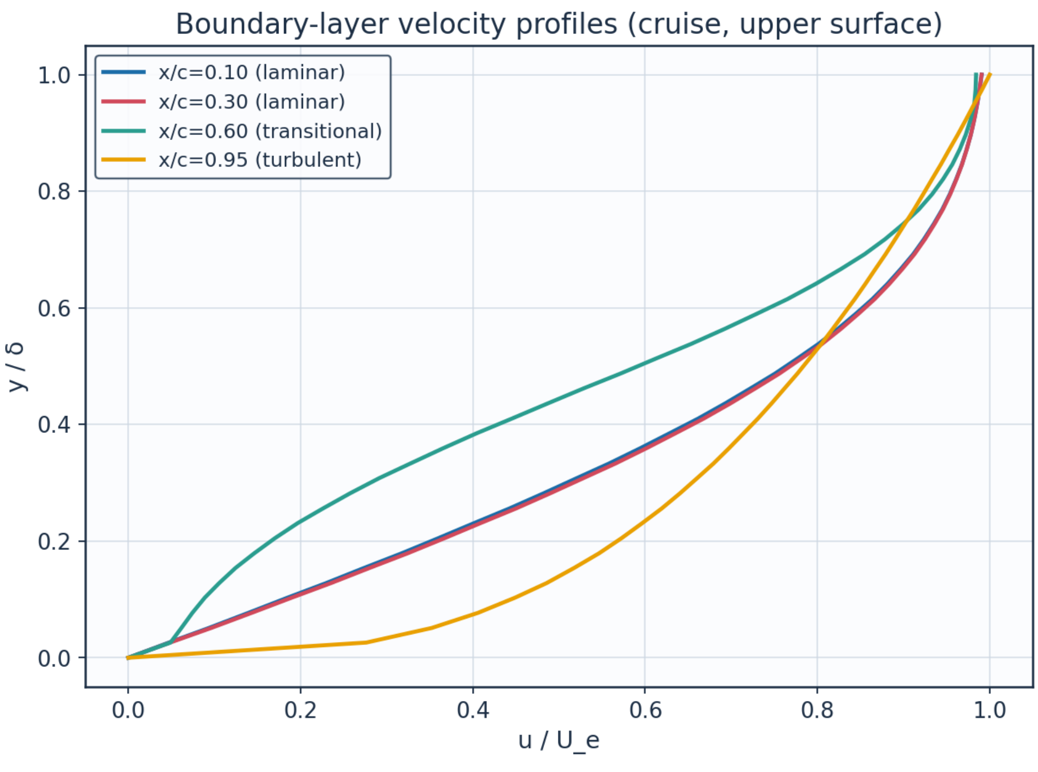


Fig. 36. Boundary-layer velocity profiles.

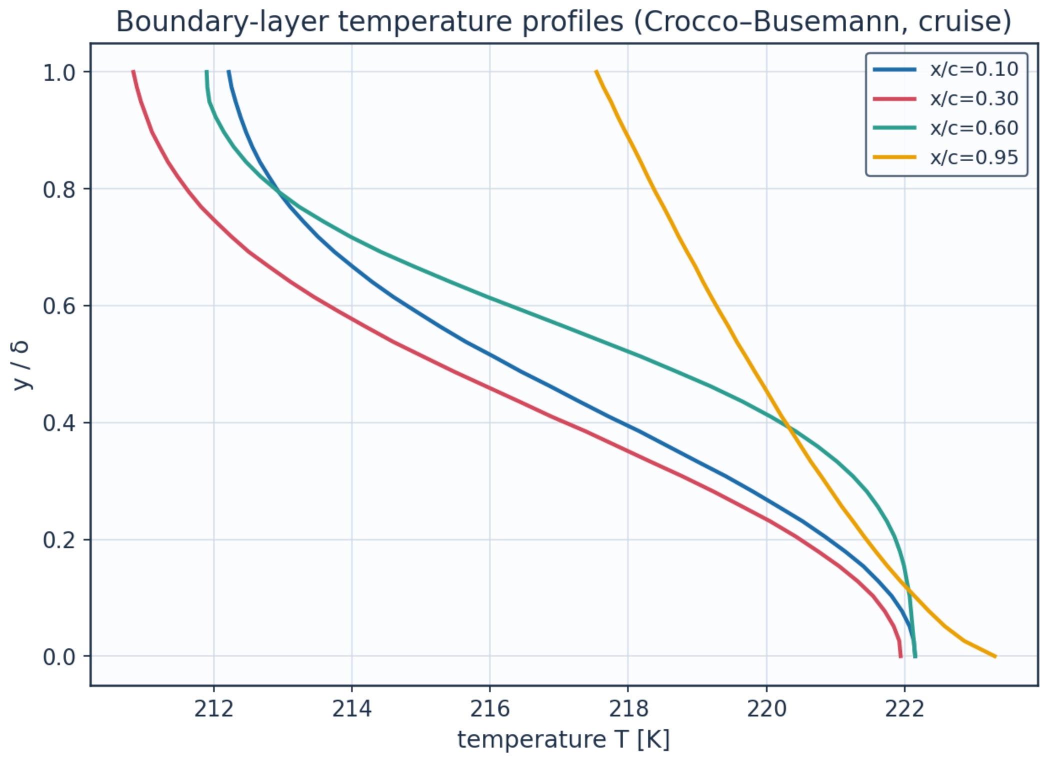


Fig. 37. Boundary-layer temperature profiles (K).

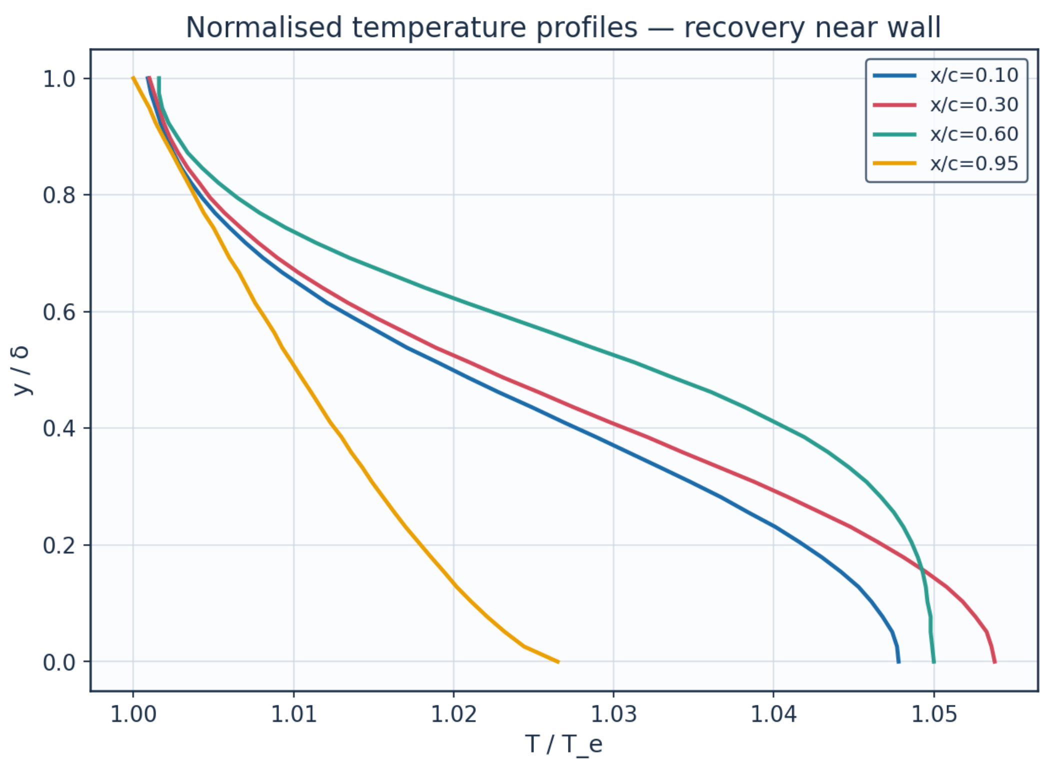


Fig. 38. Normalised temperature profiles T/T_e.

station	x_c	y_delta	y_mm	u_Ue	T_Te	T_K
x/c=0.10	0.1	0.0	0.0	0.0	1.0478	222.15
x/c=0.10	0.1	0.128	0.1423	0.2296	1.0453	221.62
x/c=0.10	0.1	0.256	0.2847	0.4424	1.0384	220.17
x/c=0.10	0.1	0.385	0.427	0.6284	1.0289	218.16
x/c=0.10	0.1	0.513	0.5694	0.7766	1.019	216.05
x/c=0.10	0.1	0.641	0.7117	0.8811	1.0107	214.29
x/c=0.10	0.1	0.769	0.8541	0.9449	1.0051	213.11
x/c=0.10	0.1	0.897	0.9964	0.978	1.0021	212.47
x/c=0.30	0.3	0.0	0.0	0.0	1.0538	221.94
x/c=0.30	0.3	0.128	0.2344	0.2352	1.0508	221.32
x/c=0.30	0.3	0.256	0.4688	0.45	1.0429	219.65
x/c=0.30	0.3	0.385	0.7032	0.6352	1.0321	217.38
x/c=0.30	0.3	0.513	0.9377	0.7811	1.021	215.03
x/c=0.30	0.3	0.641	1.1721	0.8833	1.0118	213.11
x/c=0.30	0.3	0.769	1.4065	0.9456	1.0057	211.82
x/c=0.30	0.3	0.897	1.6409	0.978	1.0023	211.11
x/c=0.60	0.6	0.0	0.0	0.0	1.05	222.15
x/c=0.60	0.6	0.128	0.446	0.1054	1.0495	222.03
x/c=0.60	0.6	0.256	0.892	0.2259	1.0475	221.61
x/c=0.60	0.6	0.385	1.338	0.4044	1.0419	220.42
x/c=0.60	0.6	0.513	1.784	0.6123	1.0313	218.18
x/c=0.60	0.6	0.641	2.23	0.7977	1.0182	215.42
x/c=0.60	0.6	0.769	2.6759	0.9177	1.0079	213.24
x/c=0.60	0.6	0.897	3.1219	0.9721	1.0028	212.15
x/c=0.95	0.95	0.0	0.0	0.0	1.0265	223.3
x/c=0.95	0.95	0.128	3.1722	0.4855	1.0202	221.94
x/c=0.95	0.95	0.256	6.3445	0.6192	1.0163	221.09
x/c=0.95	0.95	0.385	9.5167	0.7143	1.013	220.36

(columns 2-7 of 14; station repeated on each block)

station	Me_edge	H_shape	H_profile	recovery_r	delta_mm	intermittency_ gamma
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.10	0.531	2.512	2.526	0.8485	1.1103	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.30	0.563	2.49	2.504	0.8485	1.8284	0.0
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113
x/c=0.60	0.541	3.71	3.335	0.8539	3.4787	0.113

x/c=0.95	0.384	1.708	1.735	0.8961	24.7435	0.997
x/c=0.95	0.384	1.708	1.735	0.8961	24.7435	0.997
x/c=0.95	0.384	1.708	1.735	0.8961	24.7435	0.997
x/c=0.95	0.384	1.708	1.735	0.8961	24.7435	0.997

(columns 8-13 of 14; station repeated on each block)

station	state
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.10	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.30	laminar
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.60	transitional
x/c=0.95	turbulent
x/c=0.95	turbulent
x/c=0.95	turbulent
x/c=0.95	turbulent

Table 24. BL velocity/temperature profiles (bl_profiles_cruise.csv, sampled). (columns 14-14 of 14; the table is split across 3 blocks so that no cell is compressed to illegibility, with station repeated on each)

(table sampled evenly to 28 of 160 rows; full data in 04_solution/bl_profiles_cruise.csv)

10.3 Three-dimensional surface contours and vectors

The full 3-D wing surface is coloured by the predicted fields; the transition front is directly visible as the laminar-to-turbulent boundary on the intermittency contour.

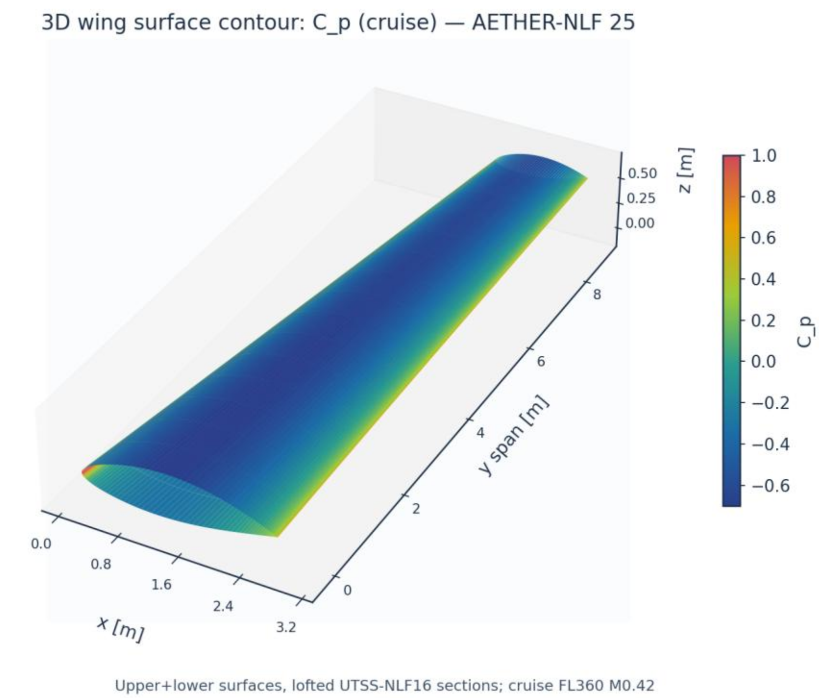


Fig. 39. 3-D wing surface contour — pressure C_p.

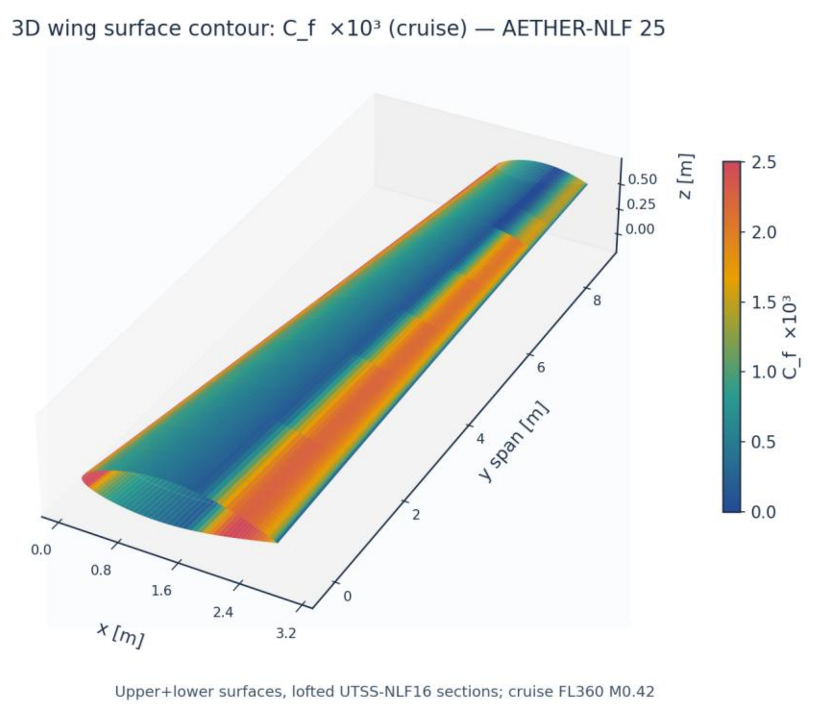


Fig. 40. 3-D wing surface contour — skin friction C_f (×10³).

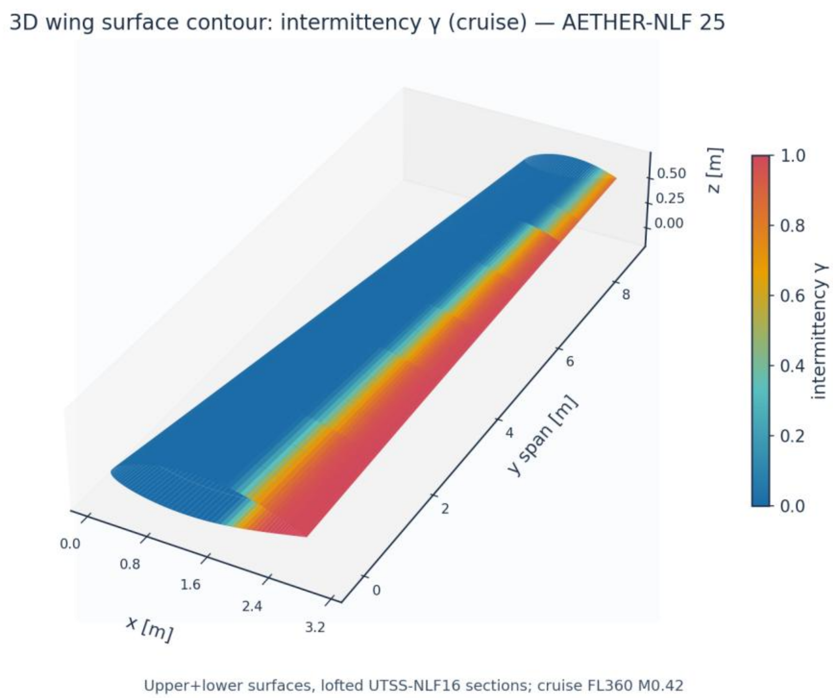


Fig. 41. 3-D wing surface contour — intermittency γ (transition front).

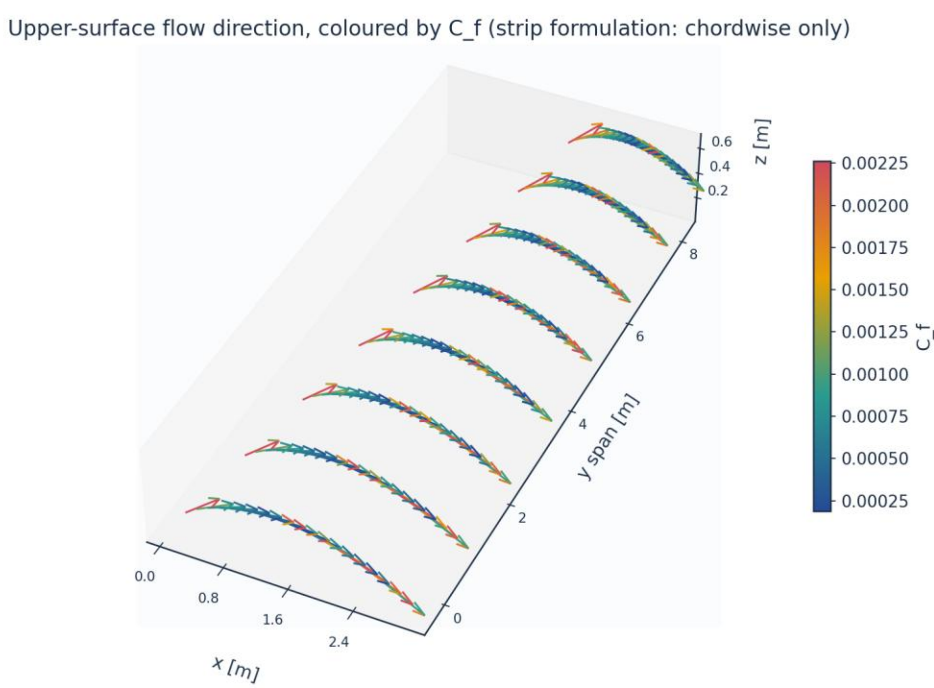


Fig. 42. Upper-surface flow direction coloured by C_f (chordwise only — the strip formulation carries no span-wise wall shear).

11. Validation and Calibration

To qualify as universal, the solver is validated against every published dataset the kernel's four branches reach — five flat plates spanning 0.03 to 6 per cent free-stream turbulence (bypass, natural and separation-induced transition), 86 aerofoil conditions on the NLF(1)-0416 section, and two swept wings from different facilities and eras — using ONE universal calibration set with no per-case re-tuning of the physics. The transition-onset momentum-thickness Reynolds number Re_θt is the metric on the plates and the transition location x_tr/c on the aerofoil and the swept wings. The skin-friction error is reported separately over the laminar run and over the turbulent run, on the stations where the measurement and the prediction are in the same state; pooled across transition it measures the onset error a second time, in the wrong units, because a plate whose onset is early by 16 % is then charged with the whole laminar-to-turbulent step in C_f over the interval between the two onsets.

case	Tu_inlet_pct	L_turb_mm	Re_theta_t_exp	Re_theta_t_pred	Re_theta_t_err_pct	Re_theta_t_exp_lo
ERCOFTAC T3A flat plate (bypass transition)	3.043	1.53	272.3	282.1	3.6	
ERCOFTAC T3A- flat plate (low-Tu bypass)	0.874	0.98	818.8	683.6	-16.5	
ERCOFTAC T3B flat plate (high-Tu bypass)	5.952	3.83	181.3	168.0	-7.3	
ERCOFTAC T3C4 flat plate (laminar separation bubble)	2.11		381.3	266.2	-30.2	309.3
Schubauer & Skramstad flat plate (natural transition)	0.03		1100.0	1100.7	0.1	

(columns 2-7 of 18; case repeated on each block)

case	Re_theta_t_exp_hi	Re_theta_t_err_bracket_pct	Re_x_tr_exp	Re_x_tr_pred	Cf_err_laminar_pct	n_pts_laminar
ERCOFTAC T3A flat plate (bypass transition)			1.348e+05	1.804e+05	3.8	4
ERCOFTAC T3A- flat plate (low-Tu bypass)			1.443e+06	1.060e+06	4.2	8
ERCOFTAC T3B flat plate (high-Tu bypass)			5.910e+04	6.405e+04	8.2	3

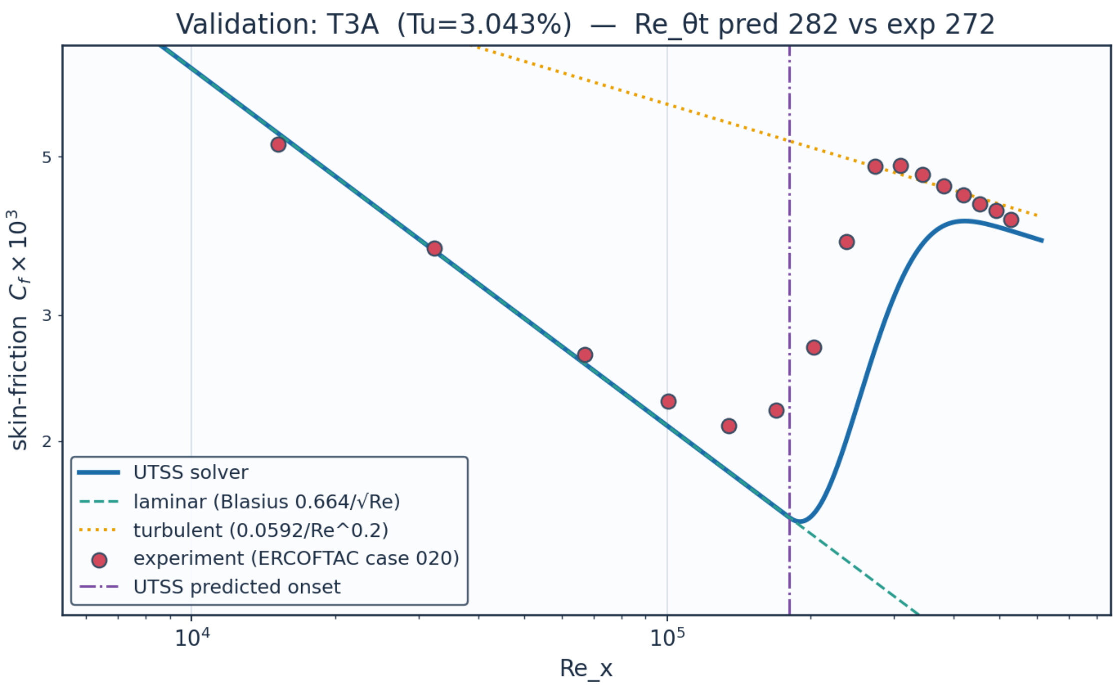
ERCOFTAC T3C4 flat plate (laminar separation bubble)	381.3	-13.9	1.785e+05	1.665e+05	21.9	9
Schubauer & Skramstad flat plate (natural transition)			2.800e+06	2.747e+06		0

(columns 8-13 of 18; case repeated on each block)

case	Cf_err_turbulent_pct	n_pts_turbulent	Cf_err_all_pts_pct	Re_theta_meas_over_blasius	mechanism
ERCOFTAC T3A flat plate (bypass transition)	4.8	3	17.2	1.117	bypass
ERCOFTAC T3A- flat plate (low-Tu bypass)		0	121.6	1.027	bypass
ERCOFTAC T3B flat plate (high-Tu bypass)	8.2	9	12.4	1.123	bypass
ERCOFTAC T3C4 flat plate (laminar separation bubble)		0	35.2	1.359	separation
Schubauer & Skramstad flat plate (natural transition)		0			TS-natural

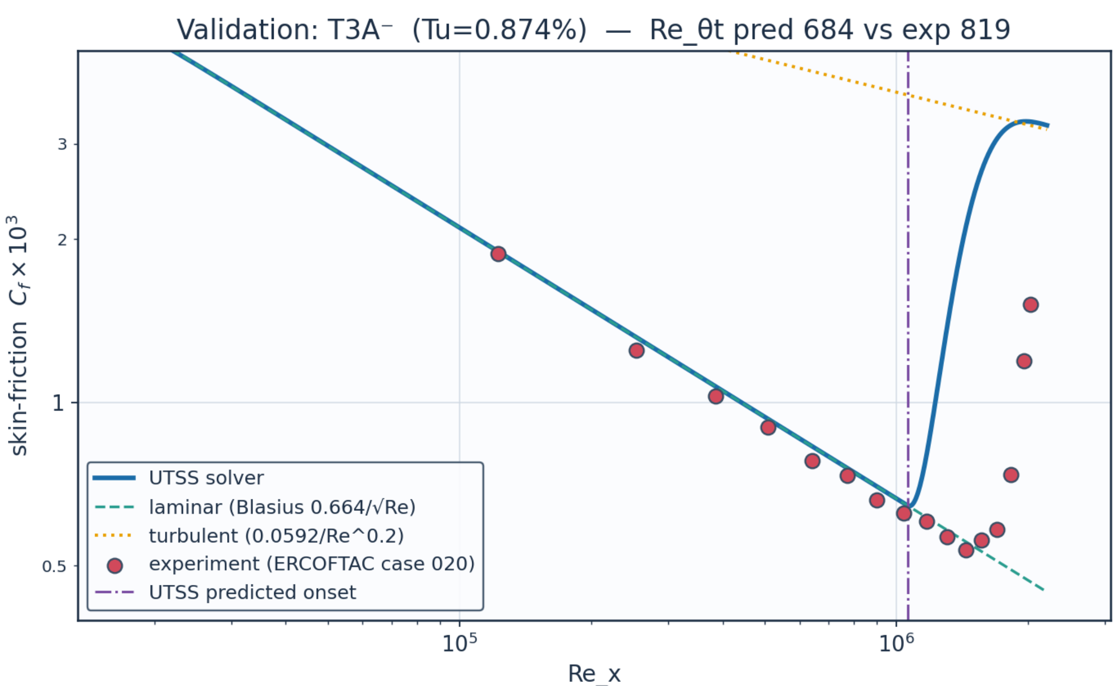
Table 25. Validation summary — predicted vs published Re_ϑt. (columns 14-18 of 18; the table is split across 3 blocks so that no cell is compressed to illegibility, with case repeated on each)

11.1 Flat plates



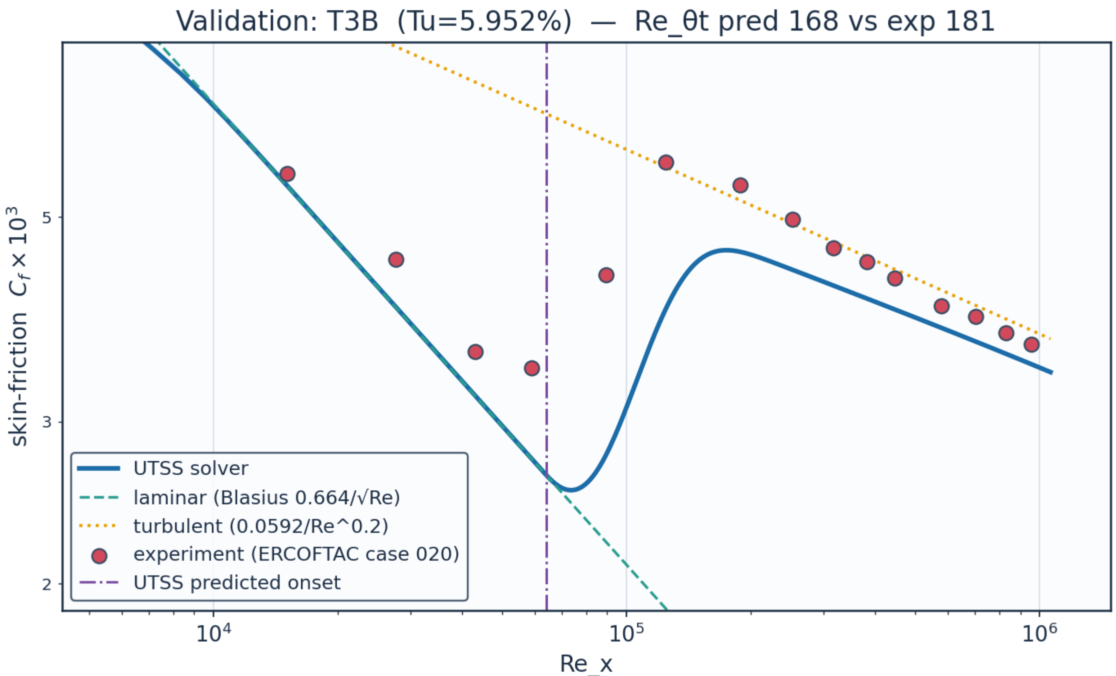
Source: Roach & Brierley (1990), ERCOFTAC T3A; Savill (1993); Langtry & Menter, AIAA J. 47(12) 2009.

Fig. 43. Validation — ERCOFTAC T3A flat plate (Tu = 3.0 %, bypass).



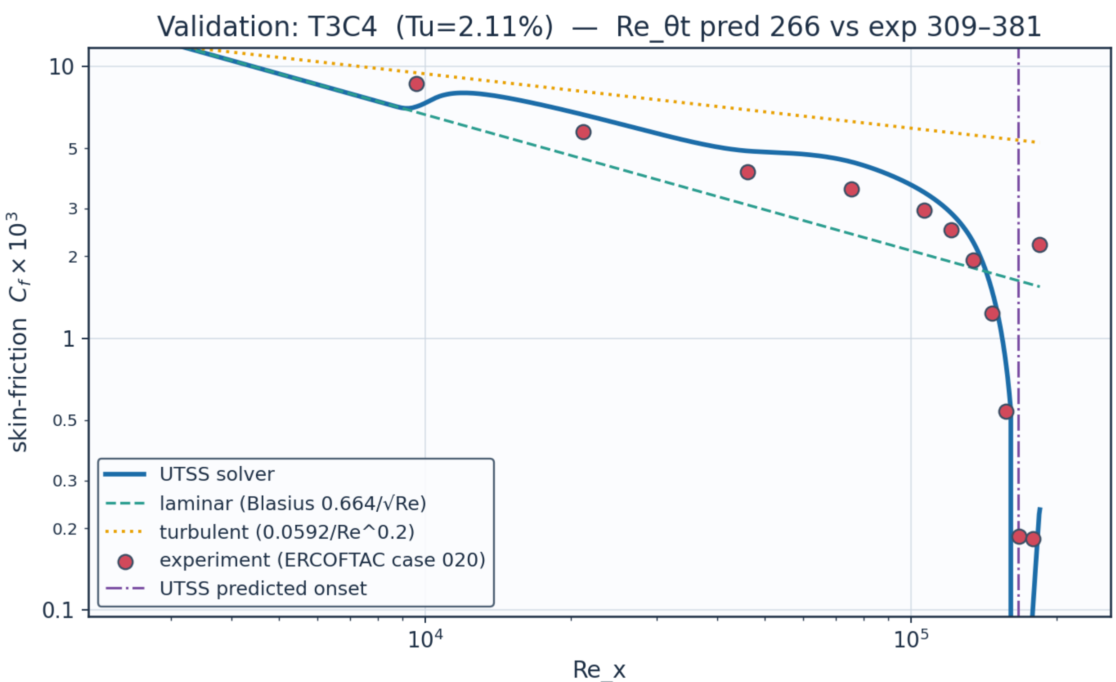
Source: Roach & Brierley (1990), ERCOFTAC T3A-; ERCOFTAC Classic Collection Case 020.

Fig. 44. Validation — ERCOFTAC T3A⁻ flat plate (Tu = 0.87 %, bypass).



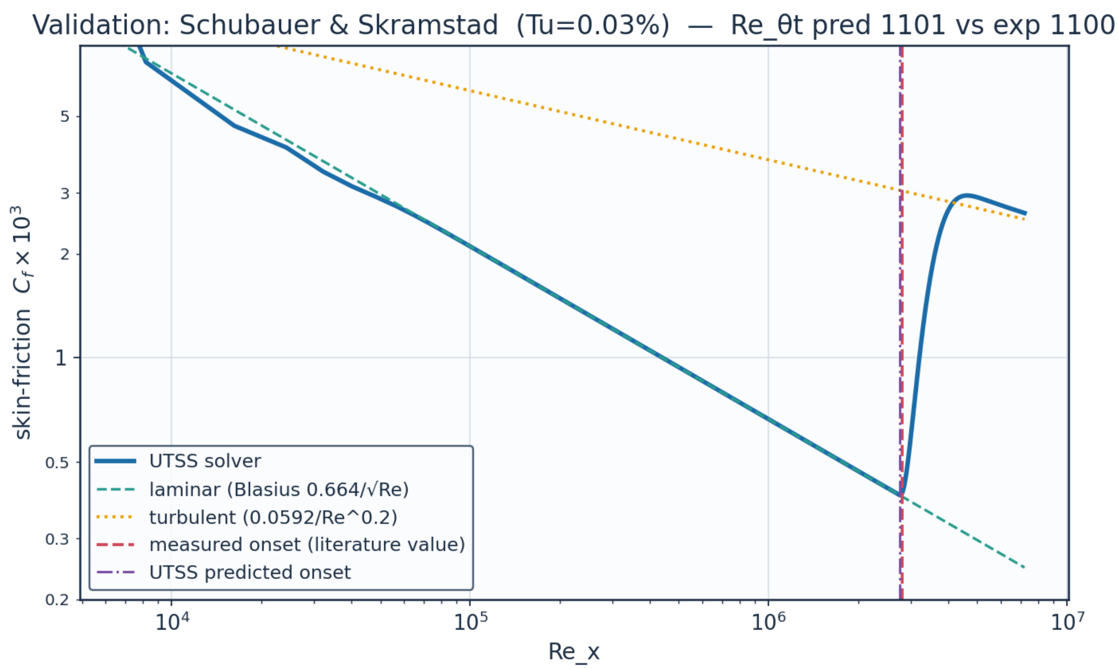
Source: Roach & Brierley (1990), ERCOFTAC T3B; Langtry & Menter, AIAA J. 47(12) 2009.

Fig. 45. Validation — ERCOFTAC T3B flat plate (Tu = 6.0 %, bypass).



Source: Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020 … (full citation in sources_and_references.csv)

Fig. 46. Validation — ERCOFTAC T3C4 flat plate (laminar separation bubble).



Source: Schubauer & Skramstad (1948), NACA Report 909.

Fig. 47. Validation — Schubauer & Skramstad natural transition (Tu = 0.03 %).

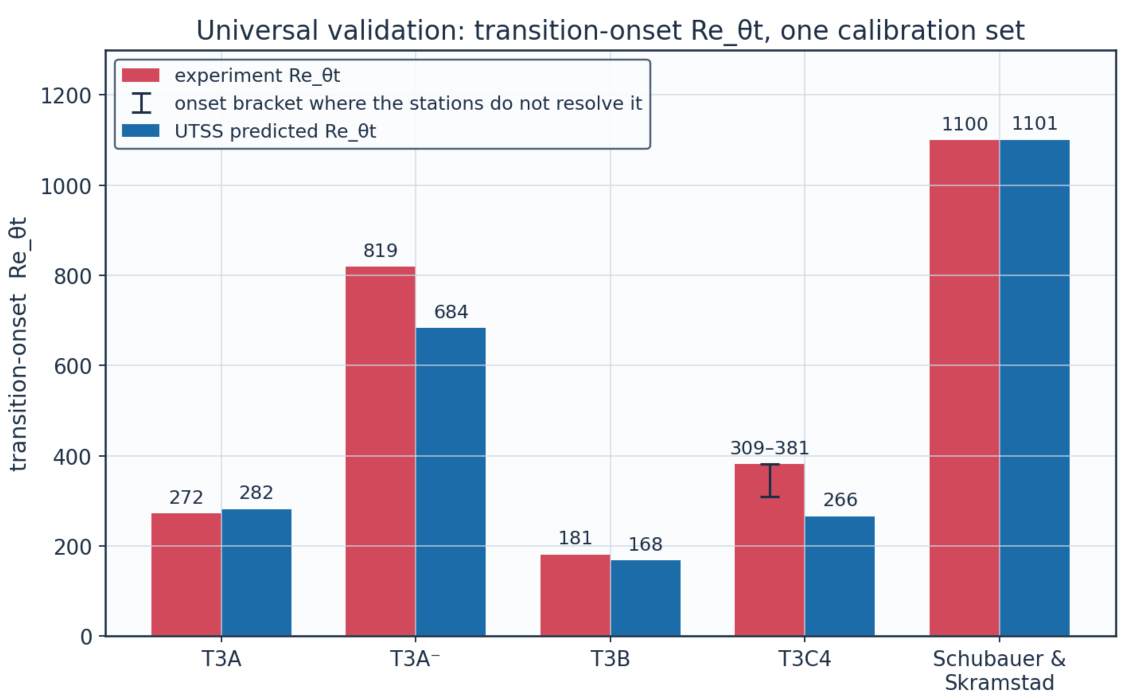


Fig. 48. Universal validation — Re_ϑt across all five plates.

11.1a Where the flat-plate residuals come from

The residuals are accounted for here, and the accounting is generated by gen_validation.py rather than argued.

The laminar branch is not where they are. Compared against the model's own marched momentum thickness at the same stations — not against flat-plate Blasius, which is the wrong reference for the one plate that has a pressure gradient — the measured laminar layer agrees to within 6 % on all 4 plates that carry C_f data, and to 2 % on T3C4. An earlier version of this report attributed the T3C4 residual to the pre-transitional thickening a laminar layer undergoes in a turbulent free stream, on the strength of the measured momentum thickness being 1.36 times Blasius at onset. That comparison was wrong: a laminar layer in an adverse gradient is thicker than Blasius for a reason that has nothing to do with turbulence, and against the march that carries the gradient the agreement is 2 %.

The T3C4 residual is in the bubble, and Table 27 localises it — but not where this study previously placed it. Three things are now established. First, the shape-factor cap is gone: $H^*(H)$ folds at $H = 4.03$ where the attached and reverse-flow branches meet, so the inversion $H = H(H^*)$ is not unique there, but a march knows which branch it is on because it arrived continuously, and taking the root nearest the previous H carries it across. The reverse branch itself was stopped at $H = 4.99$ by the continuation parameter, which was a choice and not a limit — it continues smoothly to $H = 6.41$, past the measured 5.17.

Second, and this is the substantive point: THE MOMENTUM INTEGRAL CANNOT BE CLOSED WITH THIS EXPERIMENT'S OWN DATA. Across the measured bubble, reproducing the measured $d\theta/dx = 0.00591\ m^{-1}$ from the measured $dU_e/dx = -0.200\ s^{-1}$ and $\theta = 2.70$ mm requires a shape factor of 19. The measurement itself reports 5.17, which returns $0.00202\ m^{-1}$ — short by a factor of 2.93. No integral method closed on any physical profile family reproduces this bubble, so the shape-factor cap was never the leading term, and the earlier attribution of a factor 1.32 of the shortfall to that cap was accounting for the wrong thing.

Third, onset is not resolved to a station here. It is taken as the station of minimum measured C_f, as on the other plates. On this plate C_f is 1.87×10^{-4} at $x = 1.295$ m and 1.83×10^{-4} at $x = 1.395$ m — two per cent apart, in a hot-film measurement of a quantity at its floor. They are not distinguishable, so onset is bracketed by them, Re_θ from 309.3 to 381.3, and quoting the second alone reports the end of the plateau as though it were the beginning. Against that bracket the error is -13.9 %, not -30.2 %. The point value is retained in the tables for continuity with the literature; the bracket is what the residual should be read against.

Reading the amplification rate at the marched shape factor instead of at the developed reverse-flow profile became possible once the march crossed the fold, and looks more principled because it removes a constant. It is rejected on measurement: the T3C4 bubble lengthens from the 0.055 m the shipped closure returns to 0.087, towards the measured 0.10, but the 86 NLF(1)-0416 conditions collapse from 50 to 21 inside the bracket. The reason is physical — the marched H is the integral shape factor of the whole dead-

air region, still near 3.4, while the detached shear layer riding on it is inflectional from the moment the flow leaves the wall. It is exposed as cal["bub_sigma_local"] so the test is reproducible.

Conditioning. In a decaying stream the onset threshold rises while Re_θ grows only as the square root of distance, so the two curves close at a shallow angle and the crossing is sensitive. Shifting the threshold by ±10 % moves the predicted transition location by a factor of 3.5 on T3A, 2.5 on T3A⁻ and 2.0 on T3B. A correlation accurate to ten per cent cannot locate transition to ten per cent in such a flow.

case	branch	Re_theta_t_pred	Re_theta_t_exp	err_pct	laminar_run_meas_over_march	n_laminar_stations
ERCOFTAC T3A flat plate (bypass transition)	bypass	282.1	272.3	3.6	1.058	6
ERCOFTAC T3A- flat plate (low-Tu bypass)	bypass	683.6	818.8	-16.5	0.988	8
ERCOFTAC T3B flat plate (high-Tu bypass)	bypass	168.0	181.3	-7.3	1.058	4
ERCOFTAC T3C4 flat plate (laminar separation bubble)	separation	266.2	381.3	-30.2	1.02	9
Schubauer & Skramstad flat plate (natural transition)	TS-natural	1100.7	1100.0	0.1		0

(columns 2-7 of 8; case repeated on each block)

case	location_gain_bypass_threshold
ERCOFTAC T3A flat plate (bypass transition)	3.5
ERCOFTAC T3A- flat plate (low-Tu bypass)	2.5
ERCOFTAC T3B flat plate (high-Tu bypass)	2.0
ERCOFTAC T3C4 flat plate (laminar separation bubble)	
Schubauer & Skramstad flat plate (natural transition)	

Table 26. The laminar branch against the measurements, and the sensitivity of the crossing (residual_diagnostics.csv). (columns 8-8 of 8; the table is split across 2 blocks so that no cell is compressed to illegibility, with case repeated on each)

quantity	model	measured	note
separation station x [m]	1.236	1.295	model / first station at the C_f floor
reattachment station x [m]	1.291	1.395	model / last station at the C_f floor
bubble length [m]	0.055	0.1	measured length is a lower bound
dtheta/dx across the bubble [1/m]	1.889e-03	5.906e-03	momentum integral is exact given U_e and H

shape factor at reattachment	3.4	5.17	solved on the combined attached + reverse-flow family (H to 6.41), not capped: the march crosses the fold in H*(H) by taking the root nearest the previous H
dU_e/dx, x = 1.195-1.495 m [1/s]	-0.3782	-0.38	smoothing spline against the tabulated points. Over the shorter C_f-floor interval alone the measured value is -0.20 /s, which is the figure the report's shape-factor arithmetic uses
max |U_e spline - tabulated| [m/s]	0.011	0.01	the edge velocities are tabulated to 0.01 m/s and the spline is a SMOOTHING fit weighted at half of that, not an interpolant, so a departure of order one quotation step is intended rather than an error; the worst is 1.1 steps. (This note read 'so the spline is within quotation', which the value beside it contradicted.)

Table 27. The T3C4 bubble, modelled against measured (bubble_diagnostics.csv). The momentum integral across the dead-air region is exact given U_e and H, so these are the whole of the residual.

11.2 NLF(1)-0416 aerofoil — 86 transition locations

The largest single body of evidence in this work, and genuinely out of sample. That was not always true: the anchor of the natural branch — the one quantity that branch takes from measurement — used to be chosen at whatever value put the most of these 86 predictions inside the experimental bracket, which made this a calibration set while this section, the figure below and the project README all called it otherwise. It is now set on the Schubauer & Skramstad plate alone, which it reproduces to 0.07 %, and the cost of that is reported rather than absorbed: the bracket count fell from 51 to 50 and the mean absolute error rose, which is what happens when a constant stops being fitted to the set it is scored on. The error today is 0.0386 chord (Table 28 below); this sentence used to give it as 0.0334c, which was the value at the time of that change and has since moved with corrections that have nothing to do with the anchor - a before-and-after pair frozen in the narrative stops describing the current model the moment anything else moves, so only the current figure is quoted and it is read from the table. Transition locations were digitised from Fig. 9 of the source report at four chord Reynolds numbers on both surfaces, and each condition is matched by trimming the incidence to the measured lift coefficient. The experiment brackets transition between adjacent orifices 0.05c apart, so its own uncertainty is ±0.025c and a prediction inside that band cannot be distinguished from the measurement. Both the natural (TS) and the separation-induced branches are selected on this set.

set	points	mean_abs_err_c	bias_c	within_bracket	within_bracket_pct
Upper surface	46	0.0389	-0.0102	23	50.0
Lower surface	40	0.0382	-0.0307	27	67.5
All	86	0.0386	-0.0198	50	58.1
Conditions the method accepts	84	0.0315	-0.0152	50	59.5

Table 28. NLF(1)-0416 error statistics by surface (aerofoil_nlf0416_summary.csv).

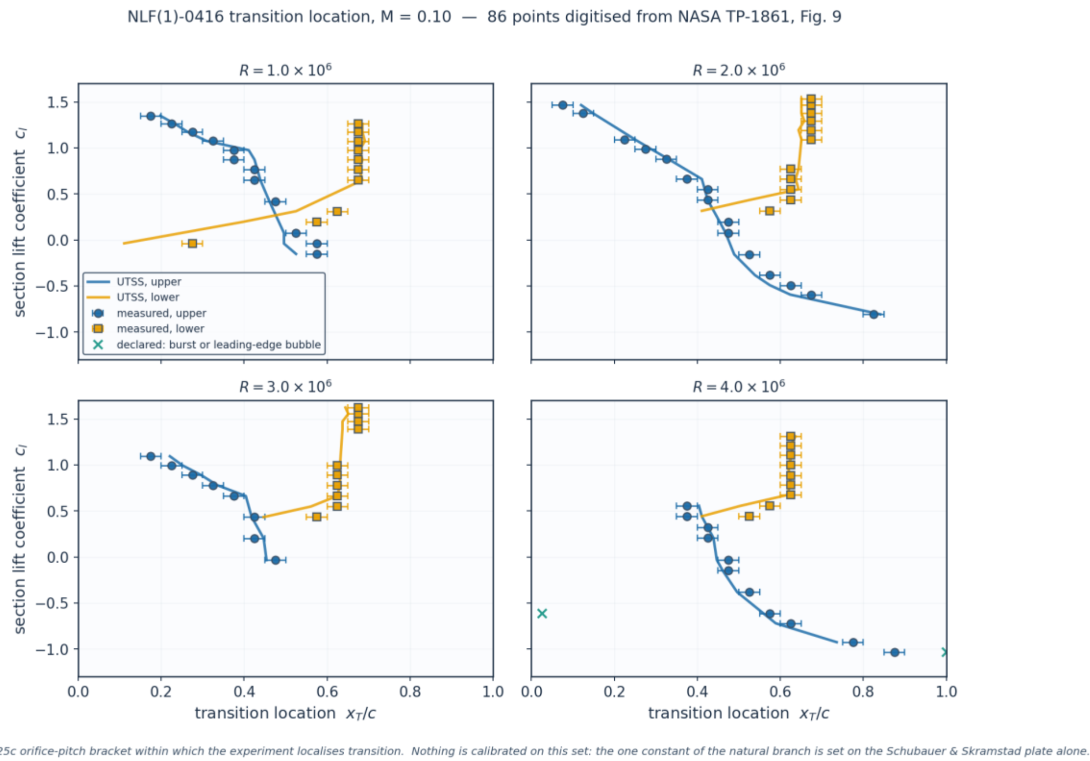


Fig. 49. Transition location against lift coefficient at four chord Reynolds numbers, measurement bars = the ±0.025c orifice bracket.

Re_c	surface	c_l_exp	alpha_deg	c_l_solver	x_tr_c_exp	x_tr_c_pred
1.0e+06	upper	1.35	6.483	1.3503	0.175	0.198
1.0e+06	upper	1.174	5.036	1.1743	0.275	0.264
1.0e+06	upper	0.976	3.413	0.9762	0.375	0.411
1.0e+06	upper	0.765	1.687	0.765	0.425	0.432
1.0e+06	upper	0.419	-1.132	0.4187	0.475	0.461
1.0e+06	upper	-0.039	-4.846	-0.0387	0.575	0.496
1.0e+06	lower	1.262	5.759	1.2623	0.675	0.685
1.0e+06	lower	1.075	4.224	1.0753	0.675	0.691
1.0e+06	lower	0.874	2.578	0.8741	0.675	0.685
1.0e+06	lower	0.651	0.757	0.6508	0.675	0.685
1.0e+06	lower	0.197	-2.935	0.1968	0.575	0.397
2.0e+06	upper	1.466	7.438	1.4661	0.075	0.12
2.0e+06	upper	1.089	4.339	1.0893	0.225	0.251
2.0e+06	upper	0.883	2.652	0.8831	0.325	0.329
2.0e+06	upper	0.551	-0.058	0.5507	0.425	0.418
2.0e+06	upper	0.198	-2.927	0.1978	0.475	0.454
2.0e+06	upper	-0.156	-5.791	-0.1552	0.525	0.489
2.0e+06	upper	-0.491	-8.514	-0.491	0.625	0.575

2.0e+06	upper	-0.807	-11.077	-0.8069	0.825	0.841
2.0e+06	lower	1.469	7.463	1.4691	0.675	0.651
2.0e+06	lower	1.294	6.022	1.2943	0.675	0.658
2.0e+06	lower	1.093	4.372	1.0933	0.675	0.651
2.0e+06	lower	0.663	0.855	0.6628	0.625	0.637
2.0e+06	lower	0.434	-1.01	0.4337	0.625	0.525
3.0e+06	upper	1.094	4.38	1.0943	0.175	0.221
3.0e+06	upper	0.89	2.709	0.8901	0.275	0.296
3.0e+06	upper	0.662	0.847	0.6618	0.375	0.404
3.0e+06	upper	0.202	-2.894	0.2018	0.425	0.447
3.0e+06	lower	1.623	8.733	1.6227	0.675	0.644
3.0e+06	lower	1.472	7.487	1.4721	0.675	0.637

(columns 2-7 of 16; Re_c repeated on each block)

Re_c	err_c	err_pct	within_bracket	degenerate	burst	le_sep
1.0e+06	0.023	12.9	True	False	False	False
1.0e+06	-0.011	-4.2	True	False	False	False
1.0e+06	0.036	9.7	False	False	False	False
1.0e+06	0.007	1.7	True	False	False	False
1.0e+06	-0.014	-3.0	True	False	False	False
1.0e+06	-0.079	-13.7	False	False	False	False
1.0e+06	0.01	1.4	True	False	False	False
1.0e+06	0.016	2.4	True	False	False	False
1.0e+06	0.01	1.4	True	False	False	False
1.0e+06	0.01	1.4	True	False	False	False
1.0e+06	-0.178	-30.9	False	False	False	False
2.0e+06	0.045	59.7	False	False	False	False
2.0e+06	0.026	11.6	False	False	False	False
2.0e+06	0.004	1.1	True	False	False	False
2.0e+06	-0.007	-1.6	True	False	False	False
2.0e+06	-0.021	-4.5	True	False	False	False
2.0e+06	-0.036	-6.8	False	False	False	False
2.0e+06	-0.05	-8.1	False	False	False	False
2.0e+06	0.016	1.9	True	False	False	False
2.0e+06	-0.024	-3.5	True	False	False	False
2.0e+06	-0.017	-2.5	True	False	False	False
2.0e+06	-0.024	-3.5	True	False	False	False
2.0e+06	0.012	2.0	True	False	False	False
2.0e+06	-0.1	-16.0	False	False	False	False
3.0e+06	0.046	26.1	False	False	False	False
3.0e+06	0.021	7.5	True	False	False	False
3.0e+06	0.029	7.8	False	False	False	False
3.0e+06	0.022	5.1	True	False	False	False
3.0e+06	-0.031	-4.6	False	False	False	False
3.0e+06	-0.038	-5.6	False	False	False	False

(columns 8-13 of 16; Re_c repeated on each block)

Re_c	x_sep_c	bubble_len_theta_s	mechanism
1.0e+06			TS-natural
1.0e+06			TS-natural
1.0e+06	0.335	208.8	separation
1.0e+06	0.356	211.3	separation
1.0e+06	0.383	213.5	separation

1.0e+06			TS-natural
1.0e+06	0.582	238.7	separation
1.0e+06	0.582	243.7	separation
1.0e+06	0.575	238.5	separation
1.0e+06	0.568	243.9	separation
1.0e+06			TS-natural
2.0e+06			TS-natural
2.0e+06			TS-natural
2.0e+06			TS-natural
2.0e+06	0.369	192.6	separation
2.0e+06	0.404	189.3	separation
2.0e+06			TS-natural
2.0e+06			TS-natural
2.0e+06			TS-natural
2.0e+06	0.589	207.7	separation
2.0e+06	0.589	222.2	separation
2.0e+06	0.582	218.7	separation
2.0e+06	0.568	205.9	separation
2.0e+06			TS-natural
3.0e+06			TS-natural
3.0e+06			TS-natural
3.0e+06	0.363	199.9	separation
3.0e+06	0.404	198.3	separation
3.0e+06	0.596	198.3	separation
3.0e+06	0.589	198.5	separation

Table 29. NLF(1)-0416 point-by-point comparison (aerofoil_nlf0416.csv, sampled). (columns 14-16 of 16; the table is split across 3 blocks so that no cell is compressed to illegibility, with Re_c repeated on each)

(table sampled evenly to 30 of 86 rows; full data in 06_validation/aerofoil_nlf0416.csv)

The separation branch claims to predict a LENGTH and not a point, and that the length scales with the disturbance environment because it is $N_{crit}\,\theta_s/\sigma_{sep}$. Both of these datasets form bubbles, at free-stream turbulence levels seventy times apart, so together they measure that claim rather than illustrate it.

dataset	Tu_pct	N_crit	n_bubbles	median_len_theta_s	min_len_theta_s	max_len_theta_s
ERCOFTAC T3C4 flat plate	2.11	0.5	1	28.1	28.1	28.1
Somers NLF(1)-0416 aerofoil	0.03	7.64	50	207.6	186.7	1380.3

(columns 2-7 of 9; dataset repeated on each block)

dataset	ratio_to_T3C4	N_crit_ratio_to_T3C4
ERCOFTAC T3C4 flat plate	1.0	1.0
Somers NLF(1)-0416 aerofoil	7.39	15.28

Table 30. Bubble length against the disturbance environment (bubble_length_scaling.csv): the separating plate at Tu = 2.11 % against every NLF(1)-0416 bubble at 0.03 %, in separation momentum thicknesses. (columns 8-9 of 9; the table is split across 2 blocks so that no cell is compressed to illegibility, with dataset repeated on each)

11.3 Swept wings — the cross-flow branch

The cross-flow coefficient is set on the first of these two experiments and nothing is calibrated on the second, which is a different facility, section and era. The branch reproduces the calibration set to 21.8 % at the frozen constant C1 = 150. On the independent set that same constant gives 51.1 %, and C1 = 200 gives 17.6 % — so the criterion has the right functional form on both wings but not one critical value that serves both. The independent set is therefore tabulated at BOTH ends of that band: reporting only C1 = 150 understates what the criterion does here, and reporting only C1 = 200 would be a per-case re-tune of the kind this work is claiming not to need. The spread between the two columns is the limitation, stated rather than averaged away. Nothing else in the model differs between the two columns.

What each experiment requires of the criterion is measured rather than asserted (Table 37). The march is run with every branch disabled so that it reaches the measured transition station, and the criterion is evaluated there. Dagenhart & Saric require a critical Re_θ2 of 153 with a 19.5 % coefficient of variation over six chord Reynolds numbers; Boltz et al. require 234 with 4.6 % over four sweep angles and a factor of three in chord Reynolds number. Each facility is therefore internally consistent — the second markedly so — and the two differ by 53 %. That is the shape of a receptivity difference rather than of a criterion with the wrong form: stationary cross-flow vortices are seeded by leading-edge roughness, and neither report documents the surface finish. THREE attempts to close the gap fail and are recorded rather than dropped. Solving the swept sections in the plane normal to the leading edge, which is the correct mean flow and had not been done, widens the gap rather than closing it — it was the strongest remaining physical candidate and it is now tested rather than argued about. Replacing the constant surrogate by the exact Falkner-Skan-Cooke factor K(λ) makes matters worse, taking the pooled coefficient of variation from 26 to 81 % and inverting the ratio between the two sets. Giving the cross-flow branch its own amplification threshold, separate from the one Mack's relation supplies for Tollmien-Schlichting waves — which is defensible, since a stationary cross-flow vortex is not seeded by free-stream turbulence — does not help either (Table 31). Sweeping that threshold from N = 2 to N = 12, with C1 refitted on the calibration set at every value so the two constants are not confounded, takes the calibration set from 17.9 to 15.6 % and the INDEPENDENT set from 55.1 to 61.9 % — away from agreement, not towards it. This report previously said it moved the independent set "only from 55 to 51 %"; the sweep is generated now, and both the size and the sign of that were wrong. The reason it cannot help is that once Re_θ2 exceeds C1 the amplification builds so quickly that the threshold is nearly redundant with C1 itself, so the two constants cannot be separated by these data — which the refitted C1 column shows directly, barely moving across the sweep.

CF_N	C1_refitted	calibration_err_pct	independent_err_pct	n_C1_evaluated
2.0	150.0	17.9	55.1	7
5.0	125.0	17.4	65.9	7
8.0	125.0	16.6	64.4	7
12.0	125.0	15.6	61.9	7

Table 31. A separate amplification threshold for the cross-flow branch, with C1 refitted on the calibration set at each value (crossflow_threshold_sweep.csv).

The eight formulations of the branch that this solver can be put into are scored against both experiments in Table 32, and not one of them reconciles the two. The pattern is the same throughout: every variant that helps the independent set costs more on the calibration set, and the three that cost nothing there help nothing here. That was a claim about what had been tried; it is a table now.

formulation	calibration_err_pct	independent_err_pct	pooled_err_pct	helps_independent_vs_shipped	costs_calibration_vs_shipped
amplification integral, C1 = 150 (shipped)	21.8	51.1	33.5	False	False
amplification integral, C1 = 200	67.5	17.6	47.5	True	True
local C1 threshold, no amplification integral	17.2	55.1	32.4	False	False
local C1 threshold, C1 = 200	62.6	22.6	46.6	True	True
exact Falkner-Skan-Cooke K(lambda) for the thickness	44.3	50.4	46.7	True	True
exact K(lambda), local C1 threshold	51.2	44.3	48.4	True	True
streamwise frame (no normal-plane transformation)	14.5	52.3	29.6	False	False
solved stationary cross-flow eigenvalue problem, N_cf = 7.0	16.7	83.7	43.5	False	False

Table 32. Every cross-flow formulation the solver exposes, scored on both swept wings (crossflow_formulations.csv).

A fourth attempt goes after the criterion's form directly, by solving the problem C1 stands in for. The Orr-Sommerfeld equation is solved on the velocity resolved along each wave-angle direction, $U_\psi = \cos\Lambda\, f' \cos\psi + \sin\Lambda\, g \sin\psi$ in units of the total edge speed; the wave angle at which the mode is stationary — $\omega_r = 0$, which is what a naphthalene visualisation can see — is found by bisection on the sign change, and the amplification a stationary packet accumulates over dx is $\omega_i\, dx/c_{gx}$ with the chordwise group velocity taken from the two derivatives the (k, ψ) parameterisation already provides. The sweep angle the similarity solution is given is the LOCAL one, between the external streamline and the chord line: the span-wise edge velocity is constant on an infinite swept wing while the chordwise one grows through the favourable run, so that angle falls from 59° at 3 % chord to 40° at 60 % on these sections, against a leading-edge value of 45°.

Two things make this harder than the two-dimensional problem and both decide the answer. The mode cannot be found by asking for the eigenvalue nearest $c = 0$: the resolved profile's edge velocity is near zero at exactly the wave angles of interest, so the discretised continuous spectrum crowds onto the physical mode rather than onto $c_r = 1$, and a nearest-root search returns growth rates three orders above anything physical. What separates them is the eigenfunction, not the eigenvalue. And the outer boundary must be far enough out for that test to mean anything. The filter measures the eigenfunction over the outer fifth of the domain and requires it to be under 2 per cent of its peak; a wave of wavenumber k decays as $\exp(-k\,y)$, so at $k = 0.1$ — the longest wave of interest — that is $\exp(-0.1\times0.8\times40) = 4$ per cent on a $y_{max} = 40\,\theta$ grid, and the filter throws the PHYSICAL mode away, leaving only short waves and putting the envelope maximum on the edge of the surviving band. At $y_{max} = 100\,\theta$ the same wave is at $\exp(-8) = 0.03$ per cent and passes, while the discretised continuous spectrum stays above 25 per cent of its peak out there and does not. This paragraph gave the two decay figures as five and 0.3 per cent; neither is what $\exp(-k\,y)$ returns where the filter looks, and the first did not even fail the 2 per cent test the argument turns on.

The result is checked against an independent stability code before it is used for anything. Dagenhart & Saric computed stationary N-factors with SALLY for three of their six conditions and tabulated them; against those three the present solve differs by 0.50 on average and 0.56 root-mean-square (Table 33), which is agreement, not calibration — nothing here is fitted to them. On the levels the two facilities require it does better than the surrogate and still not well enough. Dagenhart & Saric's transitions occur at $N_{cf} = 7.02 \pm 1.37$ and Boltz et al.'s at 5.67 ± 1.63, a ratio of 1.24 where the algebraic surrogate's critical Reynolds numbers differ by 1.53. The levels move together; the scatter does not. Within Boltz's four conditions the coefficient of variation rises from 4.6 % on the surrogate to 28.7 % here, and the reason is visible rather than statistical: N_{cf} at the measured transition falls monotonically with sweep angle, 7.58 at 20° down to 3.66 at 50°, so no single threshold can pass through all four.

dataset	sweep_deg	Re_c	x_tr_c_measured	N_cf	N_cf_SALLY	N_cf_minus_SALLY
Dagenhart & Saric (calibration)	45.0	1.920e+06	0.78	8.2		
Dagenhart & Saric (calibration)	45.0	2.190e+06	0.73	8.85		
Dagenhart & Saric (calibration)	45.0	2.370e+06	0.58	7.5	6.8	0.7
Dagenhart & Saric (calibration)	45.0	2.730e+06	0.45	6.37	6.5	-0.13
Dagenhart & Saric (calibration)	45.0	3.270e+06	0.33	5.47		
Dagenhart & Saric (calibration)	45.0	3.730e+06	0.3	5.74	6.4	-0.66
Boltz et al. (independent)	20.0	2.700e+07	0.207	7.58		

Boltz et al. (independent)	30.0	1.500e+07	0.213	6.07		
Boltz et al. (independent)	40.0	1.200e+07	0.2	5.36		
Boltz et al. (independent)	50.0	9.500e+06	0.211	3.66		

(columns 2-7 of 12; dataset repeated on each block)

dataset	N_TS	N_crit_TS	Re_theta	sweep_local_deg_at_x003	sweep_local_deg_at_x060
Dagenhart & Saric (calibration)	2.48	7.64	615.5	59.1	39.7
Dagenhart & Saric (calibration)	0.03	7.64	513.4	59.1	39.7
Dagenhart & Saric (calibration)	0.0	7.64	447.4	59.1	39.7
Dagenhart & Saric (calibration)	0.0	7.64	418.2	59.1	39.7
Dagenhart & Saric (calibration)	0.0	7.64	382.3	59.1	39.7
Dagenhart & Saric (calibration)	0.0	7.64	385.6	59.1	39.7
Boltz et al. (independent)	5.15	7.64	1476.1	18.8	18.0
Boltz et al. (independent)	2.31	7.64	1029.7	28.4	27.3
Boltz et al. (independent)	1.06	7.64	787.5	38.2	36.8
Boltz et al. (independent)	0.34	7.64	605.1	48.2	46.8

Table 33. Stationary cross-flow amplification factor at the measured transition, every branch of the kernel held off, against Dagenhart & Saric's own SALLY N-factors where they exist (crossflow_amplification.csv). (columns 8-12 of 12; the table is split across 2 blocks so that no cell is compressed to illegibility, with dataset repeated on each)

dataset	n_points	mean_N_cf	sd_N_cf	coeff_of_variation_pct	note
Dagenhart & Saric (calibration)	6	7.02	1.37	19.6	critical N this facility requires
Boltz et al. (independent)	4	5.67	1.63	28.7	critical N this facility requires
vs SALLY (three conditions)	3	0.5	0.56		mean |N - N_SALLY| and RMS in the last two columns

Table 34. What each facility requires of the amplification factor (crossflow_amplification_summary.csv).

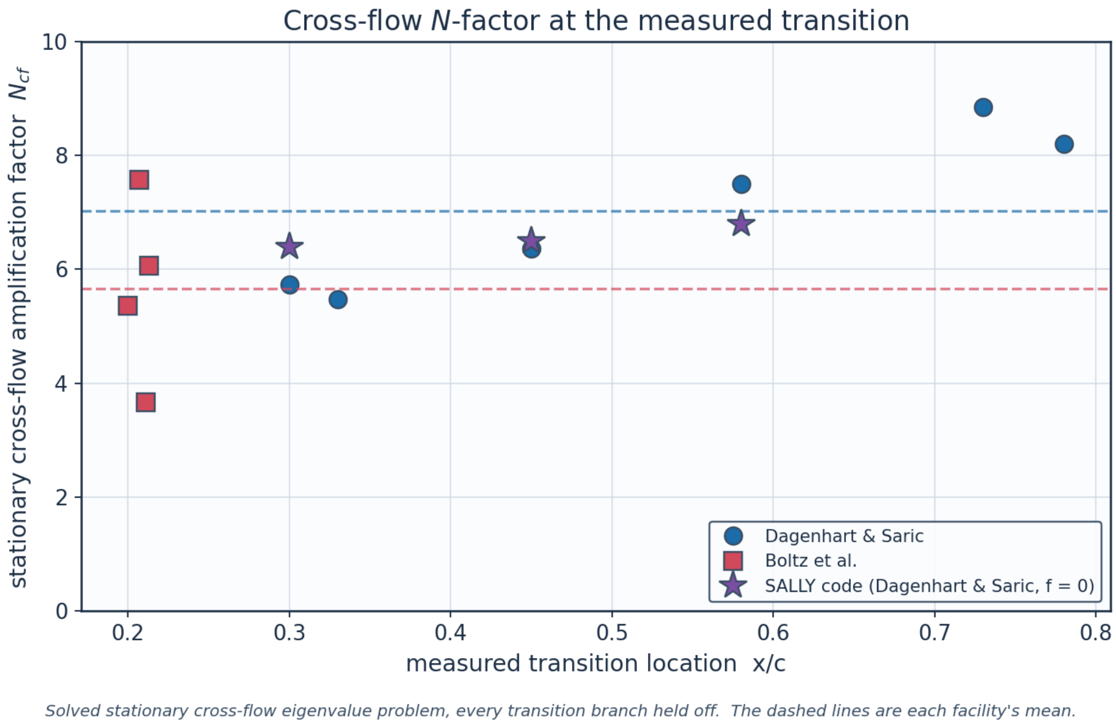
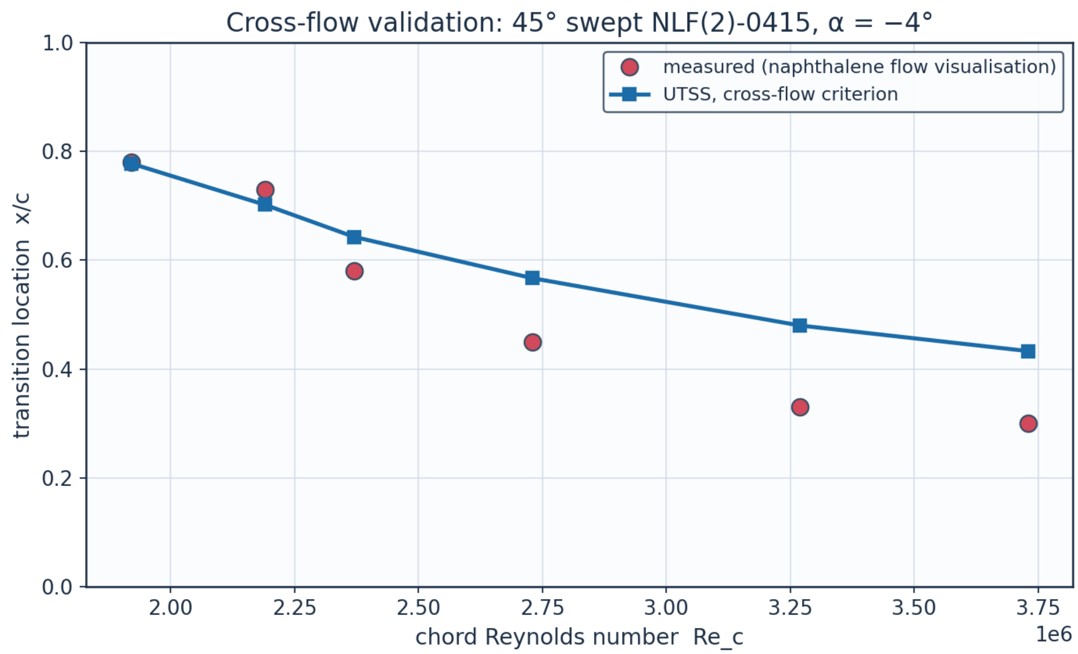


Fig. 50. Cross-flow N-factor at the measured transition. The two facilities sit on different levels; the stars are SALLY.

Put through the kernel with its threshold set on the calibration set alone — the same way C1 was — it follows the pattern every other variant follows, from the other side. It IMPROVES the calibration set, from 21.8 % to 16.7 % in transition location, and it costs the independent set 83.7 % against 51.1 %, for a pooled 43.5 % against 33.5 % (Table 32 carries it as a row like every other variant). It is therefore not adopted, for the reason the shipped branch is: a formulation that is better only on the set its one constant was fitted to has not been shown to be better. What fails is that a stationary cross-flow vortex is forced by surface roughness, and the amplification factor carries no information about how large the disturbance was when it started. Dagenhart & Saric say so themselves: “the receptivity portion of the transition process is equally important in the vortex development, growth, and eventual breakdown”, and they cite Radeztsky et al. for the finding that micron-sized roughness near the attachment line strongly influences crossflow-dominated transition. A polished natural-laminar-flow model of 1993 and an untapered wing tested in 1960 are not the same surface, and neither report gives a roughness height. Closing the gap needs an input the experiments do not contain, not a better stability calculation — and that is now a measurement rather than a conjecture.

Re_c	x_tr_c_exp	x_tr_c_pred	err_pct	Re_theta_at_onset	mechanism
1.920e+06	0.78	0.778	-0.3	577.5	separation
2.190e+06	0.73	0.702	-3.8	480.6	crossflow
2.370e+06	0.58	0.643	10.9	471.4	crossflow
2.730e+06	0.45	0.567	25.9	474.3	crossflow
3.270e+06	0.33	0.48	45.6	474.7	crossflow
3.730e+06	0.3	0.433	44.5	478.6	crossflow

Table 35. Cross-flow validation, 45° swept NLF(2)-0415 (Dagenhart & Saric — the calibration set).



Source: Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in ... (full citation in sources_and_references.csv)

Fig. 51. Transition location against chord Reynolds number, 45° swept NLF(2)-0415.

sweep_deg	alpha_deg	Re_c	x_tr_c_exp	x_tr_c_pred_C1_150	err_pct_C1_150	Re_theta2_at_onset_C1_150
20.0	0.0	2.700e+07	0.207	0.093	-55.0	154.2
30.0	0.0	1.500e+07	0.213	0.093	-56.2	154.9
40.0	0.0	1.200e+07	0.2	0.098	-51.1	161.7
50.0	0.0	9.500e+06	0.211	0.122	-42.0	162.5

(columns 2-7 of 11; sweep_deg repeated on each block)

sweep_deg	mechanism	x_tr_c_pred_C1_200	err_pct_C1_200	Re_theta2_at_onset_C1_200
20.0	crossflow	0.161	-22.4	207.1
30.0	crossflow	0.161	-24.6	208.0
40.0	crossflow	0.161	-19.7	211.6
50.0	crossflow	0.203	-3.8	213.4

Table 36. Independent swept-wing check, NACA 64(2)A015 (Boltz et al.) at both ends of the reported cross-flow band — nothing calibrated here. (columns 8-11 of 11; the table is split across 2 blocks so that no cell is compressed to illegibility, with sweep_deg repeated on each)

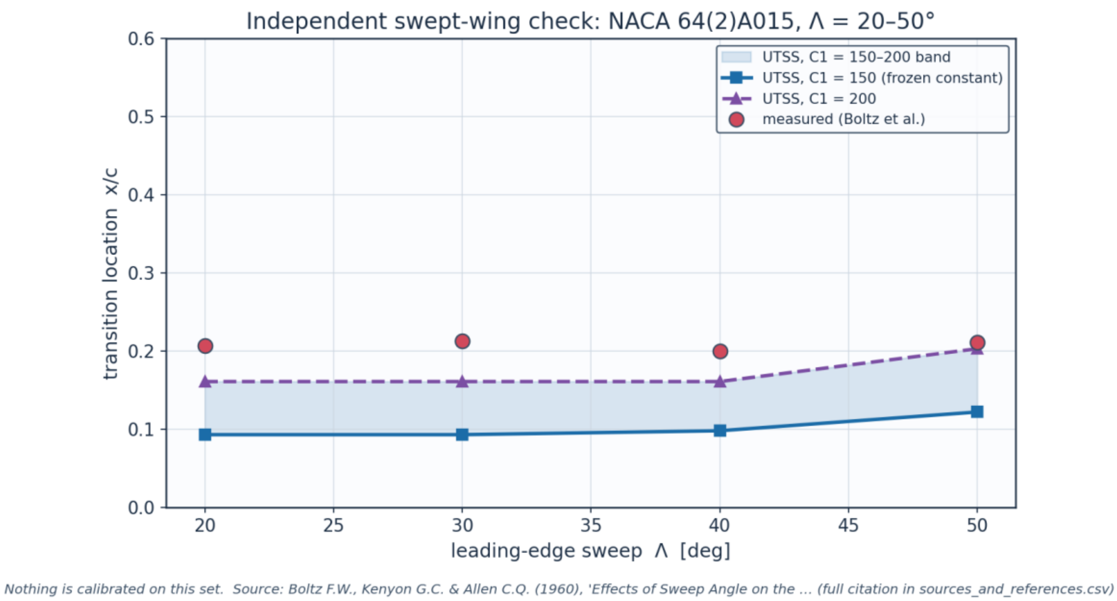


Fig. 52. Transition location against sweep angle, NACA 64(2)A015, with the C1 = 150–200 band shaded.

11.4 What the two swept-wing experiments require of the criterion

dataset	criterion	mean_critical_value	coeff_of_variation_pct	n_points
Dagenhart & Saric (calibration)	surrogate Re_theta2	153.0	19.5	6
Dagenhart & Saric (calibration)	exact Falkner-Skan-Cooke Re_cf	428.8	59.0	6
Boltz et al. (independent)	surrogate Re_theta2	233.8	4.6	4
Boltz et al. (independent)	exact Falkner-Skan-Cooke Re_cf	117.8	5.5	4
Both facilities pooled	surrogate Re_theta2	185.3	25.7	10
Both facilities pooled	exact Falkner-Skan-Cooke Re_cf	304.4	81.4	10

Table 37. What each swept-wing experiment requires of the cross-flow criterion, evaluated at the measured transition station (crossflow_criticals_summary.csv).

dataset	sweep_deg	Re_c	x_tr_c_measured	Re_theta	Re_theta2_surrogate	Re_cf_exact_FSC
Dagenhart & Saric (calibration)	45.0	1.920e+06	0.78	615.5	204.5	811.9
Dagenhart & Saric (calibration)	45.0	2.190e+06	0.73	513.4	170.6	677.3
Dagenhart & Saric (calibration)	45.0	2.370e+06	0.58	447.4	148.7	354.4
Dagenhart & Saric (calibration)	45.0	2.730e+06	0.45	418.2	139.0	281.7
Dagenhart & Saric (calibration)	45.0	3.270e+06	0.33	382.3	127.1	226.3

Dagenhart & Saric (calibration)	45.0	3.730e+06	0.3	385.6	128.2	221.1
Boltz et al. (independent)	20.0	2.700e+07	0.207	1476.1	237.3	119.8
Boltz et al. (independent)	30.0	1.500e+07	0.213	1029.7	242.0	119.6
Boltz et al. (independent)	40.0	1.200e+07	0.2	787.5	237.9	123.4
Boltz et al. (independent)	50.0	9.500e+06	0.211	605.1	217.9	108.4

Table 38. Point-by-point cross-flow criterion values at the measured transition stations (crossflow_criticals.csv).

The gap cannot be converted into a roughness ratio by this method, and saying why is firmer than attributing it to receptivity by elimination. For a roughness-seeded stationary vortex the natural currency is amplification, $\Delta N = \ln(A_{0,1}/A_{0,2})$, which is the ratio of initial amplitudes and so of effective leading-edge roughness. But the rate this branch integrates is read off a separated STREAMWISE profile and is nearly Reynolds-independent — 0.0417 at Re_θ = 200 against 0.0461 at 8000 — so N_cf is essentially σ times run length over θ, which scales as √Re_c: a property of the chord Reynolds number, not of the cross-flow instability. The two facilities do not overlap in chord Reynolds number at all — 1.92–3.73 million against 9.5–27.0 — and the N they require at their own measured stations is 0.0–11.9 and 43.1–129.2 respectively (Table 39), while the critical cross-flow Reynolds number is tight within each, at 19.5 and 4.6 per cent. The elimination behind the receptivity attribution was therefore incomplete: it had not considered that the branch's own rate carries no cross-flow physics, which is a defect of the model and not a property of the experiments.

What would close it is a defined piece of work on the method rather than a request for measurements on two wings from 1960 and 1999: the Orr-Sommerfeld problem solved on the Falkner-Skan-Cooke CROSS-FLOW profile — which stability.fsc_profile already returns — and tabulated the way the streamwise rates of Sec. 4.3 are. Both closures were compared over both facilities while this was established: the amplification integral, which is the shipped form, gives 21.8 and 51.1 per cent (read from the two tables above, where they had been left at 21.8 and 51.1), the local C1 criterion 17.2 and 55.1, pooled 33.5 against 32.4. Neither dominates, so the shipped form is kept and the comparison recorded rather than the choice asserted.

dataset	Re_theta2_mean	Re_theta2_cov_pct	N_cf_min	N_cf_max	n_points
Dagenhart & Saric (calibration)	153.0	19.5	0.0	11.92	6
Boltz et al. (independent)	233.8	4.6	43.14	129.19	4

Table 39. What each swept-wing facility requires, in the two currencies: a critical cross-flow Reynolds number, which is consistent within each facility, and an amplification factor, which is not comparable between them.

TN D-338 reports crossflow Reynolds numbers of its own: "the critical values ... for vortex formation ... range from about 135 to 190", and "the values ... for beginning transition were found to be between 190 and 260." It is tempting to note that the values this work's SURROGATE requires — 153 on Dagenhart &

Saric and 234 on Boltz et al. — fall in those two ranges respectively, and to read that as the two datasets marking two different events. That reading does not survive checking, and it is recorded here as rejected rather than dropped. The surrogate is Re_theta2 = 0.47 Re_theta sin(L), a MOMENTUM-THICKNESS quantity; what a 1960 report means by a crossflow Reynolds number is w_max delta_10 / nu, the displacement-type one. This work computes that too, in the "exact Falkner-Skan-Cooke" column of Table 37, and it gives 429 and 118 — the first far above TN D-338's transition range, the second below its vortex-formation range, and the two INVERTED against the surrogate. Two quantities on different scales landing in the right intervals is a coincidence, not corroboration.

What does survive is weaker and qualitative. TN D-338 separates two events in ONE facility by roughly a factor of 1.4, and the two facilities here differ by a factor of 1.53. A gap of that size between experiments is therefore the size of the gap a single facility reports between vortex formation and the beginning of transition, so what each experiment CALLS transition remains a candidate explanation alongside receptivity — as an argument about event definition, not as a numerical match.

One explanation can be ruled out rather than merely doubted. If the difference between the two facilities were a Reynolds-number effect the criterion is missing, the requirement would have to vary with chord Reynolds number in the same direction within a facility as it does between them. It does not. Within Dagenhart & Saric the required critical value FALLS steeply with chord Reynolds number, by 248 per decade with a correlation of -0.90 over a factor of two in Re_c; Boltz et al. sit at six times that Reynolds number and require 53 % MORE, not less, and are themselves flat across a factor of three, at +32 per decade. The between-facility offset therefore has the opposite sign to the within-facility trend, and no monotone function of Re_c can carry one set into the other. That is what leaves receptivity — the leading-edge surface finish neither report documents — as the explanation, and it is now a measurement rather than an appeal to the literature.

Two numbers in this section have to be reconciled or they look inconsistent: the band is reported as C1 = 150–200, while Table 37 says the independent facility requires a critical Re_θ2 of 234. They are different quantities. C1 does not fire the branch; it starts the amplification integral, which then decides where transition is placed, so the EFFECTIVE critical value at the predicted station is higher than C1. Measured at the four Boltz conditions it is 158 for C1 = 150 and 210 for C1 = 200 — read from the table above rather than typed, where they stood at 157 and 209 — so the integral adds about 5 %, not the 17 % that would be needed to reach 234. That residual 12 % is precisely why the upper end of the band still leaves a 17.6 % error in location on the independent set, and it is the honest reason the band is offered as a bound rather than as a value.

dataset	n_points	Re_c_min	Re_c_max	mean_ required	slope_per_ decade_Re_c	correlation
Dagenhart & Saric (calibration)	6	1.92e+06	3.73e+06	153.0	-247.7	-0.903
Boltz et al. (independent)	4	9.50e+06	2.70e+07	233.8	32.1	0.58

Table 40. The required critical value against chord Reynolds number within each facility (crossflow_reynolds_trend.csv). The trends have opposite signs to the offset between the facilities.

11.5 Ablations — what each element of the formulation is worth

Each of the three elements that distinguish this formulation is switched off in turn, everything else held fixed, and the same two datasets the natural branch reaches are re-run. The table is regenerated by gen_validation.py and is not quoted from memory.

configuration	SS_err_pct	mean_abs_err_c	within_bracket	points	within_bracket_pct
no bubble closure	0.1	0.063	18	86	20.9
one-equation laminar march	-10.0	0.0415	42	86	48.8
Drela-Giles envelope	-11.2	0.0435	46	86	53.5
full model	0.1	0.0386	50	86	58.1

Table 41. Ablation study (ablations.csv): Schubauer-Skramstad onset error and the 86 aerofoil conditions, one closure removed at a time.

Calibration. The solver is calibrated through the single constant set of Table 7. The critical amplification factor is not among the constants: it follows from the free-stream turbulence intensity by Mack's correlation, clamped to the 0.0008-0.0298 range over which that correlation is quoted. The SAME constants reproduce every dataset here — five flat plates, two swept wings and 86 aerofoil conditions — demonstrating that no case-specific physics tuning is required, which is the essence of the universality claim.

Which sets are calibration and which are validation. The constants are frozen across every case, but three of them were SET on data in these tables, and a universality claim is only worth what its out-of-sample evidence is worth, so this is stated rather than left to be inferred. N_anchor, the units offset of the amplification scale, is set on the Schubauer & Skramstad plate. tu_hist, the weight on the flow history of Tu in the Abu-Ghannam & Shaw correlation, is set on the three ERCOFTAC T3 plates, which are the only bypass data here. CF_ratio, the cross-flow surrogate, is set on the Dagenhart & Saric wing. Everything else is out of sample: ERCOFTAC T3C4, which is the only test of the separation branch; all 86 NLF(1)-0416 conditions, which are the only test of the natural branch on a real aerofoil; and the Boltz et al. swept wing, which is the only independent test of the cross-flow branch. One constant per branch, each set on the smallest dataset that determines it, with the largest dataset for each branch held back.

11.6 Sources and references

All data sources used for validation, for calibration of the closures, and for the case-study definition are recorded below.

category	item	reference
Validation	ERCOFTAC T3A flat plate	Roach P.E. & Brierley D.H. (1990), 'The influence of a turbulent free stream on zero pressure gradient transitional boundary layer development', ERCOFTAC Workshop; reproduced in Savill (1993) and Langtry & Menter, AIAA J. 47(12), 2009.

Validation	ERCOFTAC T3B flat plate	Roach & Brierley (1990), ERCOFTAC T3B (Tu=6%); Langtry & Menter, AIAA J. 47(12), 2009.
Validation	Schubauer & Skramstad natural transition	Schubauer G.B. & Skramstad H.K. (1948), 'Laminar boundary-layer oscillations and transition on a flat plate', NACA Report 909. NO MEASURED C_f DISTRIBUTION IS CARRIED for this case and there is no experiment_SS.csv: the report's transition results are figures in a 1948 scan and no reliable digitisation was available to the author, so only the onset Reynolds number is used - Re_x,tr = 2.8e6 at Tu = 0.03 %, equivalently Re_theta_t = 1100 via Re_theta = 0.664 sqrt(Re_x) - and it appears in validation_summary.csv.
Calibration	Bypass onset correlation (AGS)	Abu-Ghannam B.J. & Shaw R. (1980), 'Natural transition of boundary layers - the effects of turbulence, pressure gradient and flow history', J. Mech. Eng. Sci. 22(5).
Calibration	Laminar integral method	Thwaites B. (1949), 'Approximate calculation of the laminar boundary layer', Aero. Quarterly 1.
Calibration	Turbulent entrainment / Cf	Head M.R. (1958) entrainment method; Ludwieg H. & Tillmann W. (1950) skin-friction law.
Calibration	Intermittency / transition length	Narasimha R. (1957); Dhawan S. & Narasimha R. (1958), J. Fluid Mech. 3.
Calibration	Cross-flow criterion	Arnal D. (1984), C1 cross-flow criterion; Mayle R.E. (1991), J. Turbomachinery 113.
Validation	ERCOFTAC T3A- flat plate	Roach & Brierley (1990), ERCOFTAC T3A- (Tu=0.87%); ERCOFTAC Classic Collection Case 020.
Validation	ERCOFTAC T3C4 flat plate (separation bubble)	Coupland J. (1990), ERCOFTAC T3C4; ERCOFTAC Classic Collection Case 020, variable-pressure-gradient series.
Validation	NLF(1)-0416 aerofoil transition (86 conditions)	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9; tunnel turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), NASA TP-2328, Fig. 25.
Validation	Swept-wing cross-flow (calibration set)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.
Validation	Swept-wing cross-flow (independent set)	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g).
Calibration	Linear stability / amplification database	Orr (1907); Sommerfeld (1908); Gaster M. (1962), JFM 14, 222; Mack L.M. (1977), AGARD CP-224; Drela M. & Giles M.B. (1987), AIAA J. 25(10),

		1347.
Calibration	Stationary cross-flow stability	Cooke J.C. (1950), 'The boundary layer of a class of infinite yawed cylinders', Proc. Camb. Phil. Soc. 46, 645; Mack L.M. (1984), 'Boundary-layer linear stability theory', AGARD R-709; Dagenhart & Saric, NASA/TP-1999-209344, Tables 3-5 (SALLY N-factors at f = 0, the independent check on stability.stationary_crossflow) and Sec. 2 (Radeztsky et al. on micron-scale leading-edge roughness, the receptivity the amplification factor cannot carry).
Case study	NLF aerofoil design reference	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861 (NLF(1)-0416).
Case study	Panel method	Kuethe A.M. & Chow C.Y. (1998), 'Foundations of Aerodynamics', 5th ed., Wiley.
Case study	ISA atmosphere	ICAO Standard Atmosphere (ISO 2533:1975), FL360 properties.

Table 42. Validation, calibration and case-study sources.

dataset	source
Dagenhart & Saric 45 deg swept NLF(2)-0415 (cross-flow)	Dagenhart J.R. & Saric W.S. (1999), 'Crossflow Stability and Transition Experiments in Swept-Wing Flow', NASA/TP-1999-209344, Table 2.

Table 43. Cross-flow calibration dataset provenance.

dataset	source
Boltz, Kenyon & Allen NACA 64(2)A015 untapered wing	Boltz F.W., Kenyon G.C. & Allen C.Q. (1960), 'Effects of Sweep Angle on the Boundary-Layer Stability Characteristics of an Untapered Wing at Low Speeds', NACA TN D-338, Fig. 9(g); transition locations digitised by the present author as x/c = R_trans/R.

Table 44. Independent swept-wing dataset provenance.

dataset	source	n_points	Tu_pct	orifice_pitch
Somers NLF(1)-0416 aerofoil, Langley LTPT	Somers D.M. (1981), 'Design and Experimental Results for a Natural-Laminar-Flow Airfoil for General Aviation Applications', NASA TP-1861, Fig. 9 (transition) and Table I (coordinates); turbulence level from McGhee R.J., Beasley W.D. & Foster J.M. (1984), 'Recent Modifications and Calibration of the Langley Low-Turbulence Pressure Tunnel', NASA TP-	86	0.03	0.05

	2328, Fig. 25.			

Table 45. Aerofoil dataset provenance.

12. Contribution to Knowledge

- A single unified transition kernel (Eq. E13) that locally selects the governing mechanism among natural-TS, bypass, separation-induced and cross-flow transition by putting all four on one scale of onset progress and firing at the first to complete — reproducing all regimes with one calibration set, and continuous in the free-stream turbulence across the natural/bypass handover.
- An amplification database in place of an envelope correlation: 61,600 Orr-Sommerfeld eigenvalue solutions on the Falkner-Skan family, tabulated against shape factor, momentum-thickness Reynolds number and frequency, with the march carrying one amplification factor per physical frequency. The stability solver reproduces the Blasius neutral point at Re_theta = 200 against the accepted 200.5, and the standard Blasius eigenvalue to better than a tenth of a per cent.
- A separation-bubble closure that predicts a length rather than a point: the shear layer is carried across the dead-air region by the momentum integral with no wall stress — a step with no fitted constant, which reproduces the growth the T3C4 hot films record — and reattachment placed where the disturbance has amplified by the same N_crit used elsewhere, so the length scales with the disturbance environment.
- A two-equation laminar march whose closures are computed from the Falkner-Skan family rather than fitted, giving the shape factor a history. On the 86 aerofoil conditions it raises the number of predictions inside the experimental bracket from 42 to 50, improving both surfaces at once.
- Regime coverage with one constant set: transition-onset Re_theta_t predicted to +3.6 % on T3A, -16.5 % on T3A-, -7.3 % on T3B, -30.2 % on T3C4, +0.1 % on Schubauer & Skramstad, spanning 0.03-6 % free-stream turbulence intensity.
- An aerofoil validation built for this work: 86 transition locations digitised from Fig. 9 of NASA TP-1861 for the NLF(1)-0416 section, both surfaces, four chord Reynolds numbers and lift coefficients from -1.03 to +1.62, with nothing calibrated on them. Mean error 0.039 chord, and 50 of the 86 predictions fall inside the +/-0.025c bracket within which the experiment itself localises transition.
- A robust, panel-method-cost (<1 s) 3-D capability via a span-wise strip formulation with built-in cross-flow, suitable for design-loop use where RANS/LES are impractical.
- An end-to-end, auditable workflow (geometry → mesh → setup → solution → post-processing → validation) producing a complete engineering output set (CSVs, curves, metrics, contours, temperature profiles, 3-D contours and vectors).
- Quantified NLF benefit for the case vehicle: 55 % laminar flow and 50 % viscous drag reduction relative to a fully-turbulent wing at the cruise design point.

13. Conclusions

The UTSS universal transition & skin-friction solver predicts boundary-layer transition over the three-dimensional NLF wing of the AETHER-NLF 25 and the dependent engineering quantities (skin friction, laminar extent, boundary-layer growth, separation margin, profile drag) at panel-method cost. The unified four-mechanism kernel, validated with a single calibration set against five flat plates, two independent swept-wing experiments and 86 aerofoil conditions, spans 0.03-6 % free-stream turbulence intensity. At cruise the wing achieves 55 % laminar flow and a 50 % viscous-drag reduction versus a turbulent wing; at the higher-turbulence climb condition the solver switches to the bypass route and predicts early transition, demonstrating regime coverage across the flight envelope.

Two limitations bound that claim and are stated here rather than left to be discovered. The cross-flow critical constant does not transfer between facilities: the two independent swept-wing experiments support the functional form of the criterion, and the measured data collapse on it, but reproducing the second requires a critical value 53 % larger than the first. That is now known not to be a defect of the criterion's form: the stationary cross-flow eigenvalue problem has been solved and checked against Dagenhart & Saric's own SALLY N-factors, and it brings the two levels closer — N_cf = 7.0 against 5.7, a ratio of 1.24 where the critical Reynolds numbers differ by 1.53 — while scattering worse within each facility, so it is not adopted. What the branch is missing is a receptivity input — the roughness height that seeds a stationary cross-flow vortex — which neither report gives, so no stability calculation performed here can supply it. And the method is an attached-flow formulation, so it has an incidence envelope — but that envelope is asymmetric and is set by the section, not by a round number. On the NLF(1)-0416 it declares its first condition at -9.5° of incidence, where the upper-surface layer separates within two per cent of chord, and it handles the same section down to -12.0° without complaint; at positive incidence nothing in the 86 conditions, which reach +8.7°, is declared. 2 of the 86 fall outside it and the method says so rather than returning a location. An earlier version of this report quoted ±8.5°, which was a plotting threshold rather than a measured envelope, and a later one added that a leading-edge bubble does not appear on the case-study section itself until about +14°; that is not so — it appears at +6°, and above it the separation station is not even monotone in incidence, alternating between a leading-edge bubble and a trailing-edge one. A single positive-incidence bound is therefore not a meaningful thing to quote for that section, and the evidence for the asymmetry is the aerofoil set, where it is measured.

Appendix A. Complete Generated-Output Inventory

Every file generated for this case study, by folder:

Total generated data/figure files: 127 (CSV: 63, PNG: 62, NPZ: 2).

file	type
01_geometry/airfoil_UTSS-NLF16.csv	CSV
01_geometry/geometry_definition.csv	CSV

01_geometry/wing_planform.csv	CSV
01_geometry/wing_sections_3d.csv	CSV
01_geometry/drawings/dwg_01_airfoil_section.png	PNG
01_geometry/drawings/dwg_02_planview.png	PNG
01_geometry/drawings/dwg_03_front_side.png	PNG
01_geometry/drawings/dwg_04_orthographic.png	PNG
01_geometry/drawings/dwg_05_isometric.png	PNG
01_geometry/drawings/dwg_06_section_BB.png	PNG
02_mesh/bl_normal_grid.csv	CSV
02_mesh/mesh_independence.csv	CSV
02_mesh/mesh_metrics.csv	CSV
02_mesh/surface_mesh_nodes.csv	CSV
02_mesh/plots/mesh_01_surface.png	PNG
02_mesh/plots/mesh_02_bl_normal.png	PNG
02_mesh/plots/mesh_03_independence.png	PNG
03_model_setup/calibration_constants.csv	CSV
03_model_setup/flow_conditions.csv	CSV
03_model_setup/material_properties.csv	CSV
03_model_setup/solver_settings.csv	CSV
04_solution/aero_polar.csv	CSV
04_solution/bl_profiles_cruise.csv	CSV
04_solution/field_pressure_climb.csv	CSV
04_solution/field_pressure_cruise.csv	CSV
04_solution/integrated_forces.csv	CSV
04_solution/laminar_bucket_edge.csv	CSV
04_solution/nlf_vs_turbulent.csv	CSV
04_solution/spanwise_distribution.csv	CSV
04_solution/squire_young_station_sensitivity.csv	CSV
04_solution/squire_young_station_summary.csv	CSV
04_solution/surface_climb_lower.csv	CSV
04_solution/surface_climb_upper.csv	CSV
04_solution/surface_cruise_lower.csv	CSV
04_solution/surface_cruise_upper.csv	CSV
04_solution/transition_length_sensitivity.csv	CSV
04_solution/transition_summary.csv	CSV
05_postprocessing/csv_plots/aero_polar.png	PNG
05_postprocessing/csv_plots/calibration_constants.png	PNG
05_postprocessing/csv_plots/climb_Cf.png	PNG
05_postprocessing/csv_plots/climb_Cp.png	PNG
05_postprocessing/csv_plots/climb_Retheta.png	PNG
05_postprocessing/csv_plots/climb_gamma.png	PNG
05_postprocessing/csv_plots/climb_theta_H.png	PNG
05_postprocessing/csv_plots/compare_cruise_climb_Cf.png	PNG
05_postprocessing/csv_plots/conditions_compare.png	PNG
05_postprocessing/csv_plots/cruise_Cf.png	PNG
05_postprocessing/csv_plots/cruise_Cp.png	PNG
05_postprocessing/csv_plots/cruise_Retheta.png	PNG
05_postprocessing/csv_plots/cruise_gamma.png	PNG
05_postprocessing/csv_plots/cruise_theta_H.png	PNG
05_postprocessing/csv_plots/geo_airfoil.png	PNG
05_postprocessing/csv_plots/geo_planform.png	PNG
05_postprocessing/csv_plots/geo_sections_3d.png	PNG
05_postprocessing/csv_plots/mesh_independence.png	PNG
05_postprocessing/csv_plots/mesh_spacing.png	PNG
05_postprocessing/csv_plots/mesh_yplus.png	PNG
05_postprocessing/csv_plots/nlf_vs_turbulent.png	PNG

05_postprocessing/csv_plots/spanwise_transition.png	PNG
05_postprocessing/csv_plots/table_geometry_definition.png	PNG
05_postprocessing/csv_plots/table_integrated_forces.png	PNG
05_postprocessing/csv_plots/table_material_properties.png	PNG
05_postprocessing/csv_plots/table_mesh_metrics.png	PNG
05_postprocessing/csv_plots/table_solver_settings.png	PNG
05_postprocessing/csv_plots/transition_summary_bar.png	PNG
05_postprocessing/three_d/td_Cf.png	PNG
05_postprocessing/three_d/td_Cp.png	PNG
05_postprocessing/three_d/td_gamma.png	PNG
05_postprocessing/three_d/td_skinfriction_vectors.png	PNG
05_postprocessing/contours/contour_Cp_climb.png	PNG
05_postprocessing/contours/contour_Cp_cruise.png	PNG
05_postprocessing/contours/contour_speed_climb.png	PNG
05_postprocessing/contours/contour_speed_cruise.png	PNG
05_postprocessing/contours/vectors_climb.png	PNG
05_postprocessing/contours/vectors_cruise.png	PNG
05_postprocessing/profiles/bl_temperature_profiles.png	PNG
05_postprocessing/profiles/bl_temperature_ratio.png	PNG
05_postprocessing/profiles/bl_velocity_profiles.png	PNG
06_validation/ablations.csv	CSV
06_validation/aerofoil_nlf0416.csv	CSV
06_validation/aerofoil_nlf0416_source.csv	CSV
06_validation/aerofoil_nlf0416_summary.csv	CSV
06_validation/bubble_diagnostics.csv	CSV
06_validation/bubble_length_scaling.csv	CSV
06_validation/crossflow_amplification.csv	CSV
06_validation/crossflow_amplification_summary.csv	CSV
06_validation/crossflow_criticals.csv	CSV
06_validation/crossflow_criticals_summary.csv	CSV
06_validation/crossflow_formulations.csv	CSV
06_validation/crossflow_receptivity.csv	CSV
06_validation/crossflow_receptivity_summary.csv	CSV
06_validation/crossflow_reynolds_trend.csv	CSV
06_validation/crossflow_threshold_sweep.csv	CSV
06_validation/experiment_T3A.csv	CSV
06_validation/experiment_T3AM.csv	CSV
06_validation/experiment_T3B.csv	CSV
06_validation/experiment_T3C4.csv	CSV
06_validation/residual_diagnostics.csv	CSV
06_validation/solver_SS.csv	CSV
06_validation/solver_T3A.csv	CSV
06_validation/solver_T3AM.csv	CSV
06_validation/solver_T3B.csv	CSV
06_validation/solver_T3C4.csv	CSV
06_validation/sources_and_references.csv	CSV
06_validation/swept_wing_crossflow.csv	CSV
06_validation/swept_wing_independent.csv	CSV
06_validation/swept_wing_independent_source.csv	CSV
06_validation/swept_wing_source.csv	CSV
06_validation/transition_length_forms.csv	CSV
06_validation/transition_length_measured.csv	CSV
06_validation/validation_summary.csv	CSV
06_validation/plots/val_SS.png	PNG
06_validation/plots/val_T3A.png	PNG
06_validation/plots/val_T3AM.png	PNG

06_validation/plots/val_T3B.png	PNG
06_validation/plots/val_T3C4.png	PNG
06_validation/plots/val_aerofoil_nlf0416.png	PNG
06_validation/plots/val_combined_Re_theta_t.png	PNG
06_validation/plots/val_crossflow_amplification.png	PNG
06_validation/plots/val_swept_crossflow.png	PNG
06_validation/plots/val_swept_independent.png	PNG
07_equations/equations_index.csv	CSV
solver/amplification_db.npz	NPZ
solver/crossflow_db.npz	NPZ
assets/banner.png	PNG
assets/headline.csv	CSV
assets/social-preview.png	PNG

Table 46. Generated-output inventory.

Appendix B. Parameter / Metric CSVs Rendered as Figures

For completeness, every remaining parameter and metric CSV is also rendered as a clean figure so that all generated CSV data appear in plotted form (the same data also appear as native tables earlier).

Geometry definition (geometry_definition.csv)

parameter	value	unit
Aircraft	AETHER-NLF 25	-
Wing reference area S	33.150	m^2
Wing span b	17.00	m
Aspect ratio AR	8.72	-
Root chord c_root	2.600	m
Tip chord c_tip	1.300	m
Taper ratio	0.500	-
Mean aerodynamic chord MAC	2.022	m
Leading-edge sweep	12.0	deg
Dihedral	4.0	deg
Tip washout (twist)	-3.0	deg
Section	UTSS-NLF16	-
Section max thickness	16.0	% chord
Max-thickness location	42.2	% chord
Trailing-edge included angle	26.8	deg
Section c_l, 2-D unswept (alpha = 1.5 deg, M = 0.42)	0.517	-

Fig. 53. geometry_definition.csv, rendered as a figure.

Mesh metrics (mesh_metrics.csv)

metric	value	unit
Surface streamwise nodes	261	-
Wall-normal layers (reconstruction)	44	-
First-cell height y1	7.80	micron
Target wall y+	0.80	-
Wall-normal growth ratio	1.12	-
Min streamwise spacing	0.151	1e-3 c
Max streamwise spacing	12.217	1e-3 c
Max streamwise spacing at MAC	24.705	mm
LE clustering ratio	81.1	-
BL grid outer extent	8.43	mm
Peak turbulent C_f (cruise, both surfaces)	0.00209	-
Friction velocity u_tau at that peak	4.010	m/s
Max cell aspect ratio (at MAC)	3169	-
Discretisation type	surface panel distribution + wall-normal reconstruction stack (no volume mesh is generated)	-

Fig. 54. mesh_metrics.csv, rendered as a figure.

Air / material properties (material_properties.csv)

property	value	unit
Working fluid	air (ideal gas)	-
Specific gas constant R	287.05	J/kg.K
Ratio of specific heats gamma	1.40	-
Prandtl number Pr	0.72	-
Sutherland reference mu0	1.716e-05	Pa.s
Sutherland reference T0	273.15	K
Sutherland constant S	110.4	K
Recovery factor, laminar (Pr^1/2)	0.849	-
Recovery factor, turbulent (Pr^1/3)	0.896	-
Cruise dynamic viscosity	1.422e-05	Pa.s
Cruise density	0.36392	kg/m^3

Fig. 55. material_properties.csv, rendered as a figure.

Solver configuration (solver_settings.csv)

setting	value
Inviscid method	Constant-strength vortex-panel (Kuethe-Chow)
Kutta condition	Enforced at sharp trailing edge
Surface panels	260 (cosine-clustered)
Compressibility (pressure)	Karman-Tsien correction on C_p
Compressibility (boundary layer)	closures evaluated at Eckert's reference temperature
Laminar BL closure	two-equation march (momentum + kinetic energy); H*, I and d from the Falkner-Skan family
Transition model	UTSS unified 4-mechanism kernel
TS / natural	e^N, one factor per physical frequency, growth rates from the tabulated Orr-Sommerfeld database
bypass	Abu-Ghannam & Shaw at the flow-history-averaged Tu
separation	laminar bubble closed by the amplification integral across the dead-air region
cross-flow	C1 on Re_theta2, closed by an amplification integral
Intermittency closure	Narasimha universal (gamma)
Transition length	Dhawan & Narasimha at constant spot formation rate, stated in Re_theta: Re_lambda = (9/0.664^1.5) Re_theta_t^1.5, which is their published Re_lambda = 9 Re_x_t^0.75 wherever Re_theta = 0.664 sqrt(Re_x) and is the form that stays exact in a pressure gradient (cal['len_re_x'] = True recovers the published form)
Turbulent BL closure	Head entrainment + Ludwieg-Tillmann Cf
Laminar separation criterion	Thwaites lambda <= -0.09
Turbulent separation flag	H > 2.6
Drag integration	Squire-Young far-wake, evaluated at 0.98c
Marching scheme	explicit trapezoidal, arc-length stepping with sub-stepping on the kinetic-energy equation
Convergence tol (panel)	1e-10 (direct solve)
Wall y+ target	0.80 (reconstruction grid; no volume mesh is solved on it)

Fig. 56. solver_settings.csv, rendered as a figure.

Integrated forces (integrated_forces.csv)

quantity	value	unit
Section lift coefficient Cl	0.4962	-
Section profile drag Cd	0.00473	-
Section Cd (counts)	47.3	counts
Section L/D	104.9	-
Section lift-curve slope a0 (panel)	8.111	1/rad
Zero-lift incidence a_L0 (panel)	-2.185	deg
Quarter-chord sweep	9.89	deg
Wing C_L (lifting line, taper + washout + sweep)	0.2548	-
Wing C_L / section c_l	0.513	-
Span efficiency e (lifting line)	0.8718	-
Wing induced drag C_Di (lifting line)	0.00272	-
Wing lift (lifting line)	23603.0	N
Wing C_L for level flight at MTOW	0.8998	-
Incidence for that C_L (lifting line)	7.55	deg
Dynamic pressure q	2794.6	Pa
Reynolds number Re_MAC	6.414e+06	-
Mach number	0.42	-

Fig. 57. integrated_forces.csv, rendered as a figure.